



CHAPTER ATA 28 FUEL SYSTEM

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

Contents

Contents	3
List of Figures	Error! Bookmark not defined.
Fuel System – General	4
Fuel Storage - Fuel Tanks	6
Fuel Tanks – Component Description	7
Vents	13
Vents- Component Description	15
Distribution – Engine Fuel Feed System	18
Engine Fuel Feed System – Component Description	20
Auxiliary Pump Manual Operation – Electrical Schematic	26
APU Fuel Feed System	27
APU Fuel Feed system – Component Description	28
Fuel Transfer System	29
Fuel Transfer System – Component Description	32
Fuel Transfer Operation – Electrical Schematic	35
Refuel/Defuel System	39
Refuel/ Defuel System – Component Description	44
Refuel/Defuel System – Controls & Indications	47
Preselect Refueling Operation	48
Preset Refueling – Electrical Schematic	50
Fuel Indicating System	54
Fuel Indicating System – Component Description	57
Fuel Quantity Indication – Electrical Schematic	63

Fuel System – General

Introduction

Fuel is contained in two integral main wing tanks designated No. 1 (Left) and No. 2 (Right). The fuel system provides for indicating, storing, venting, fuel feeding and scavenging, refueling/defueling, and transferring. Only tank to tank transfer is available; there is no engine cross feed capability. The aircraft may be gravity or pressure refueled.

General Description

[Refer Figure 28-1](#)

Each wing tank includes a surge bay and a collector bay. The left tank supplies fuel to the left engine and APU. The right tank supplies fuel to the right engine.

There are three drain valves at the lowest points of each tank to permit checks for water accumulation and to drain residual fuel from the tanks when required.

Normal engine feed is by means of an ejector pump in each collector bay. The engine Fuel Metering Unit (FMU) supplies the motive flow.

No electrical power is required for normal engine feed operation. In the event of an ejector pump failure, the auxiliary AC electrical pump in the collector bay is selected ON to supply engine fuel feed.

Four scavenge ejector pumps transfer fuel inboard to the collector bay. This maintains a constant level in the collector bay to ensure engine fuel feed regardless of aircraft attitude and minimizes the quantity of unusable fuel.

A low fuel level sensor is in each collector bay to provide a caution indication to the crew.

Pressure refuelling and defueling is accomplished through a single point refuel/defuel adaptor in the right engine nacelle and is controlled through an adjacent refuel/defuel control panel. Both automatic and manual refuelling

modes are possible.

Selected and actual fuel tank quantities are displayed on the control panel. The Fuel Quantity Computer (FQC), under the center aisle floorboards, controls the logic during the automatic refuel/defuel process. Refuel operations can be accomplished using aircraft battery power.

A high level sensor shutoff system ensures the maximum allowable tank capacity is not exceeded.

As an alternative to pressure refueling and defueling, each tank can be filled through an over wing filler port.

Each fuel tank is vented through a surge bay in the outboard section of the main tank. Fixed vent lines and float vent valves between the fuel tanks and the surge bays provide adequate capacity for in flight venting. During pressure refueling, tank venting is supplemented by a high capacity vent valve which is opened by refuel pressure.

The Fuel Quantity Gauging System (FQGS) accurately measures the quantity of fuel in the main tanks. It supplies the fuel quantity data to the flight compartment and the refuel/defuel panel for display.

Fuel quantity may also be checked on the ground by use of the magnetic dipsticks on the lower surface of the wing. The maximum usable fuel capacity of the aircraft is approximately 5,338 kg (11,724 lb).

A wing-to-wing fuel transfer system is provided to correct lateral fuel imbalance. The auxiliary pump in the donor tank transfers the fuel to the receiving tank.

The FUEL CONTROL TRANSFER panel in the flight compartment controls the operation of the fuel transfer system. The maximum allowable tank to tank imbalance is 272 kg (600 lb).

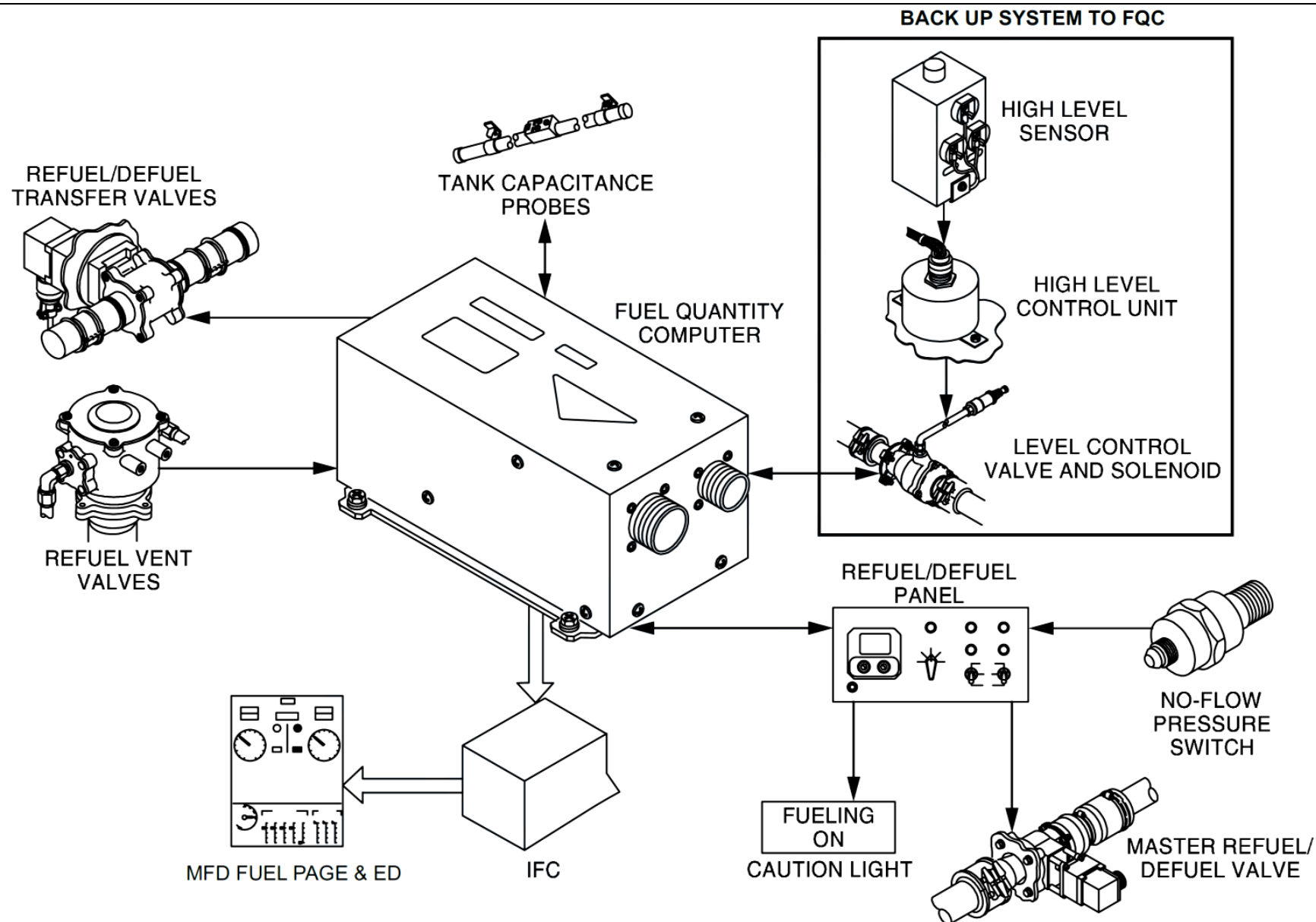


Figure 28-1 Fuel System Block Diagram

Fuel Storage - Fuel Tanks

Introduction

The fuel tanks contain the fuel to be used by the aircraft engines and optional Auxiliary Power Unit (APU). The fuel tanks have the necessary components to move the fuel from the main tanks to the collector bays for use by the fuel distribution system.

General Description

[Refer Figures 28-2 & 28-3](#)

Fuel is stored in two wing tanks which are integral parts of the aircraft structure. The left (No.1) tank supplies fuel to the left (No.1) engine and the auxiliary power unit (APU). The right (No.2) tank supplies fuel to the right (No.2) engine. Each wing tank includes a surge bay and a collector bay. There are vents in the fuel storage system to prevent over-pressurization of the fuel tanks.

The fuel tanks include the components that follow:

- Flapper Check Valves
- Motive Flow Lines
- Motive Flow Check Valves
- Scavenge Flow Lines
- Forward Scavenge Ejector
- Aft Scavenge Ejector
- Mid Wing Scavenge Ejector
- Outboard Scavenge Ejector
- Gravity Refill Cap

- Water Drain Valve

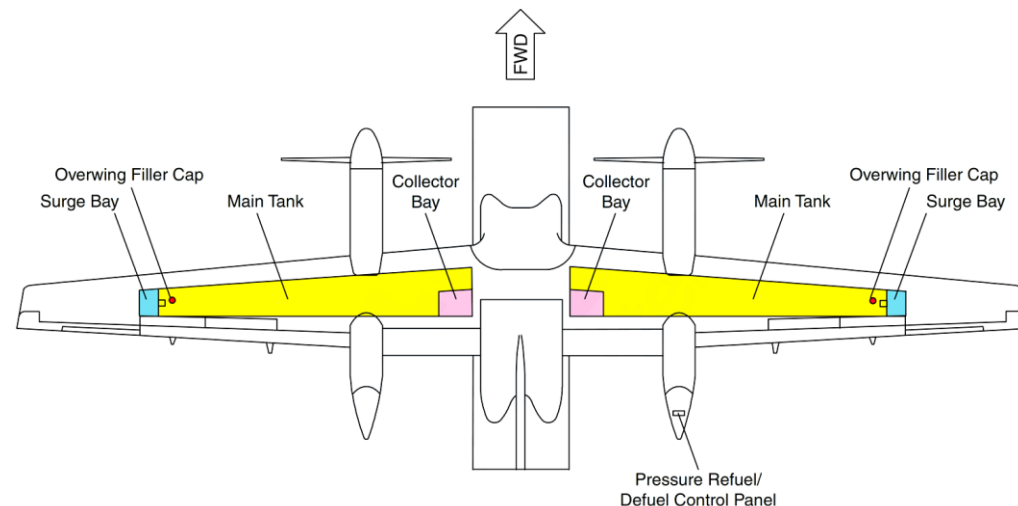


Figure 28-2 Fuel Tanks

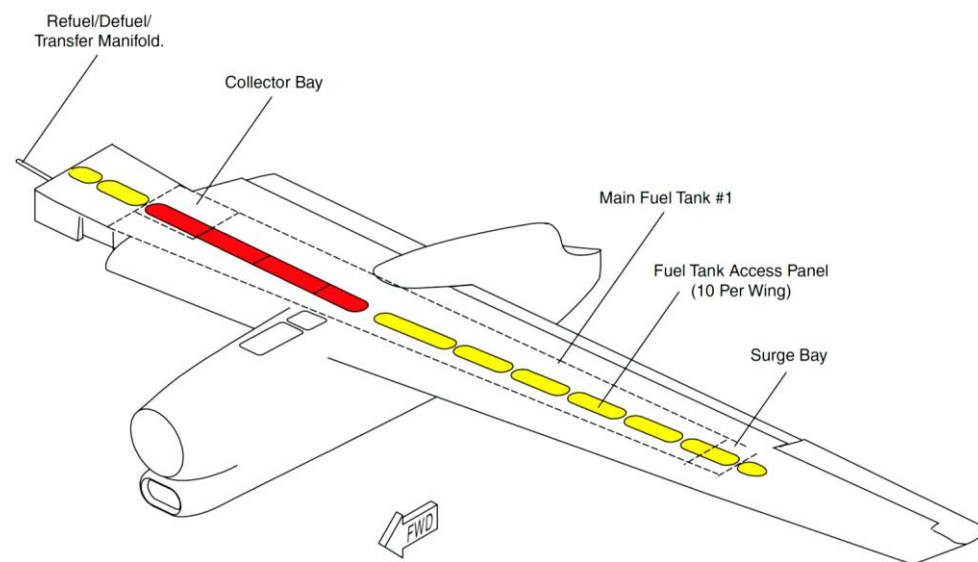


Figure 28-3 Fuel Tank – Sections

Detailed Description

There are two fuel tanks in the aircraft; one in each wing. Each fuel tank is formed by the structure of the wing itself. This is known as an integral wet wing tank.

Each tank is divided into three sections:

- the main tank
- the surge bay
- the collector bay

The main tank extends from the surge bay to the fuselage and collector bay. The collector bay is located at the inboard and aft part of the wing tank. The surge bay is located between the two ribs inboard of the aileron.

The inboard end of the main fuel tank is the wing rib at WS 42 and the outboard end is the wing rib at WS 407. The fuel tanks are formed by the front and rear wing spars, and the upper and lower wing skins. The main tanks and the collector bays are the storage areas for the fuel.

The total volume of the two wing tanks is 1,785 US gal (6757 L), 12,138 lb at 6.8 lb/gal (5505.7 kg at 0.816 kg/L). The total usable fuel from the tanks is 5,318 kg (11,724 lbs). The maximum lateral imbalance permitted between tanks is 272 kg (600 lbs).

The surge bay is used for main tank venting and fuel recovery. Two outboard float vent valves and one inboard vent line control the pressure between the related surge bay and main tank. The two float vent valves, located near the top of the tank, open and close depending on the fuel level in the main tank.

The collector bay supplies engine fuel regardless of the aircraft attitude. Fuel tank scavenging takes place using scavenge ejector pumps in each tank which draw fuel from tank low points, to keep the collector bay full. A primary ejector pump in the collector bay then provides a constant low-pressure fuel supply to the engine. High-pressure motive flow is used to operate the

scavenge and primary ejector pumps.

The related No.1 TANK and No.2 TANK FUEL LOW caution light will come on if:

- PARK BRAKE is off
- related collector bay level drops below approximately 150 kg (305 lbs)
- related engine is operating.

Fuel Tanks – Component Description

Flapper Check Valves

[Refer Figure 28-4](#)

The flapper check valves are simple mechanical valves that let fuel flow inboard only. There are 36 flapper check valves, 18 in each wing. They are located at the bottom of the wing ribs and along the forward and side walls of the collector bay.

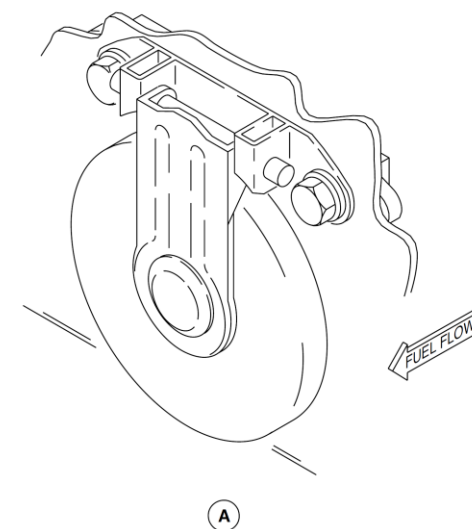
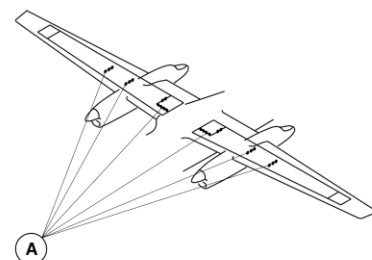


Figure 28-4 Flapper Check Valves

Motive Flow Check Valves

[Refer Figure 28-5](#)

There is one motive flow check valve in each tank. Each motive flow check

valve is located in its related tank, downstream of each engine main motive flow line. The check valves prevent back-flow through the motive flow lines.

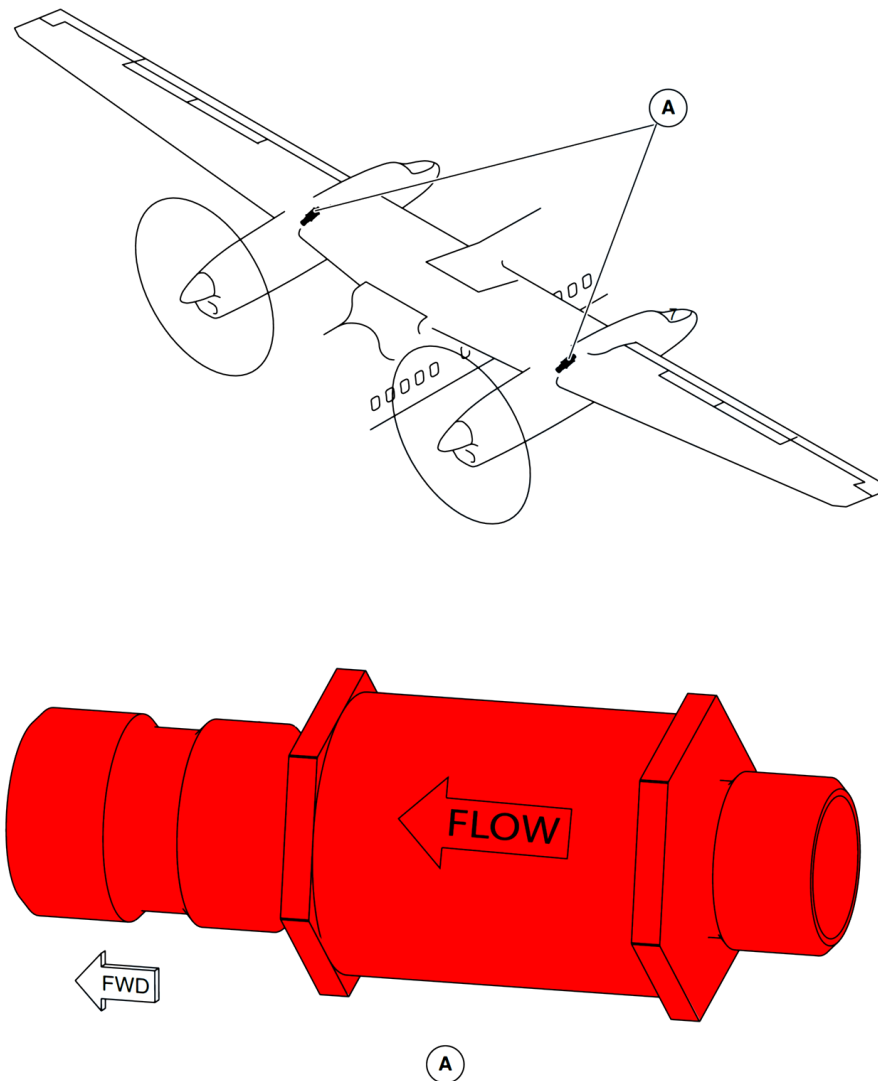


Figure 28-5 Motive Flow Check Valve

Motive Flow Lines

[Refer Figure 28-6](#)

The engine fuel pumps supply motive flow as a low volume, high pressure fuel flow. Motive flow is pumped to the scavenge ejectors and to the engine fuel feed system primary ejectors through the motive flow lines.

Scavenge Flow Lines

[Refer Figure 28-6](#)

The scavenge flow lines are used to transfer fuel from the main tanks to the collector bays.

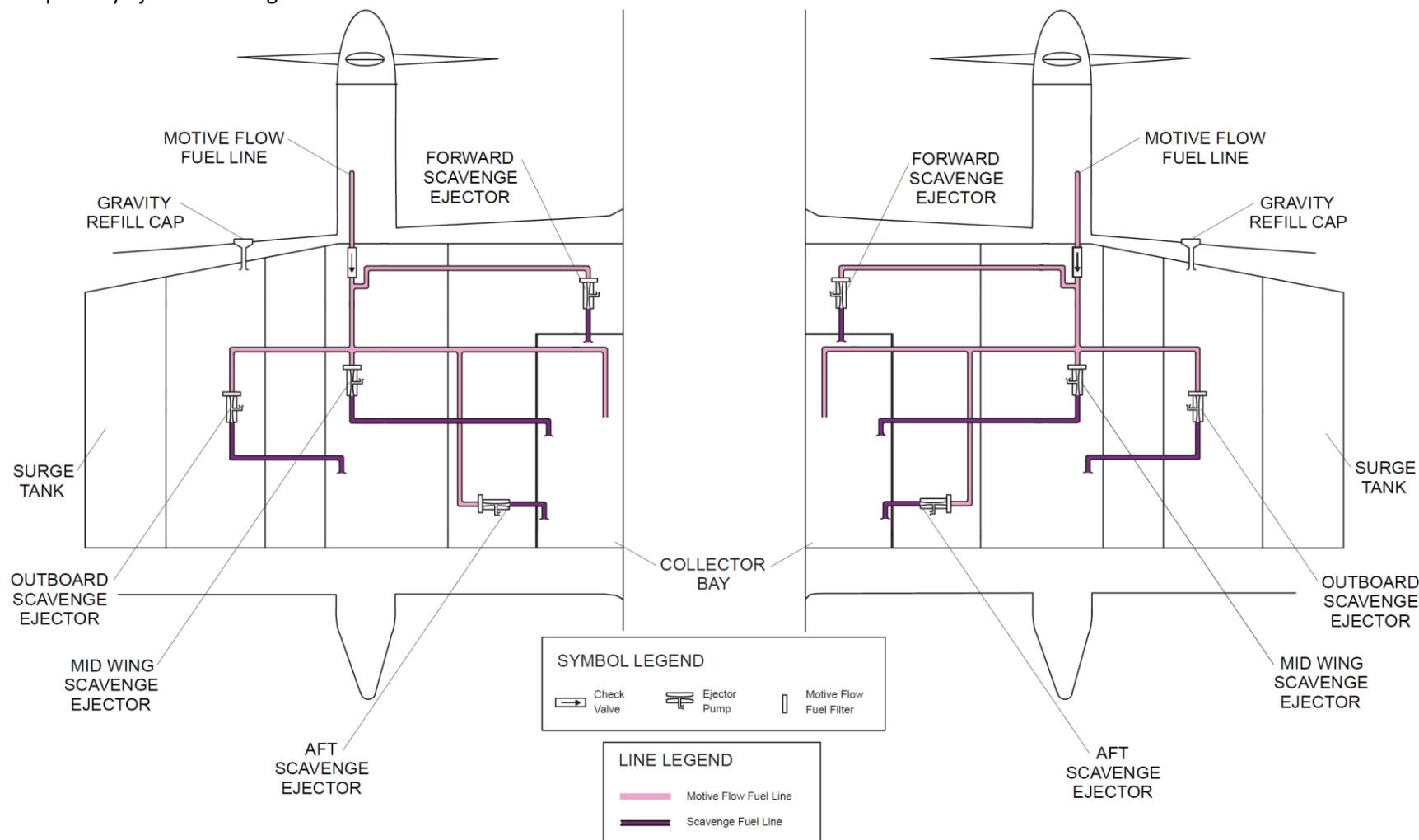


Figure 28-6 Motive Flow and Scavenge System

Forward Scavenge Ejector

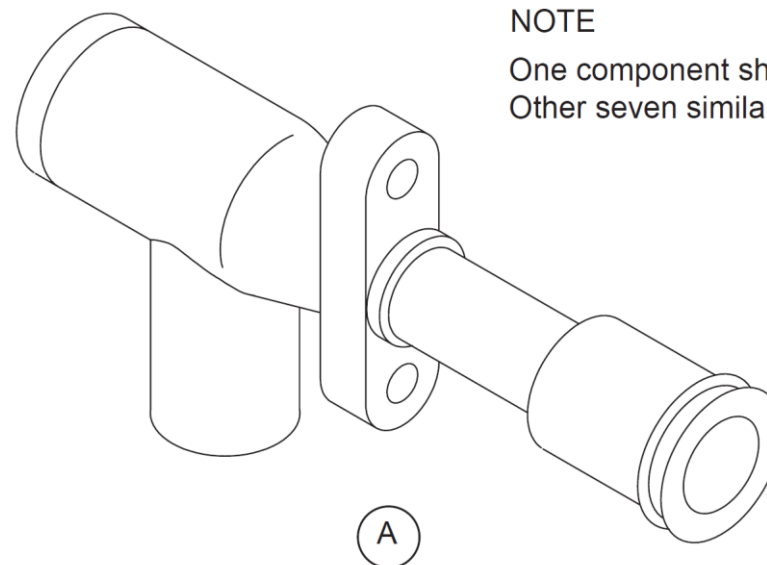
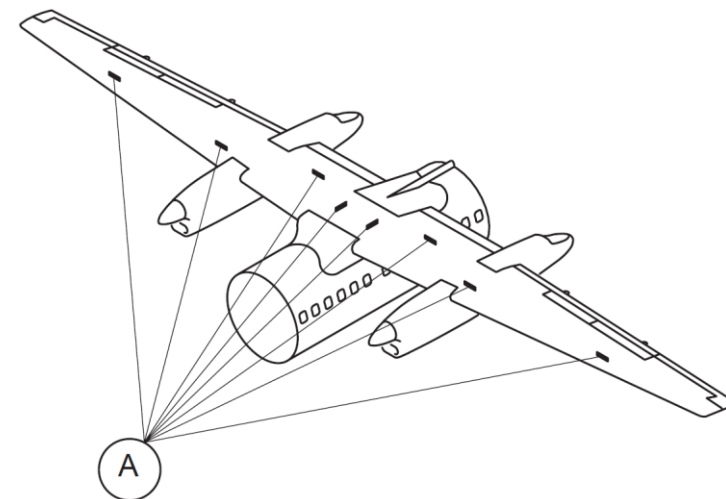
[Refer Figures 28-6, 28-7 & 28-8](#)

There is one forward scavenge ejector in each tank. The ejectors are located in the forward inboard area of the wings. Motive flow fuel is pumped to the forward scavenge ejector through the motive flow lines. The low volume, high pressure motive flow causes the ejector to discharge a high volume, low pressure flow into the scavenge lines. They supply fuel to the collector bays through the scavenge flow lines. A 250 micron filter at the motive port of the ejector protects the small nozzle opening.

Aft Scavenge Ejector

[Refer Figures 28-6, 28-7 & 28-8](#)

There is one aft scavenge ejector in each tank. The ejectors are located in the aft inboard area of the wings. Motive flow is pumped to the scavenge ejectors through the motive flow lines. The low volume, high pressure motive flow causes the ejector to discharge a high volume, low pressure flow into the scavenge lines. The aft scavenge ejectors supply fuel to the collector bays through the scavenge flow lines. A 250 micron filter at the motive port of the ejector protects the small nozzle opening.



NOTE

One component shown.
Other seven similar.

Figure 28-7 Scavenge Ejector Pump

Mid Wing Scavenge Ejector

[Refer Figures 28-6, 28-7 & 28-8](#)

There is one mid wing scavenge ejector in each tank. The ejectors are located between the outboard scavenge ejectors and the collector bays. Motive flow is pumped to the scavenge ejectors through the motive flow lines. The low volume, high pressure motive flow causes the ejector to discharge a high volume, low pressure flow into the scavenge lines. The mid wing scavenge ejectors supply fuel to the collector bays through the scavenge flow lines. A 250 micron filter at the motive port of the ejector protects the small nozzle opening.

Outboard Scavenge Ejector

[Refer Figures 28-6, 28-7 & 28-8](#)

There is one outboard scavenge ejector in each tank. The scavenge ejectors are located in the outboard area of the wings. Motive flow is pumped to the scavenge ejectors through the motive flow lines. The low volume, high pressure motive flow causes the ejector to discharge a high volume, low pressure flow into the scavenge lines. The outboard scavenge ejectors supply fuel to the inboard section of the wing. A 250 micron filter at the motive port of the ejector protects the small nozzle opening.

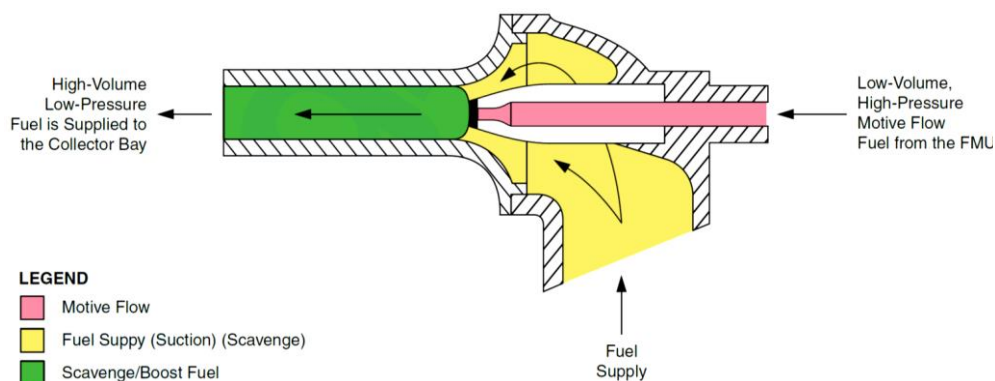


Figure 28-8 Ejector Pump

Gravity Refill Cap

[Refer Figure 28-9](#)

There is one gravity refill cap installed at the outboard section of each wing. The stainless steel gravity refill caps are lightning proof and are installed on an access panel flush with the wing surface.

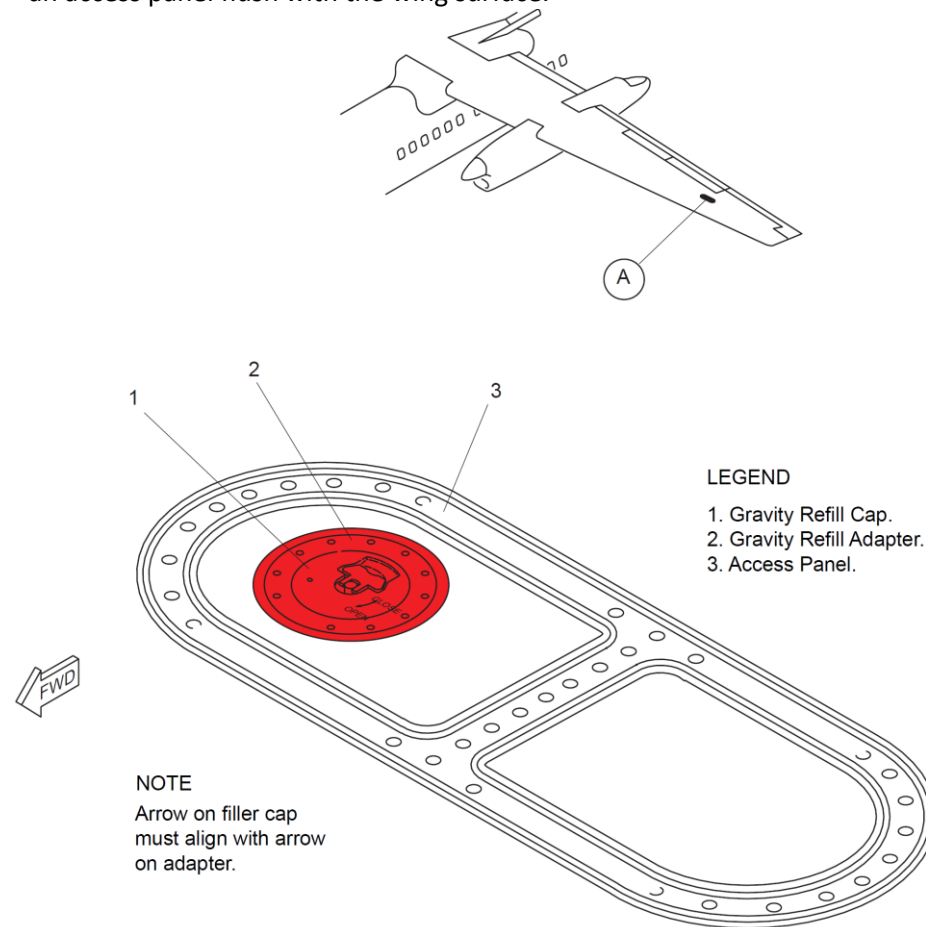


Figure 28-9 Gravity Refill Cap

Water Drain Valve

[Refer Figure 28-10 & 28-11](#)

There are six water drain valves installed on the aircraft, three in each wing. They are located at the lowest points of the surge bay, main tank and collector bay to minimize the amount of trapped fuel. The drains may be opened to check for, and drain, any water accumulation in the tank. The valves are also used to drain residual fuel from the tanks when necessary.

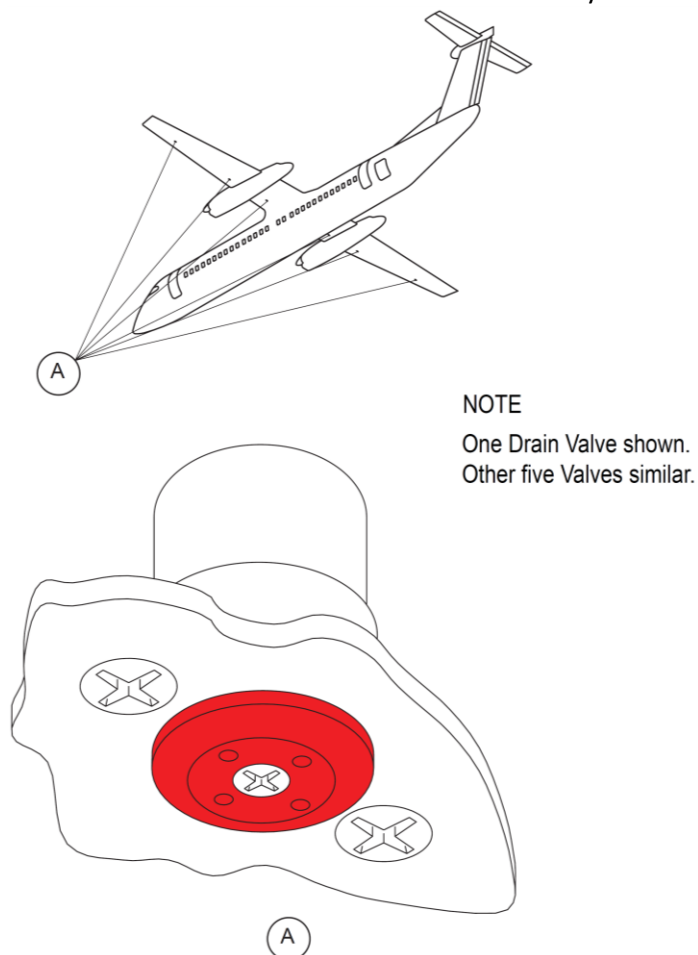


Figure 28-10 Water Drain Valve

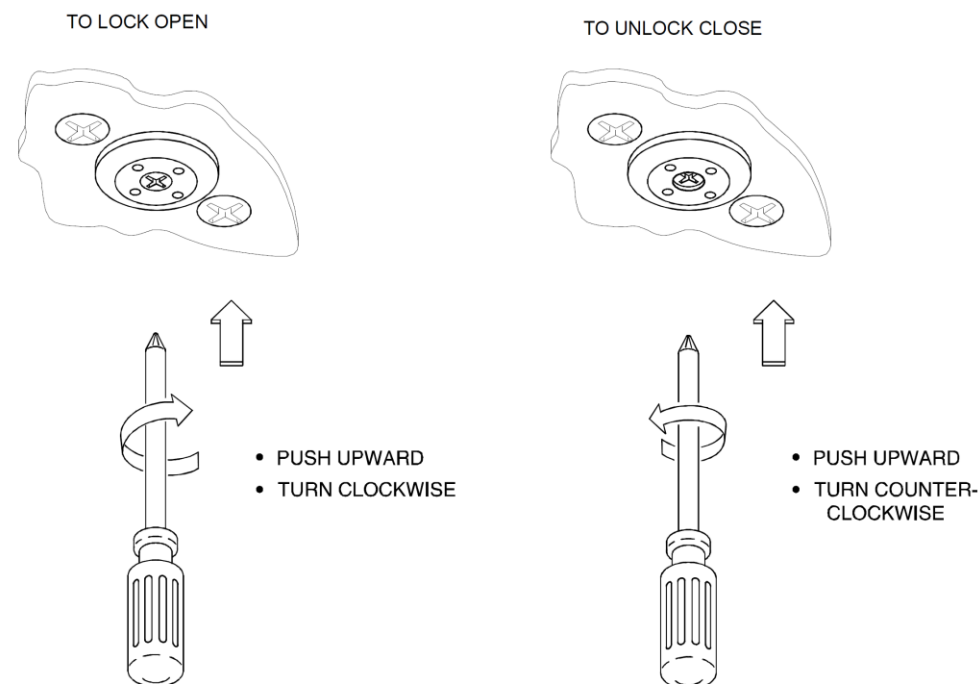


Figure 28-11 Water Drain Valve Operation

Introduction

The vent system connects each fuel tank to ambient air pressure. This equalizes the pressure inside the tank with the pressure outside the tank and supplies sufficient venting of the fuel tanks for any aircraft flight attitude. Venting of the fuel tanks also prevents any structural damage to the aircraft during refueling or defueling.

General Description

The vent system lets air enter the fuel tanks as the fuel is supplied to the engines, or during aircraft defueling. This prevents structural damage to the tanks from an excessively low internal pressure. The vents also let air out of the tanks when the aircraft is being refueled to prevent the tanks from over pressurizing.

The vent system includes the components that follow:

- Inboard Vent Line
- Outboard Vent Line
- Vent Float Valves
- NACA Vent Inlet

Detailed Description

[Refer Figures 28-12 & 28-13](#)

The vent system keeps the wing tank pressure within safe limits for all aircraft operations and flight maneuvers. It allows unrestricted air flow between the tank and ambient air. Each fuel tank is vented to its related surge bay through an inboard vent line. The surge bay itself is vented to the outside through two NACA vents located on the underside of the wing.

Each surge bay also has two (forward and aft) outboard vent lines with vent float valves. The vent float valves close the outboard vent lines when the fuel level rises above the opening of the vent lines. During a climb, the aft vent float valve closes and venting is supplied by the forward outboard vent line.

Vents

During a descent, the forward vent float valve closes and venting is supplied by the aft outboard vent line.

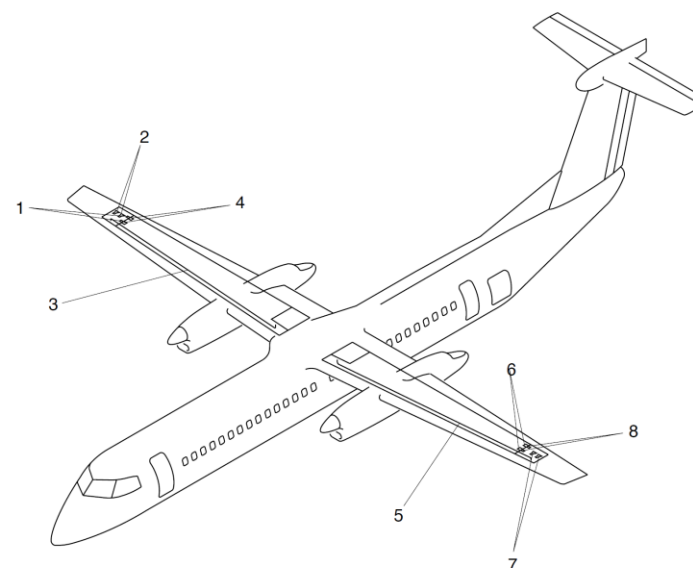


Figure 28-12 Vent System

LEGEND

1. Outboard Vent Lines.
2. NACA Vents.
3. Inboard Vent Line.
4. Vent Float Valves (2).
5. Inboard Vent Line.
6. Vent Float Valves (2).
7. NACA Vents.
8. Outboard Vent Lines.

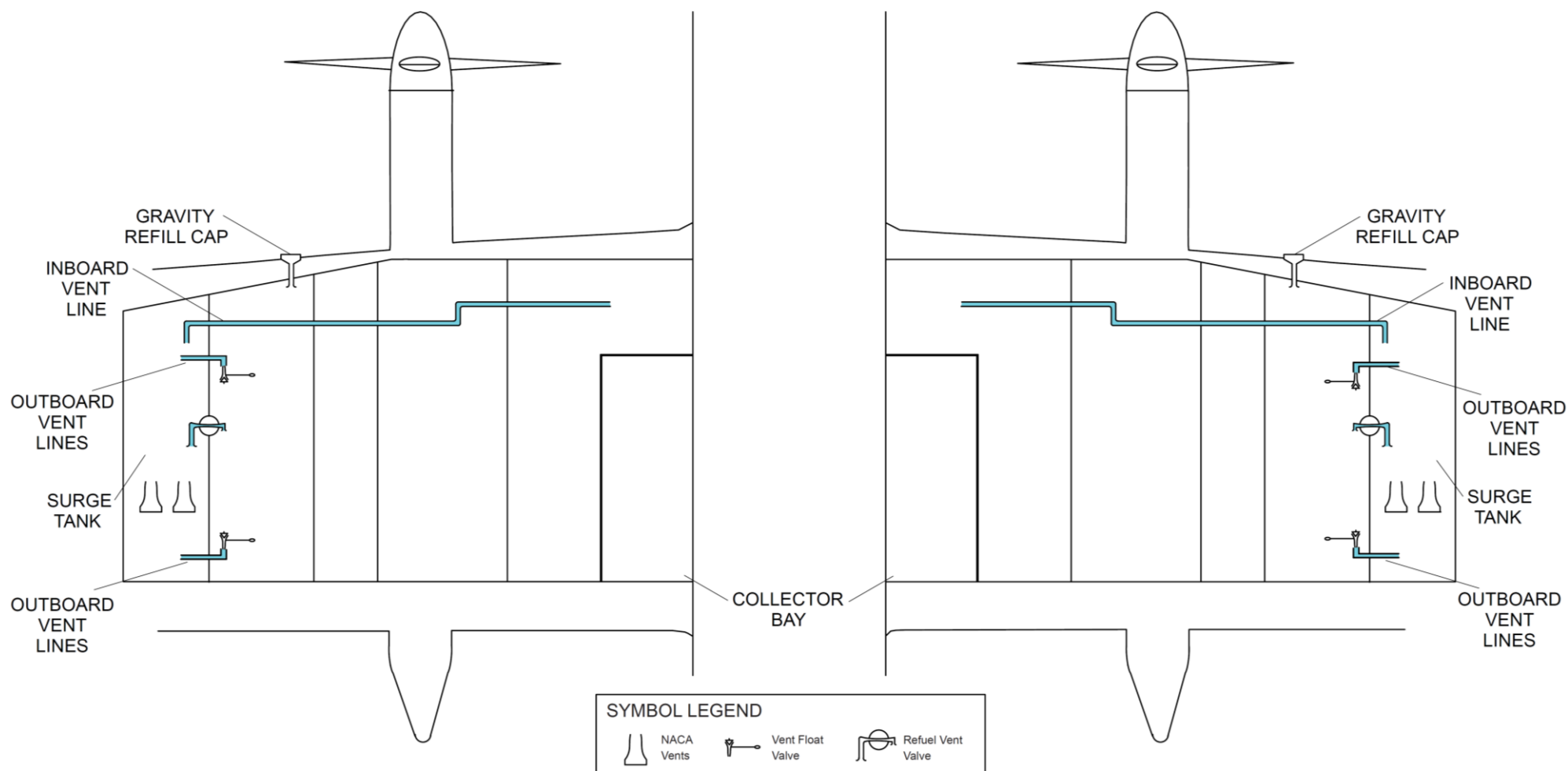


Figure 28-13 Vent System

Vents- Component Description

Inboard Vent Line

[Refer Figure 28-13](#)

An inboard vent line is routed from the surge bay along the forward wing spar to the inboard part of the main fuel tank. This line supplies venting when the fuel is moving outboard or inboard in the tank, as in climb, uncoordinated flight maneuvers, accelerated rolls, refueling and defueling. The line is completely unobstructed in order to supply continuous overpressure protection for the wing tank.

The outboard end of the vent line extends down to within 0.5 in. (12.7 mm) of the bottom of the surge bay. Due to the slight pressurization of the surge bay excess fuel in the bottom of the bay is scavenged back into the main tank.

Outboard Vent Line

[Refer Figure 28-13](#)

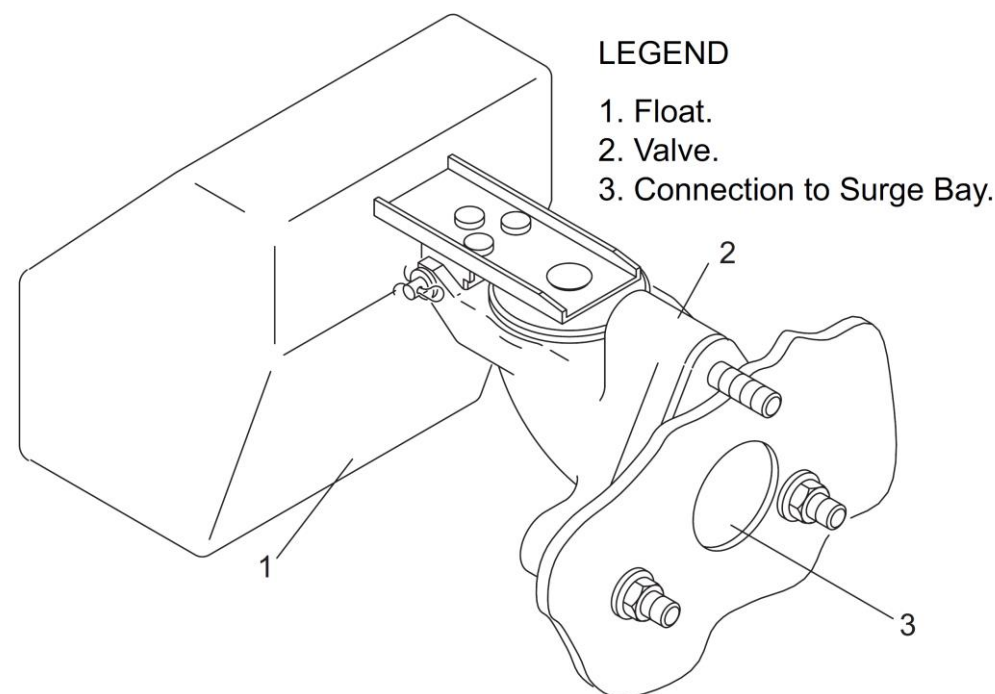
There are two outboard vent lines in each fuel tank connecting the main fuel tank to the surge bay. The vent lines extend down to within 0.5 in. (12.7 mm) of the bottom of the surge bay. As the surge bay is pressurized in flight, through the NACA vents, any fuel in the surge bay is scavenged by the outboard vent lines back into the main tank.

The vent float valves, located near the top of the tank, open and close the outboard vent lines depending on the fuel level in the main tank.

Vent Float Valves

[Refer Figure 28-14 & 28-15](#)

Two vent float valves are located near the top of the rib dividing the main tank from the surge bay (WS 407). The float valve closes the outboard vent line if the fuel level in the tank rises above the opening of the vent valve. The outboard vent lines located on the forward part of the fuel tanks, supply venting during climbs. The vent lines located on the aft part of the tanks supply venting in descents.



LEGEND

1. Float.
2. Valve.
3. Connection to Surge Bay.

Figure 28-14 Vent Float Valve

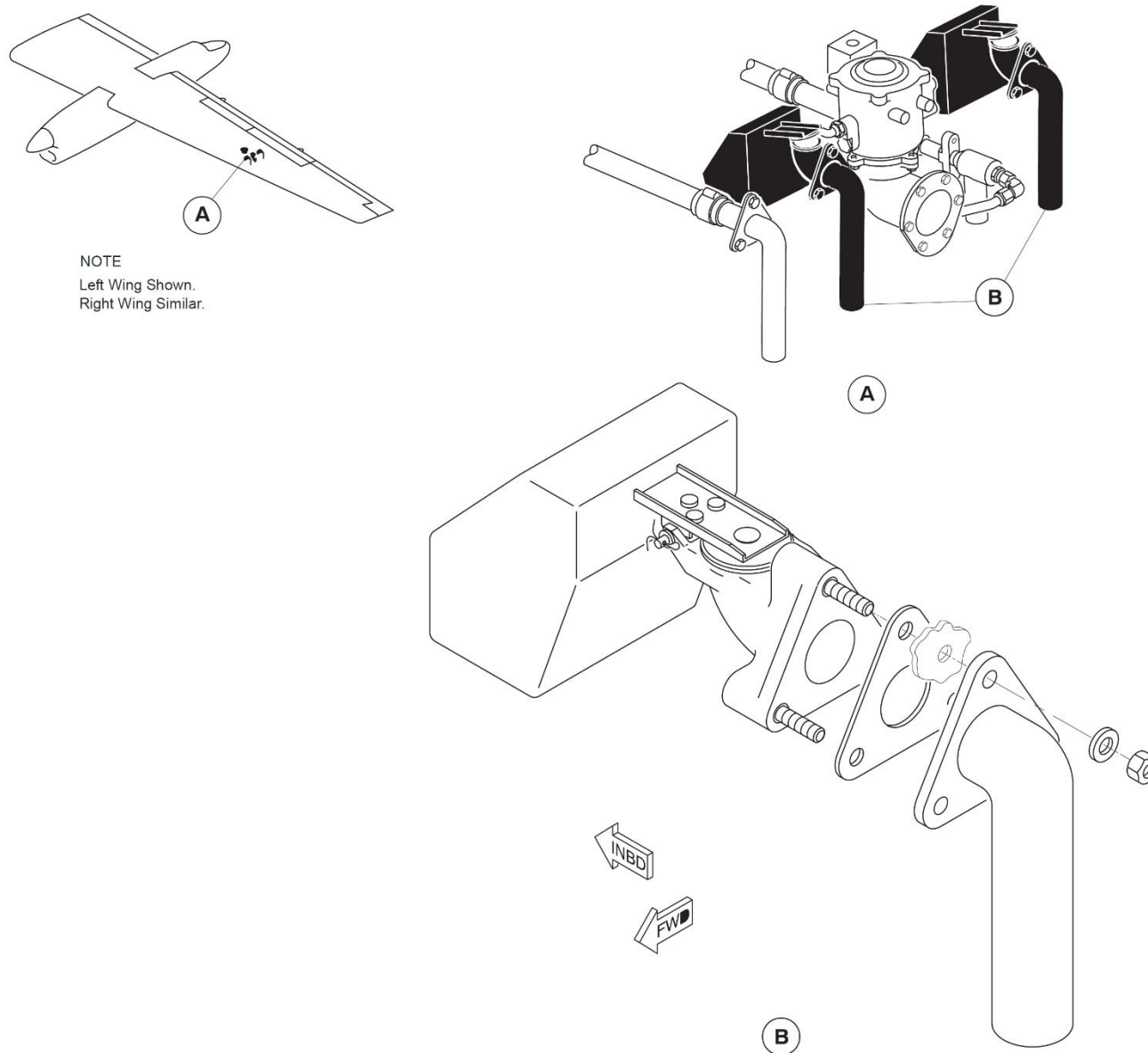


Figure 28-15 Vent Float Valve & Stand Pipe

Inlet NACA Vents

[Refer Figure 28-16](#)

The surge bay is vented overboard through dual NACA vents. The vents are located on the lower front edge of each wing. They are ice-free ram air vents

which keep a slight positive pressure in the fuel tank while in flight. The NACA vents are connected to standpipes which extend up to the top rear of the surge bay. These standpipes prevent any fuel which may accumulate in the bottom of the surge bay from spilling overboard.

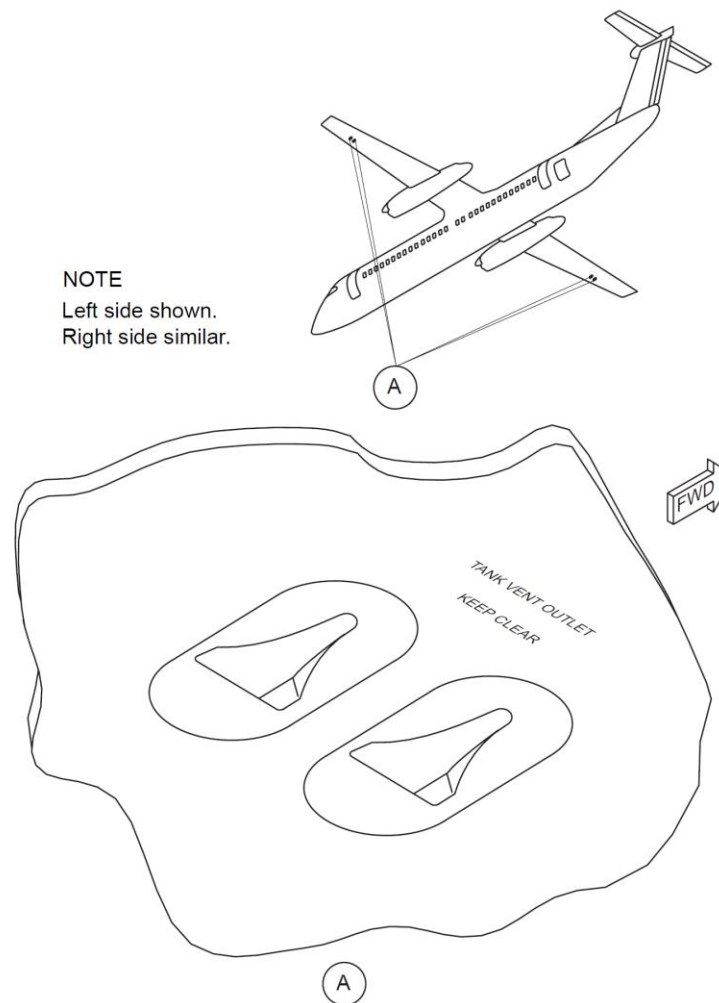
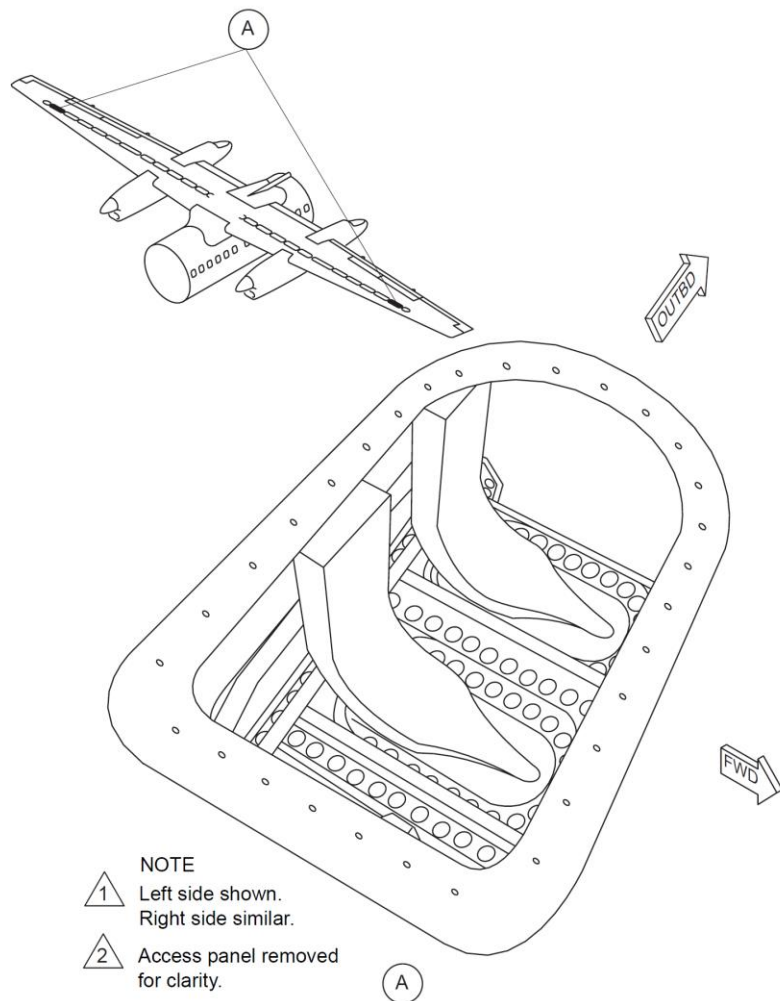


Figure 28-16 Surge Bay NACA Vents (Interior & Exterior View)

Distribution – Engine Fuel Feed System

Introduction

The engine fuel feed system supplies pressurized fuel from the tanks to the engines. The pressure and flow rate of the fuel is sufficient to satisfy all engine operational requirements.

General Description

Each individual tank supplies fuel to its related engine only. Engine feed from each tank to its related engine includes an engine feed pumping system housed in the collector bay, and a fuel emergency shut-off valve.

The engine fuel feed system includes the components that follow:

- Primary Ejector Pump
- Engine Feed Lines
- Inlet Strainer
- Engine Feed Shut-off Valve
- Auxiliary Pump
- Auxiliary Pump Pressure Switch

Detailed Description

[Refer Figures 28-17 & 28-18](#)

A primary ejector pump located in each collector bay supplies fuel to its related engine. Each primary ejector pump uses high pressure motive flow fuel to transfer fuel from the collector bay to the engine. The engine driven fuel pump in the Fuel Metering Unit (FMU) of each engine supplies the motive flow fuel. An electrically driven auxiliary pump is installed in each collector bay for use in the event of a primary ejector pump failure. Either the primary ejector pump or the auxiliary pump can supply the required amount of engine fuel feed.

There is an engine feed shut-off valve located on the rear spar of each wing. The valve is used to stop the fuel supply to the engine in the event of a fire.

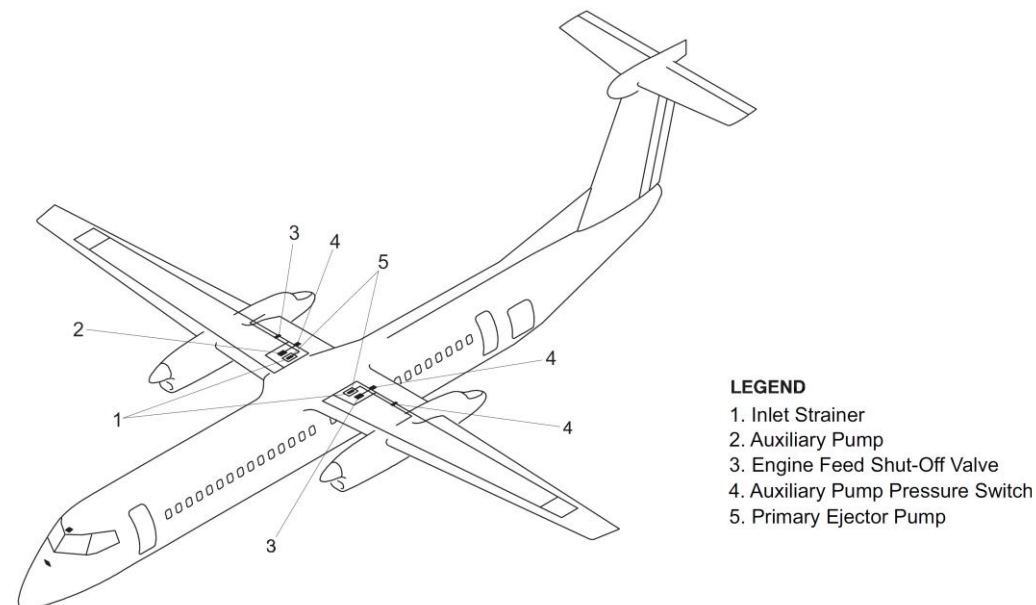


Figure 28-17 Engine Fuel Feed System

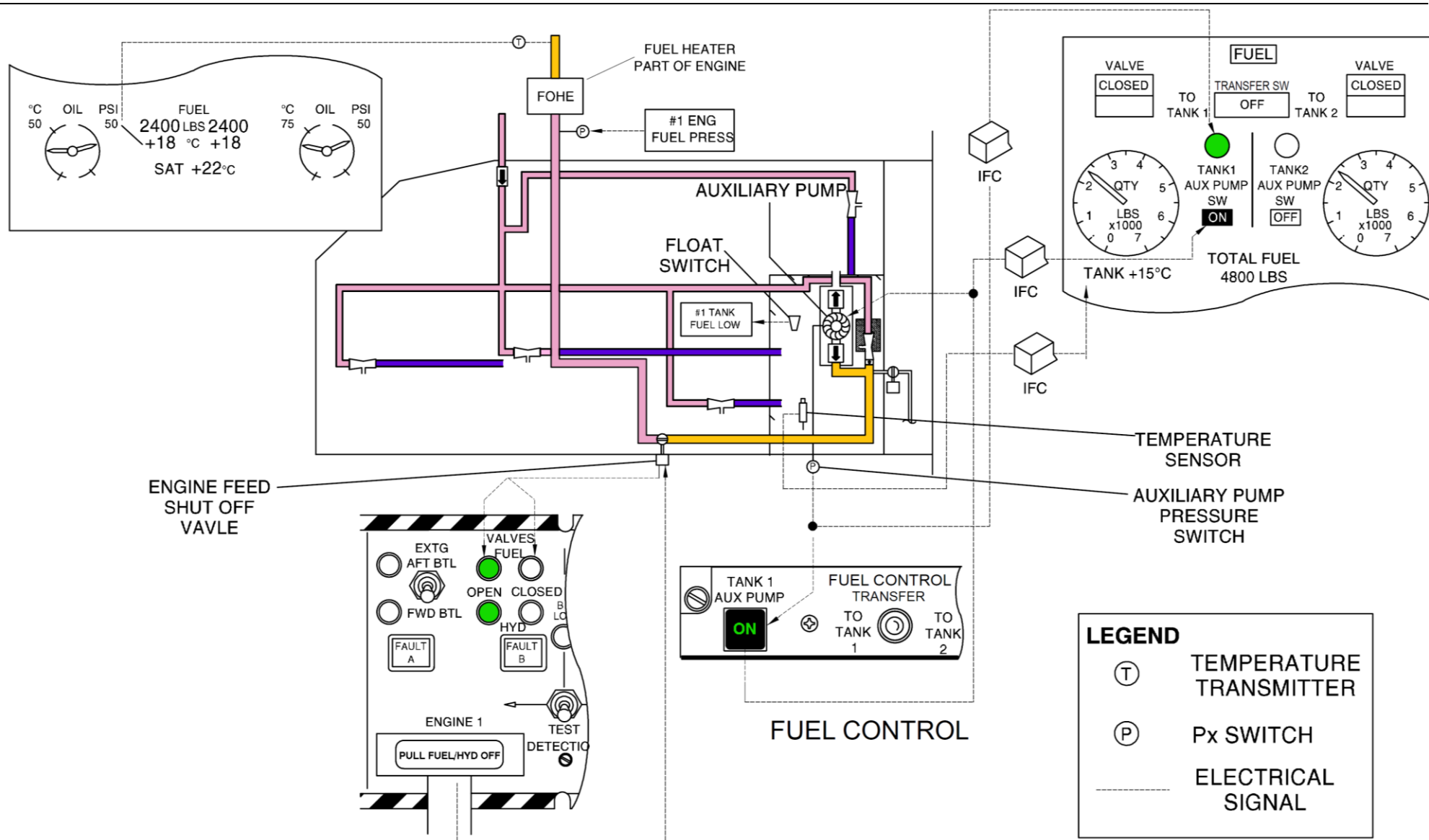


Figure 28-18 Engine Fuel Feed System

Engine Fuel Feed System – Component Description

Primary Ejector Pump

[Refer Figure 28-19](#)

There is one primary ejector pump located in each collector bay. The primary ejector pump delivery rate is always greater than the engine-driven pump demand. This makes sure that there is positive pressure to the engine-driven pump to prevent the possibility of pump cavitation.

If the inlet boost pressure drops below 5.5 psi (38 kPa), a pressure switch at the high-pressure pump inlet sends a signal to turn on the related ENG FUEL PRESS caution light in the flight compartment.

The primary ejector pump is a venturi type with no moving parts. The pump gets its motive flow fuel from the engine driven fuel pump. This high pressure motive flow fuel is fed directly into the ejector throat, causing fuel to be drawn from the collector bay and delivered to the engine feed line. A check valve prevents fuel from flowing back into the ejector pump.

Engine Feed Lines

[Refer Figure 28-18](#)

The engine feed lines transfer the fuel from the primary ejector pumps or the auxiliary pumps to the engines.

Inlet Strainer

[Refer Figure 28-19](#)

There are two inlet strainers; one for each primary ejector pump. Each inlet strainer is installed around the fuel inlet port of its related primary ejector pump. The inlet strainers help prevent foreign objects from entering the fuel flow.

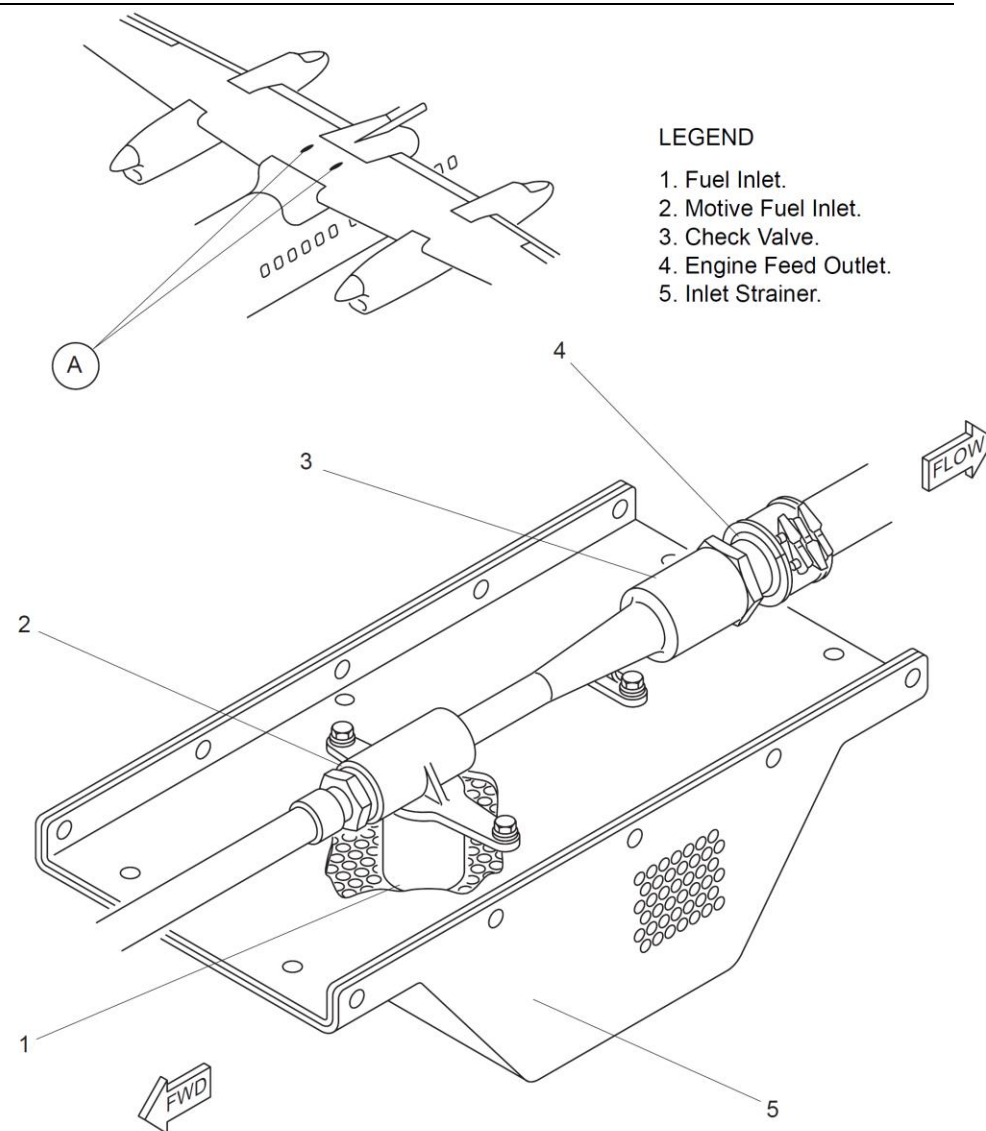


Figure 28-19 Primary Ejector Pump & Inlet Strainer

Engine Feed Shut-off Valve

[Refer Figure 28-20](#)

The engine feed shut-off valve is installed inside the tank on the rear spar. The valve is used to stop the fuel supply to the engine in the event of a fire. The valve has an electrically driven actuator which operates an open-closed ball valve positioned in the engine fuel feed line. The actuator is electrically powered by a 28 V dc motor which is installed on the dry side of the spar and splined to the valve. The actuator has physical and electrical position indication and may be replaced without changing the valve or draining the fuel tank.

The engine feed shut-off valve is controlled and operated by the PULL FUEL/HYD OFF handle located in the flight compartment, on the Engine Fire Protection panel. One green light and one white light for each tank are located on the Engine Fire Protection Panel to show the valve position. The green light comes on to show that the feed valve is open, and the white light comes on to show that the valve is closed. The battery bus supplies the electrical power to the engine feed shut-off valve.

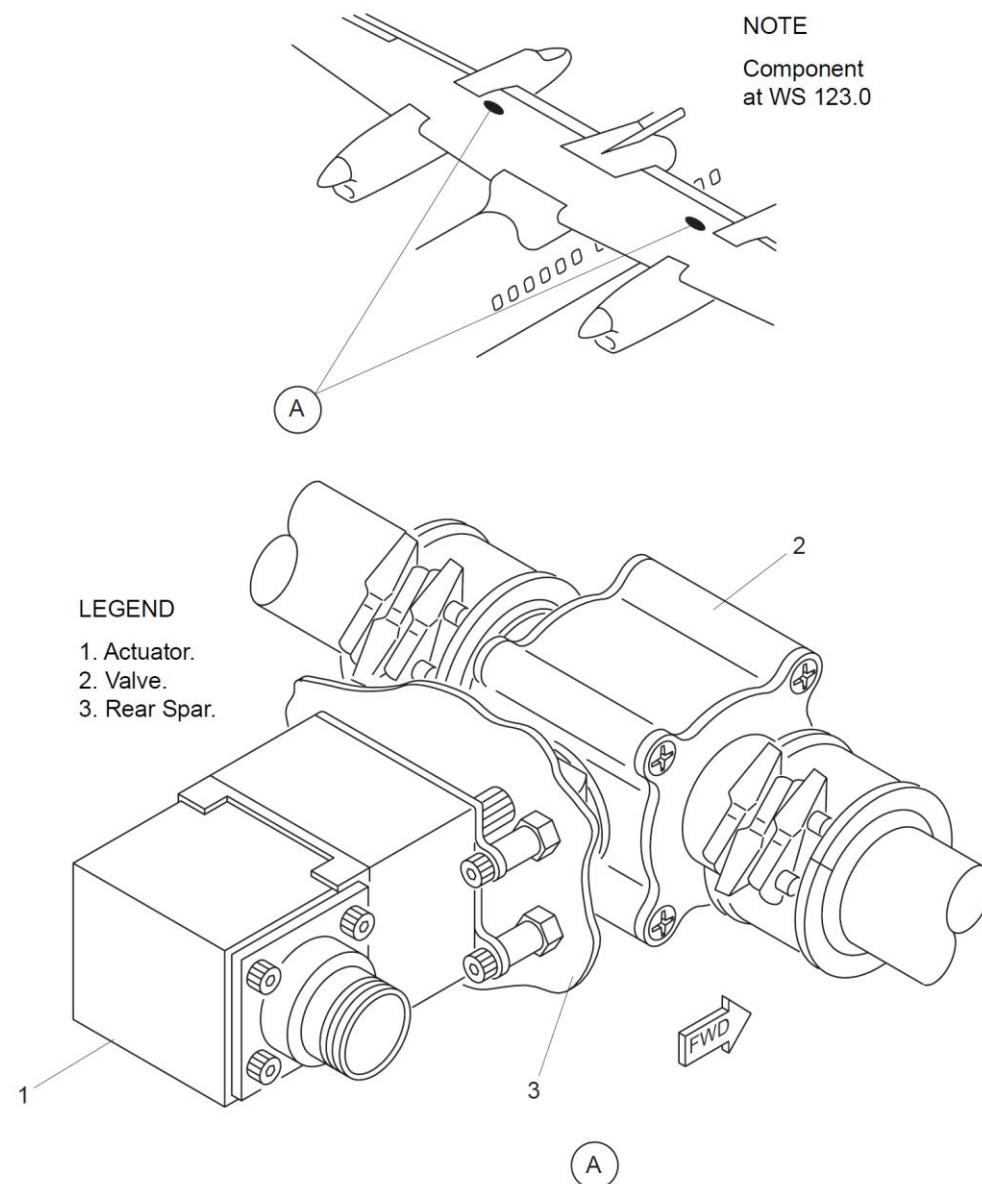


Figure 28-20 Engine Fuel Feed Shut-off Valve

Auxiliary Pump

Refer Figure 28-21

The auxiliary pump is installed on the wing lower skin inside the collector bay of each main tank, adjacent to the primary ejector pump. The auxiliary pump is a submerged, electrically operated, centrifugal pump. It is connected in parallel with the primary ejector pump. Each auxiliary fuel pump is powered by its related 115 Vac variable frequency power bus and is protected by a 5 A circuit breaker (B1 for FUEL AUX PMP1 and E1 for FUEL AUX PMP2). The canister-type pumps are accessible from the bottom surface of the wing, and permit removal of the pumping element without entering or draining the tank. There are two outlet ports, one for engine feed and one for fuel transfer. There is a check valve in each outlet port.

Refer Figure 28-22

If a primary ejector pump does not supply the minimum required fuel pressure, the related auxiliary pump serves as a backup source for pressurized engine fuel feed. When the related AUX PUMP switchlight on the FUEL CONTROL TRANSFER panel is pushed, the auxiliary pump turns on (Refer to SDS 28-23-00 Fuel Transfer). After the pump starts, the related ENG FUEL PRESS caution light goes out. The auxiliary pumps are used for take-off, landing and during operation with low fuel levels.

The auxiliary pumps start automatically for fuel transfer. The auxiliary pumps also start automatically for pressure defueling if a variable-frequency ac power source is available on the ground.

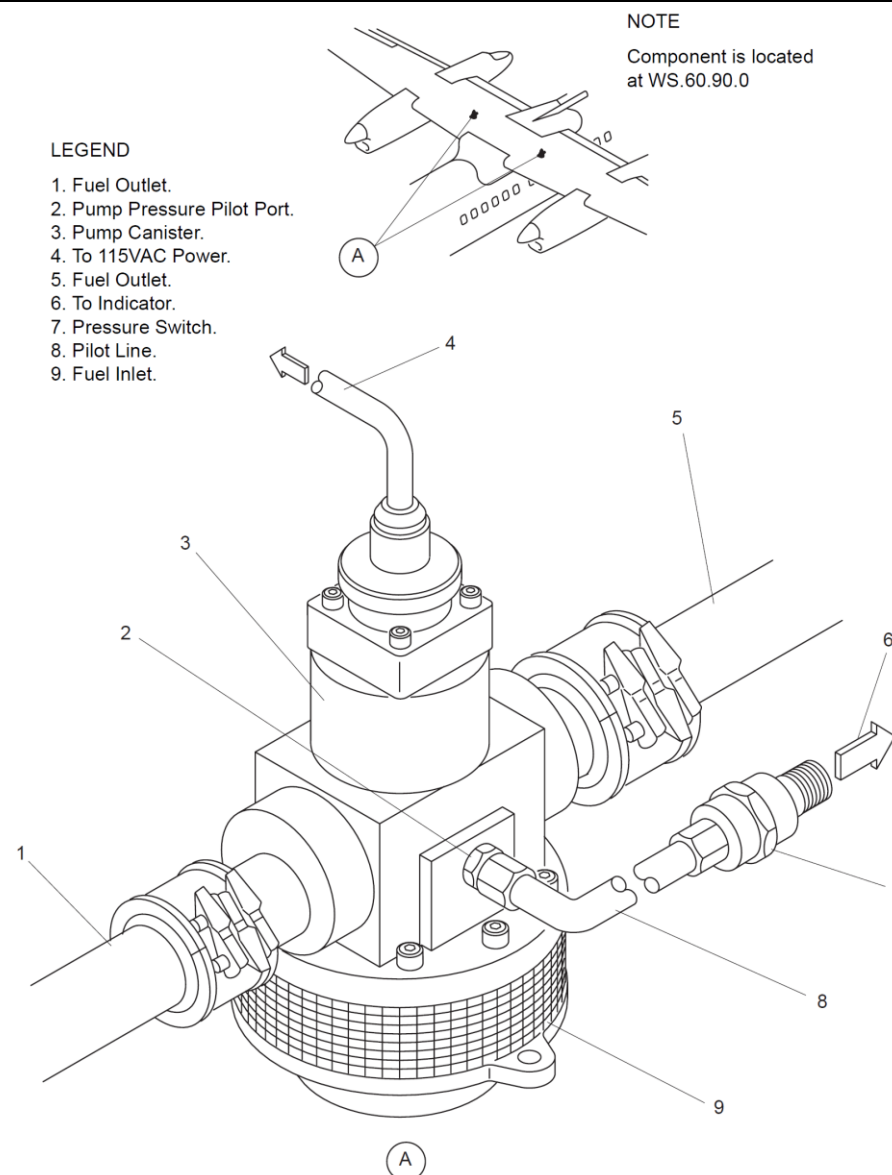


Figure 28-21 Auxiliary Pump

LEGEND

1. # 1 Engine Fuel Pressure (Amber).
2. # 2 Engine Fuel Pressure (Amber).
3. # 1 Tank Fuel Low (Amber).
4. # 2 Tank Fuel Low (Amber).
5. Fueling On (Amber).

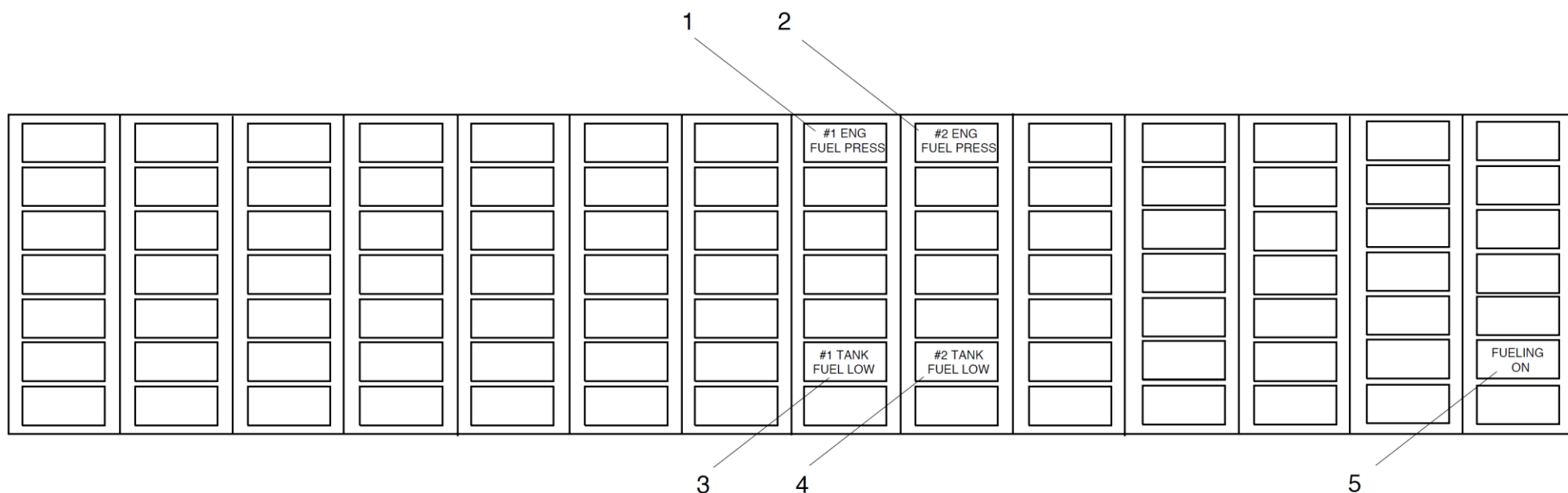


Figure 28-22 Fuel System Caution Lights

Auxiliary Pump Pressure Switch

[Refer Figure 28-23 & 28-24](#)

There are two auxiliary pump pressure switches, one connected to each auxiliary pump. The pressure switch closes when 4 psi (28 kPa) of fuel pressure is supplied. The auxiliary pump pressure switch then sends a signal to the FUEL CONTROL TRANSFER panel in the flight compartment. This signal causes the related ON switchlight segment for TANK1 or TANK2 to turn green.

[Refer Figure 28-25](#)

The auxiliary pump pressure switches, through the IFC, will also display pump operation on the MFD Fuel page, in the form of green round lights.

LEGEND

1. Tank #1 and Tank #2 Auxiliary Pump Switch Lights.
2. Fuel Transfer Selector Switch.

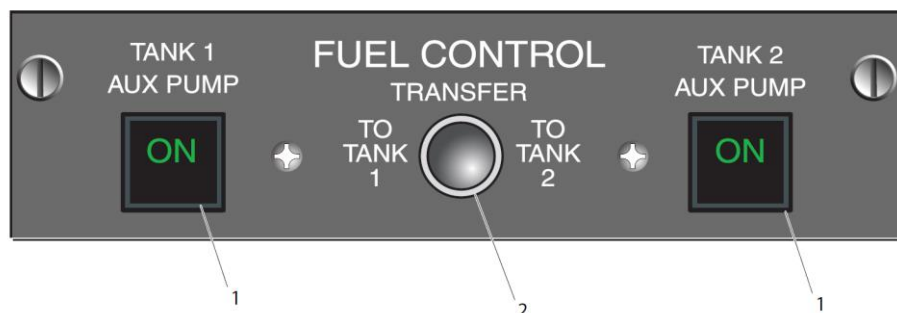


Figure 28-23 Fuel Control Transfer Panel

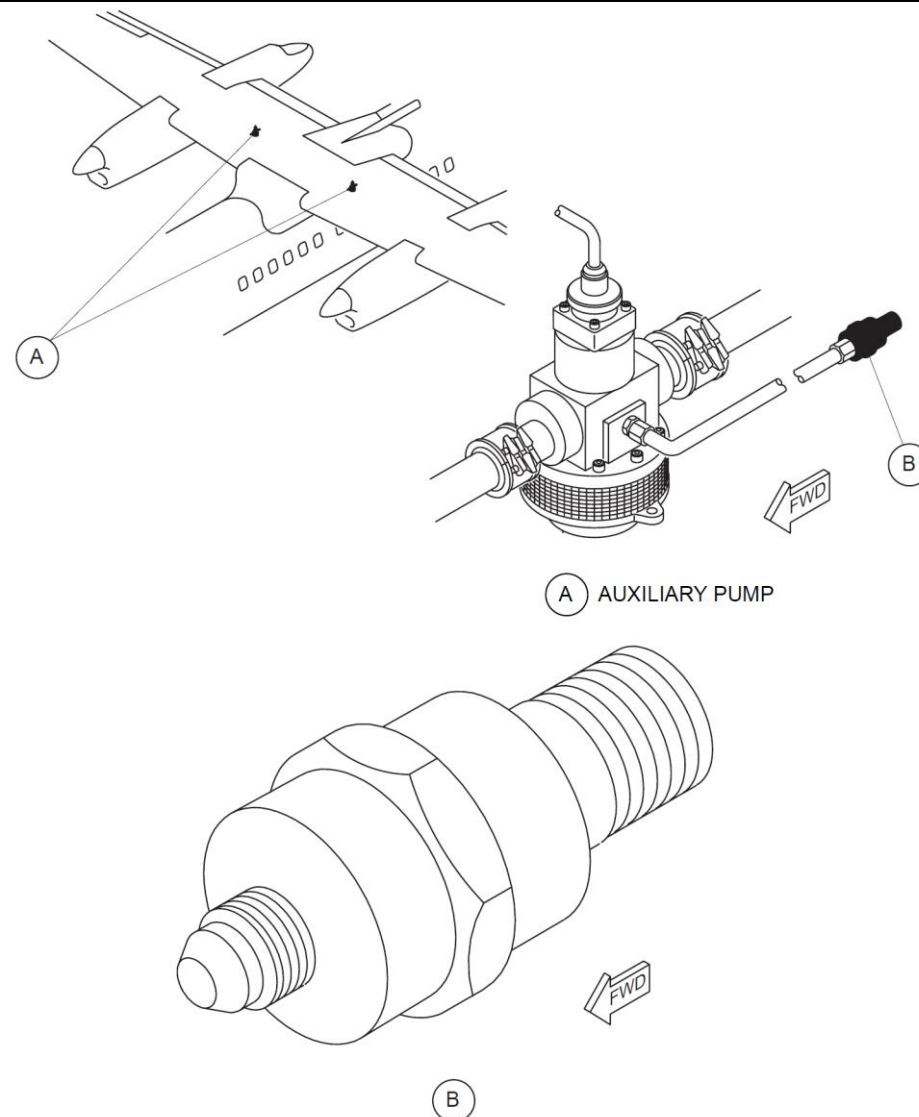


Figure 28-24 Auxiliary Pump Pressure Switch

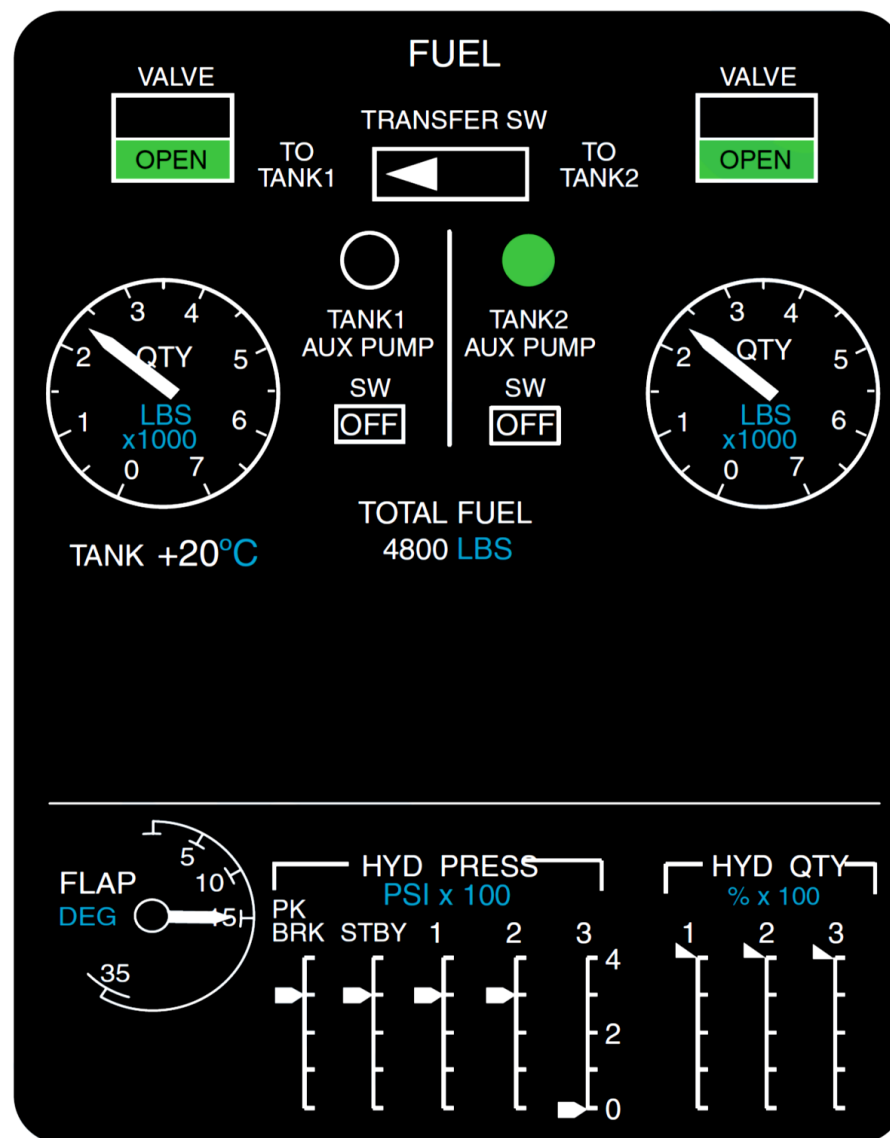


Figure 28-25 MFD Fuel Page

Auxiliary Pump Manual Operation – Electrical Schematic

[Refer Figure 28-26](#)

The left fuel AUX tank operates with:

- A selection of the TANK 1 AUX PUMP switchlight to the ON position, providing a ground for relay K1

- Energizing K1 provides power from Left 115 VAC bus to the auxiliary pump.
- The relay K1 at pin X1 is hot all times from the three REFUEL circuit breakers (F4, M3 and M4).

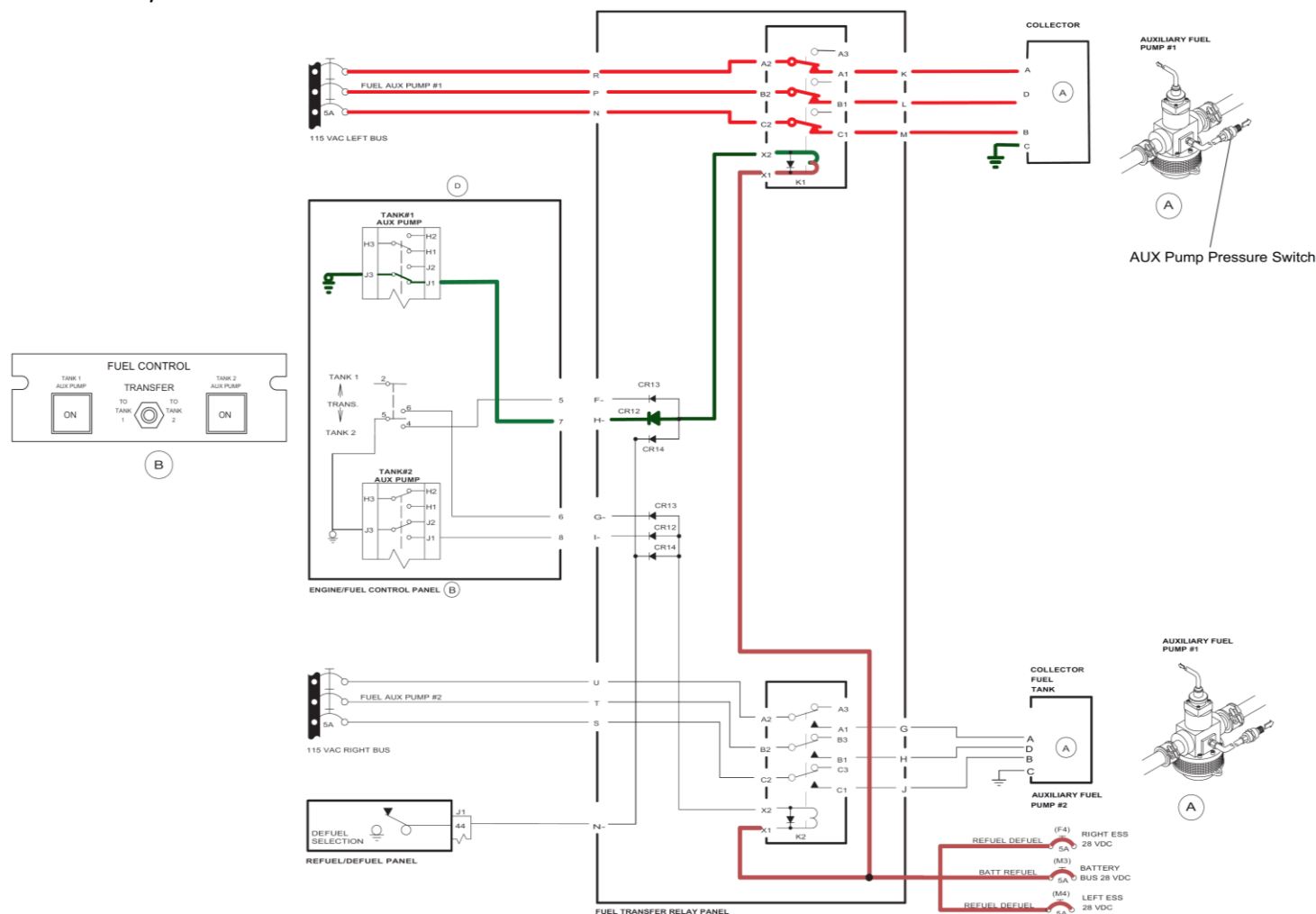


Figure 28-26 Auxiliary Pump Manual Operation – Electrical Schematic

APU Fuel Feed System

Introduction

The Auxiliary Power Unit fuel feed system supplies a flow of fuel from the left (No.1) fuel tank to the Auxiliary Power Unit (APU).

General Description

Fuel for APU operation is supplied by gravity feed from the No.1 fuel tank through piping in the dorsal fairing to the APU fuel enclosure. Fuel supply is controlled by the APU PWR switchlight located in the flight compartment, on the APU CONTROL Panel. The APU is located in the rear equipment bay.

The APU fuel feed system includes the component that follows:

- APU shut-off Valve

Detailed Description

[Refer Figure 28-27](#)

The fuel supply system is interconnected with the APU start/stop circuits and when the APU speed reaches 3%, the fuel flows to the unit. The fuel system is de-energized to shut down the APU. The system is electrically powered by 28 V dc from the right main bus through a 5 A APU FUEL circuit breaker. The APU fuel supply system has a fuel shut-off valve, related open/close indicator lights, a check valve, and piping to supply the fuel to the fuel enclosure on the APU. A check valve is installed in the fuel supply line, downstream of the APU shut-off valve. The check valve gives thermal relief for the downstream line.

LEGEND

1. APU Shut-Off Valve
2. APU Feed line

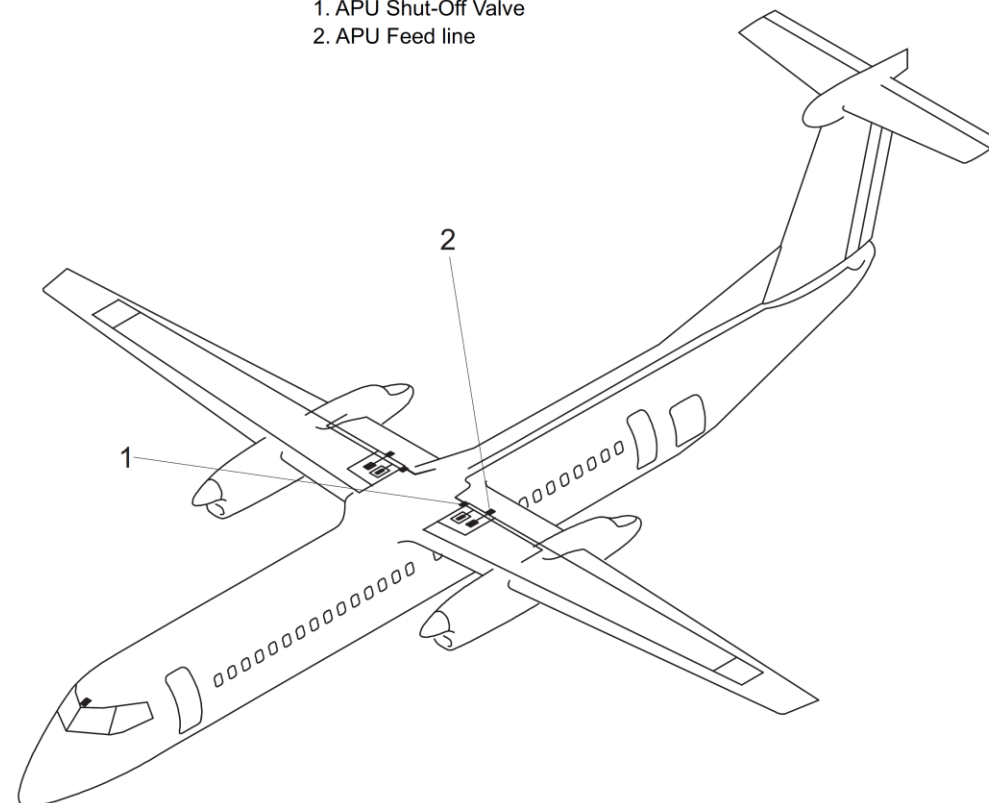


Figure 28-27 APU Fuel Feed

APU Fuel Feed system – Component Description

APU Shut-off Valve

[Refer Figure 28-28](#)

The APU shut-off valve gives a positive fuel stop in the supply line to the APU. The valve has an electrical actuator that operates an open/close type ball valve. The ball is enclosed in a tee-shaped fitting, located inside the fuel tank and installed onto the rear spar. The actuator is located externally to the fitting. The shut-off valve has a manual override and indicator arm. Valve operation is controlled by the APU PWR switchlight located in the flight compartment, on the APU CONTROL Panel. The valve position is shown by a green FUEL VALVE OPEN light and a white FUEL VALVE CLOSED light on the APU Fire Protection Panel on the overhead console.

The APU shut-off valve closes and stops the fuel flow to the APU under these conditions:

- the APU PWR switchlight is selected OFF
- fire is sensed in the APU / tail cone area
- the APU fire extinguisher switch (EXTG) on the Fire Protection Panel is selected ON
- the Weight on Wheels (WOW) switch is open (aircraft off ground).

CAUTION: APU will shut down if the fire test button is pushed when APU is running.

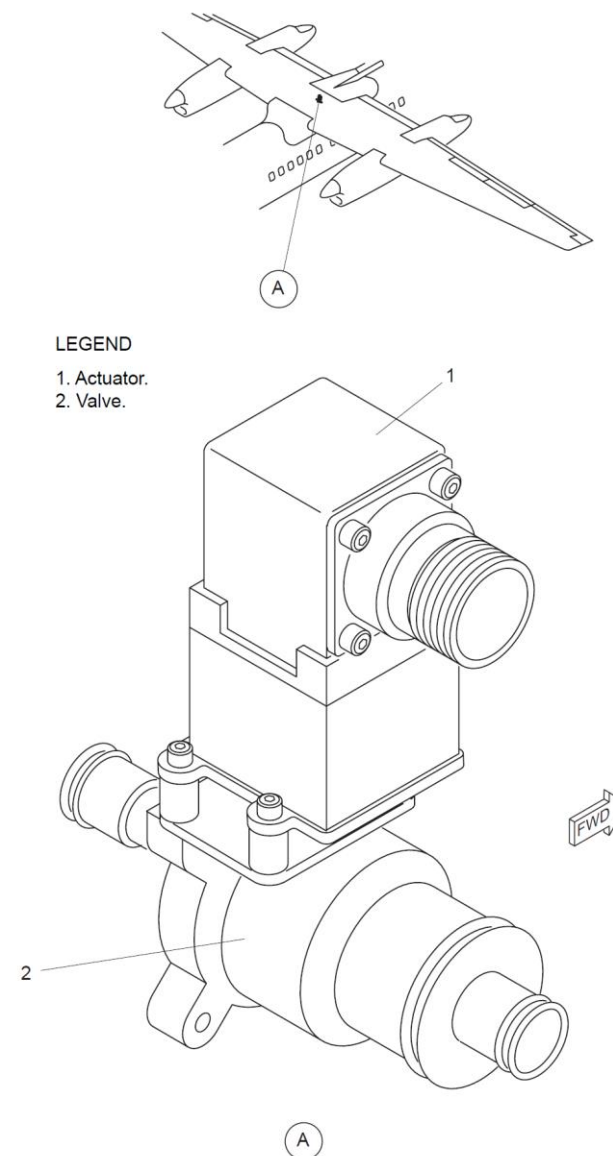


Figure 28-28 APU Shut-Off Valve

Fuel Transfer System

Introduction

The fuel transfer system moves fuel from one tank to the other to correct fuel imbalances or for fuel management. The system can empty one fuel tank for maintenance without defueling the aircraft.

General Description

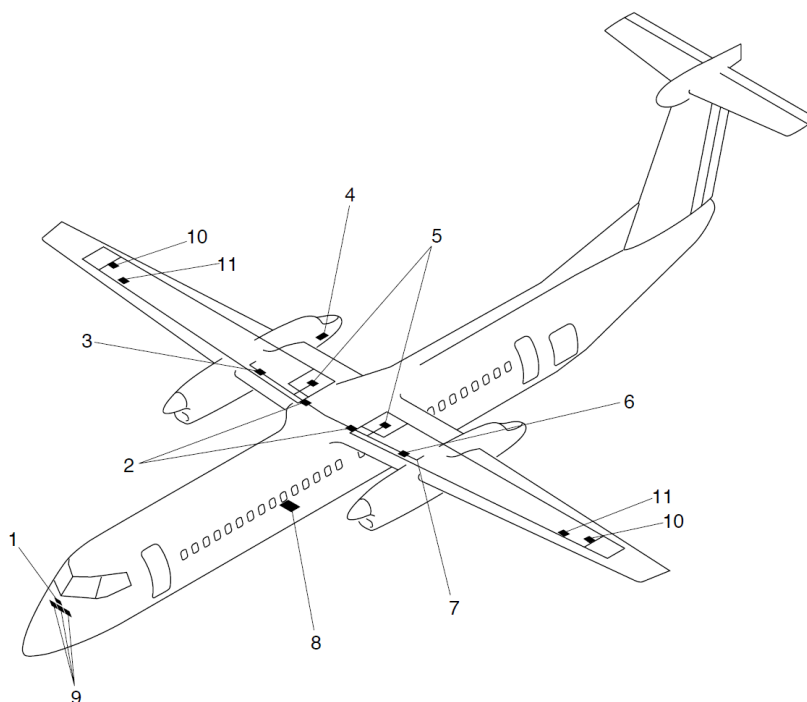
[Refer Figure 28-29](#)

Fuel can be transferred when the aircraft is in the air or on the ground. Fuel transfer is controlled through the FUEL CONTROL TRANSFER panel. The fuel transfer system shares some of its components with the refuel/defuel system. Fuel transfer cannot occur during the refuel/defuel process.

The fuel transfer system uses the auxiliary pump, part of the engine feed system, and the refuel/defuel/transfer manifold. Fuel is transferred by the auxiliary pump of the donor tank. Fuel transfer data is shown in the flight compartment.

The fuel transfer system includes the components that follow:

- Fuel Transfer Shut-off Valves
- Level Control Shut-off Valves
- Level Control Solenoid
- Fuel Control Transfer Panel



LEGEND

1. Fuel Control (Transfer) Panel.
2. Transfer Shut-Off Valve.
3. Level Control Shut-Off Valve.
4. Refuel/Defuel Panel.
5. Auxiliary Pump.
6. Level Control Shut-Off Valve.
7. Refuel/Defuel/Transfer Manifold.
8. Fuel Quantity Computer.
9. EIS Displays.
10. Refuel Vent Valve and High Level Sensor.
11. Gravity Refill Cap.

Figure 28-29 Fuel Transfer System

Detailed Description

[Refer Figure 28-30 & 28-31](#)

The fuel transfer system is used to correct lateral fuel imbalance, either in flight or when the aircraft is on the ground. The system is also used for fuel management with one engine inoperative. The fuel transfer system is controlled by a three position TRANSFER switch on the FUEL CONTROL TRANSFER panel in the flight compartment. The panel has a control switch and lights to indicate the correct operation of various components of the

system.

Fuel flows from the auxiliary pump of the donor tank through a transfer line into the receiver tank. A restrictor in the transfer line limits the fuel flow used for transfer. This makes sure there is sufficient fuel flow to the engine. Once selected, fuel transfer will continue until the TRANSFER switch is set to the center position, or until the high level sensor unit in the receiver tank senses an overfill condition.

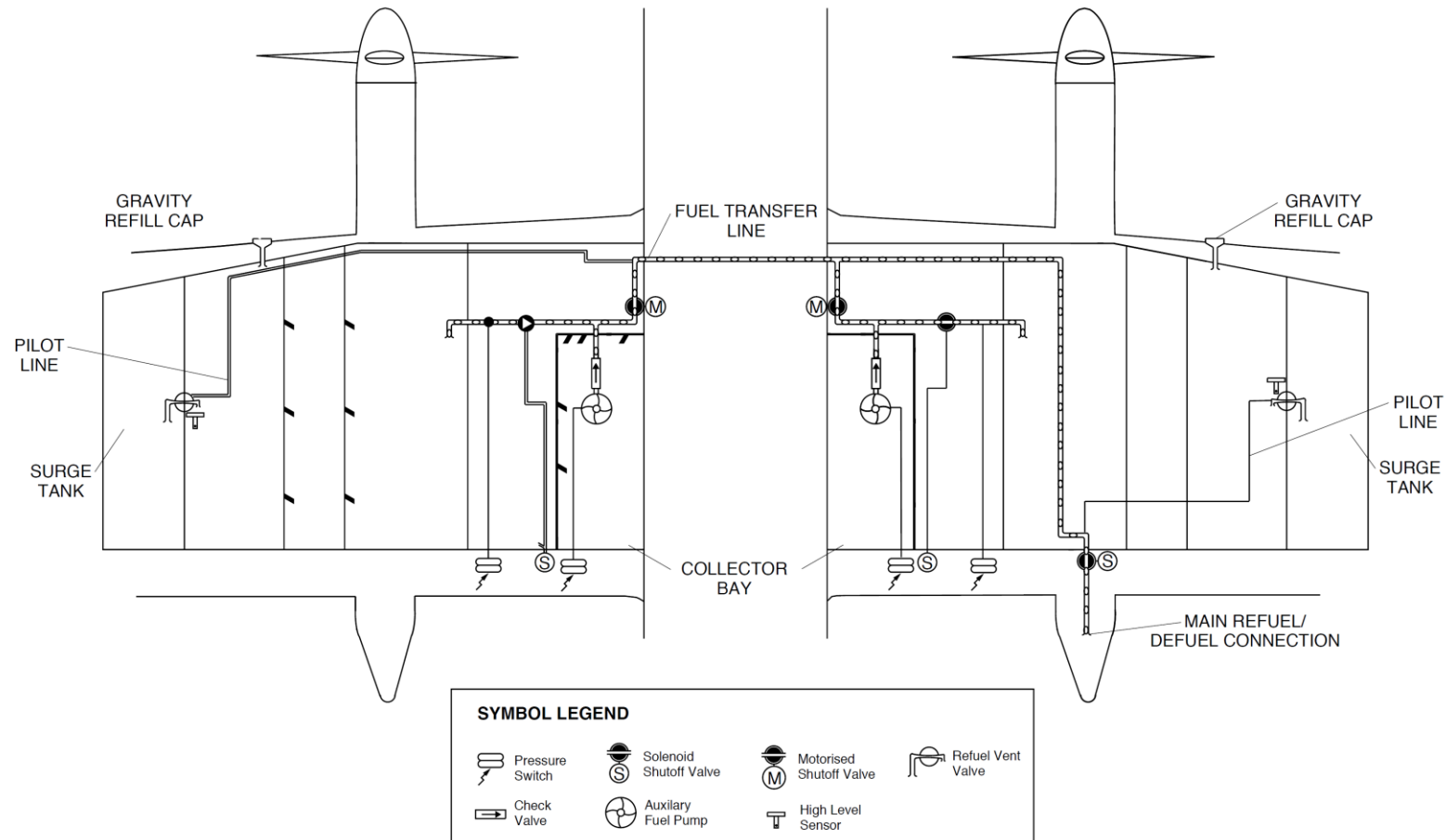


Figure 28-30 Fuel Transfer Schematic

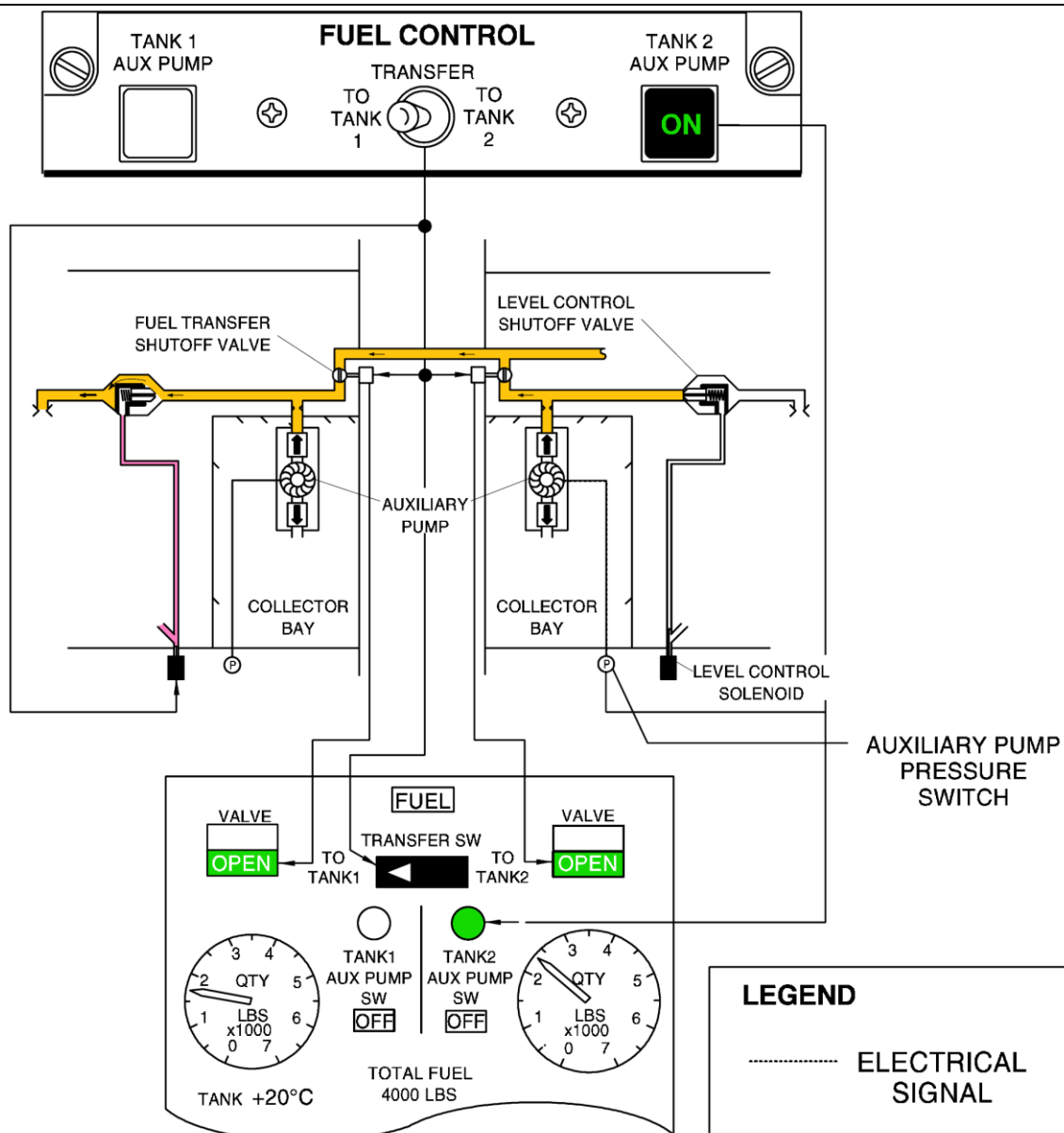


Figure 28-31 Fuel Transfer Schematic

Fuel Transfer System – Component Description

Fuel Transfer Shut-off Valves

[Refer Figure 28-32](#)

There is one fuel transfer shut-off valve in each wing. Each fuel transfer valve is located inside the related wing tank, with the valve actuator located externally on Wing Station (WS) 42. The fuel transfer shut-off valves open during refuelling, defueling or fuel transfer. The valves are 28 V dc motor operated valves and have position indicators on the body of the valves. In refuel or defuel manual mode, the valves are controlled by the refuel/defuel control panel in the right nacelle. In refuel or defuel pre-select mode, the valves are controlled by the Fuel Quantity Computer (FQC). In transfer mode, the valves are controlled by the TRANSFER switch on the FUEL CONTROL TRANSFER panel.

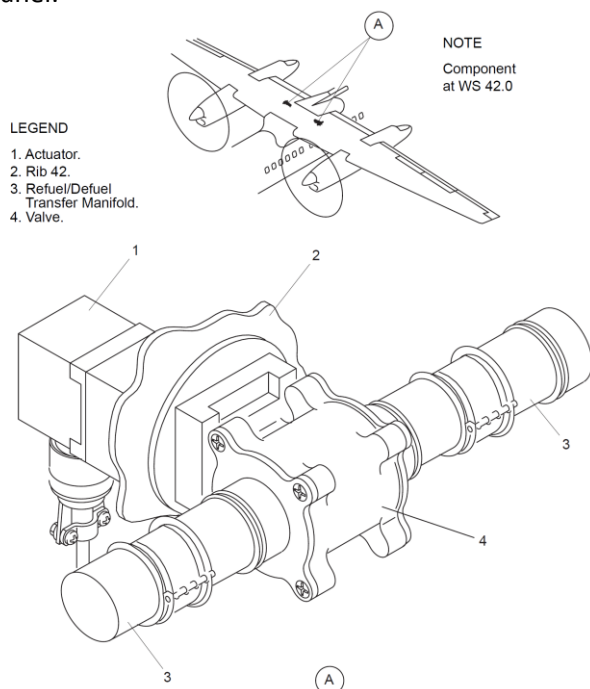


Figure 28-32 Fuel Transfer Shut-Off Valve

Level Control Shut-off Valves

[Refer Figures 28-33 & 28-34](#)

There is one level control shut-off valve in each wing. The level control shut-off valves let fuel flow during transfer and refuel/defuel operations. The valves are hydro-mechanically controlled by the level control solenoids. The valves are kept closed by springs and fuel pressure on the backs of the main poppets. The valves are kept closed by springs and fuel pressure on the backs of the main poppets.

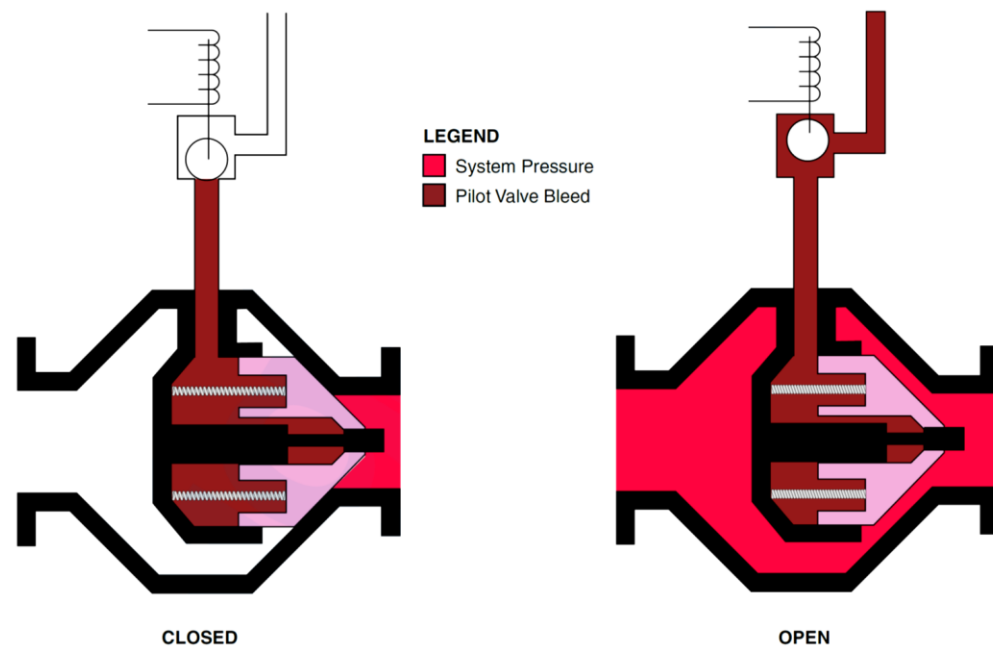


Figure 28-33 Level Control Shut-Off Valve Operation

Level Control Solenoid

[Refer Figure 28-34](#)

There is one level control solenoid in each wing. A pilot line connects each level control solenoid to its related level control shut-off valve. The solenoids are installed on the aft face of the rear wing spar. The Fuel Quantity Computer (FQC) controls the solenoids. The high level unit can also control the solenoids when the fuel level in the tanks is high. When the level control solenoid is de-energized, there is fuel pressure in the pilot line. This lets the fuel pressure at the inlet of the valve be applied to the valve's control chamber to keep the valve closed. When the level control solenoid is energized, the pilot line opens to release the pressure in the control chamber and the level control shut-off valve opens.

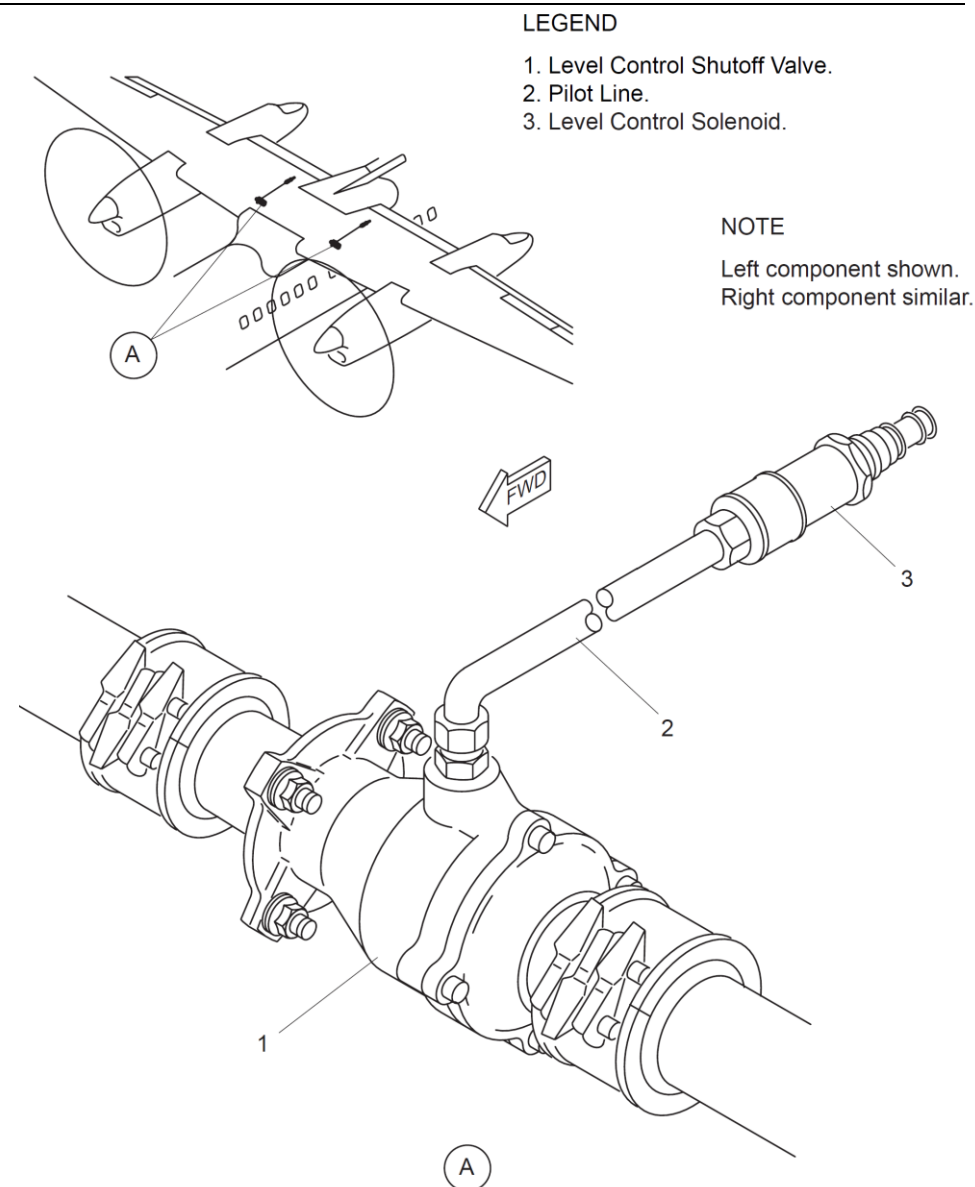


Figure 28-34 Level Control Shut-Off Valve & Solenoid

Fuel Control Transfer Panel

[Refer Figures 28-35 & 28-36](#)

The FUEL CONTROL TRANSFER panel is located in the flight compartment on the engine instrument panel. The FUEL CONTROL TRANSFER panel has a TRANSFER switch and two switchlights. The TRANSFER switch is used to set the direction of fuel transfer (TO TANK 1 or TO TANK 2) by operating the auxiliary pump of the donor tank. A signal from the operating auxiliary pump causes the related ON switchlight segment to turn green. Fuel transfer indications are shown on the FUEL CONTROL TRANSFER panel and the Fuel Page of the Multi-Function Display (MFD).

LEGEND

1. Tank #1 and Tank #2 Auxiliary Pump Switch Lights.
2. Fuel Transfer Selector Switch.

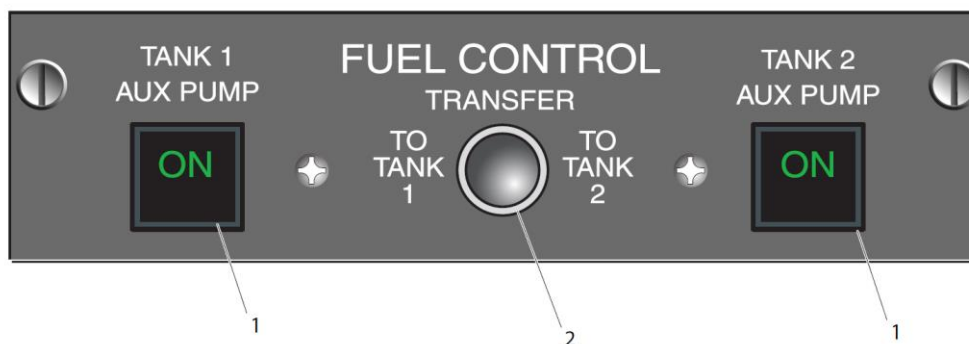
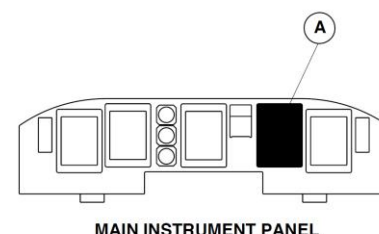


Figure 28-35 Fuel Transfer Control Panel



LEGEND

1. Left/Right Fuel Tank Quantity.
2. Fuel Temperature in Left Tank.
3. Left/Right Fuel Transfer Valve Position.
4. Fuel Transfer Switch Indication.
5. Pressure State in the Left/Right Auxiliary Fuel Pumps.
6. Total Fuel Quantity in Tanks.

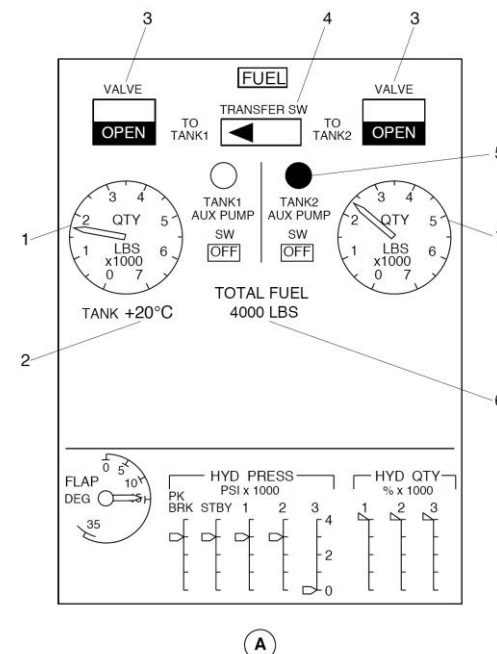


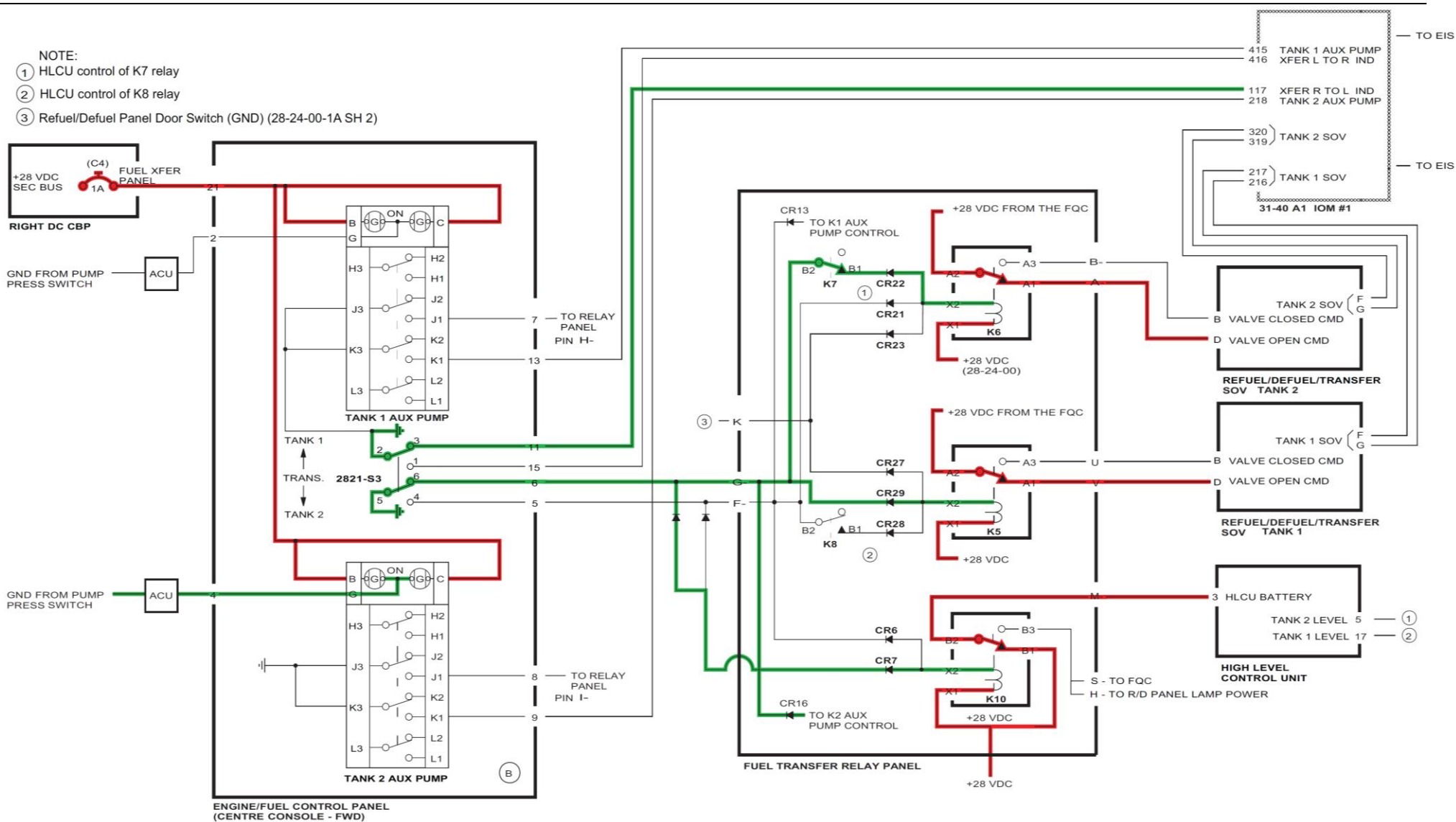
Figure 28-36 Fuel Transfer Page in Multi-Function Display

Fuel Transfer Operation – Electrical Schematic

[Refer Figure 28-37 \(Sheet 1 of 3\)](#), [28-37 \(Sheet 2 of 3\)](#) & [28-37 \(Sheet 3 of 3\)](#)

The selection of the TRANSFER switch to the TO TANK 1 position on the FUEL CONTROL panel provides ground signals from the TRANSFER switch to close the relays in the Fuel Transfer relay panel that follow:

- K2 (which energizes No.2 Aux pump), shown on Sheet 3
- K3 (which causes the Level Control solenoid to open), shown on sheet 2
- K5 (which opens the No.1 Refuel/ Defuel/Transfer Shut-Off Valve), shown on Sheet 1
- K6 (which opens the No.2 Refuel/ Defuel /Transfer Shut-Off Valve), shown on Sheet 1
- K7, (which energizes the level control solenoid valve, from HLCU, shown on Sheet 2
- K8 (which energizes No.2 Aux pump), shown on Sheet 2
- K10 (which energizes HLCU), shown on Sheet 1.
- The relay K2 receives 28 VDC from the left essential bus and a ground signal from the TRANSFER switch through diode CR16.
- The closed relay sends three phases AC from the right AC bus to the No.2 Aux Pump which causes the Aux Pump to operate.
- The rise in Aux Pump output fuel pressure closes the auxiliary pump pressure switch.
- The closed auxiliary pump pressure switch supplies a ground signal to the IOM No.1 and the Advisory Control Unit (ACU).
- The relay K3 receives 28 VDC from the FQC and a ground signal from the TRANSFER switch which energizes the relay closed.
- The closed relay sends 28 VDC from the FQC to the tank 2 Level Control solenoid which causes the solenoid to energize open.
- The open solenoid allows the Level Control valve back pressure to be bled out which allows the Aux pump pressure to open the Level Control valve and transfer the fuel. The closed relay K7 supplies the ground signal from the TRANSFER switch to energize the relay K6 closed.
- The relay K5 receives 28 VDC from the left essential bus and a ground signal from the TRANSFER switch to energize the relay K5 closed.
- The closed relays K6 and K5 send 28 VDC (valve open) signals from the Fuel Quantity Computer (FQC) to the two Refuel/Defuel/ Transfer SOV pins “D” which energize open the SOV and allows the fuel to transfer.
- Relay K10 receives 28 VDC from the left essential bus and a ground signal from the TRANSFER switch.
- The closed K10 relay provides 28 VDC from the left essential bus to energize the HLCU at pin 3. The not-full signal from the HLCU supplies ground signals to close the relays K7 and K8.



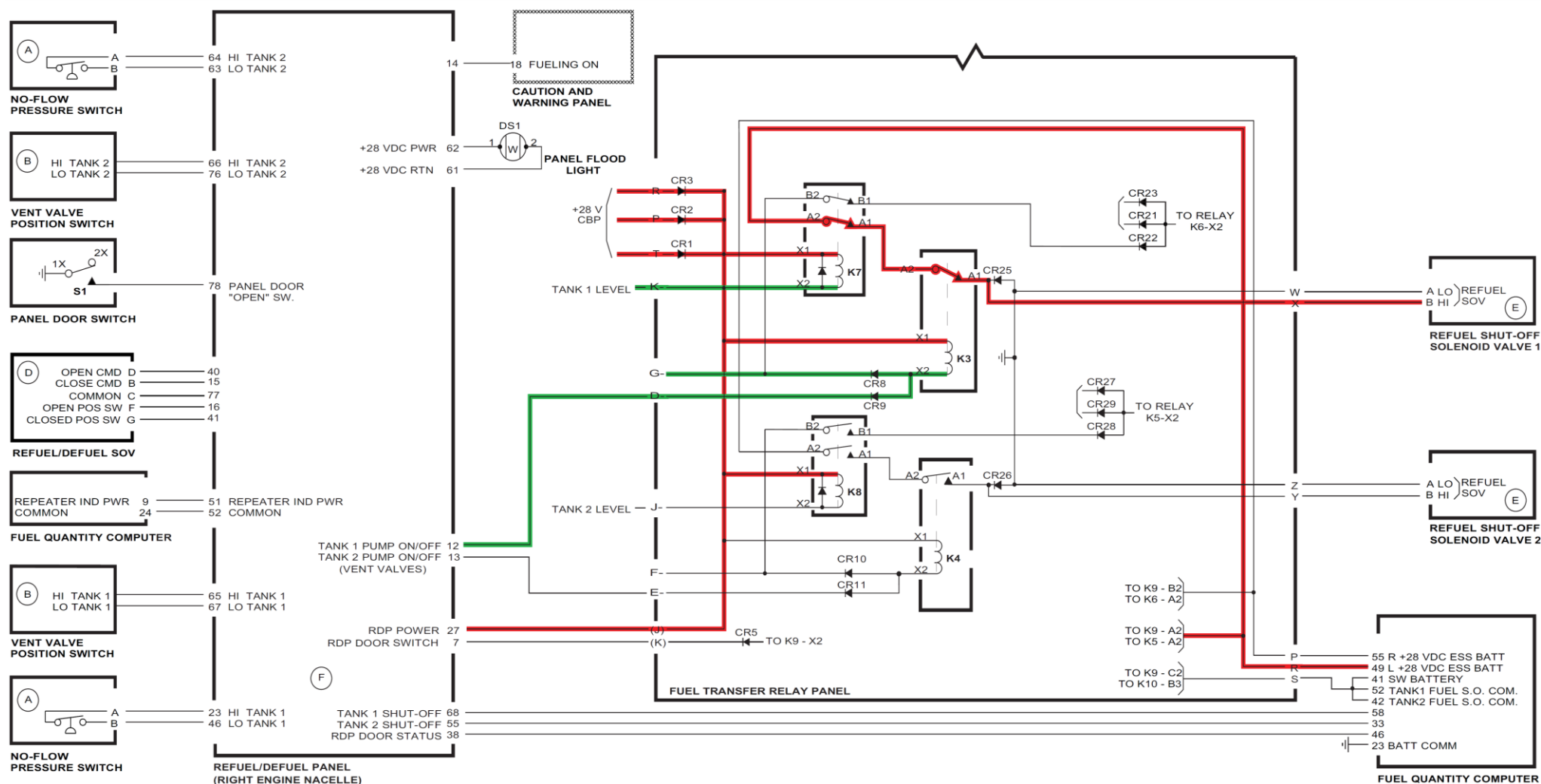


Figure 28-37 (Sheet 2 of 3) Fuel Transfer Operation – Electrical Schematic

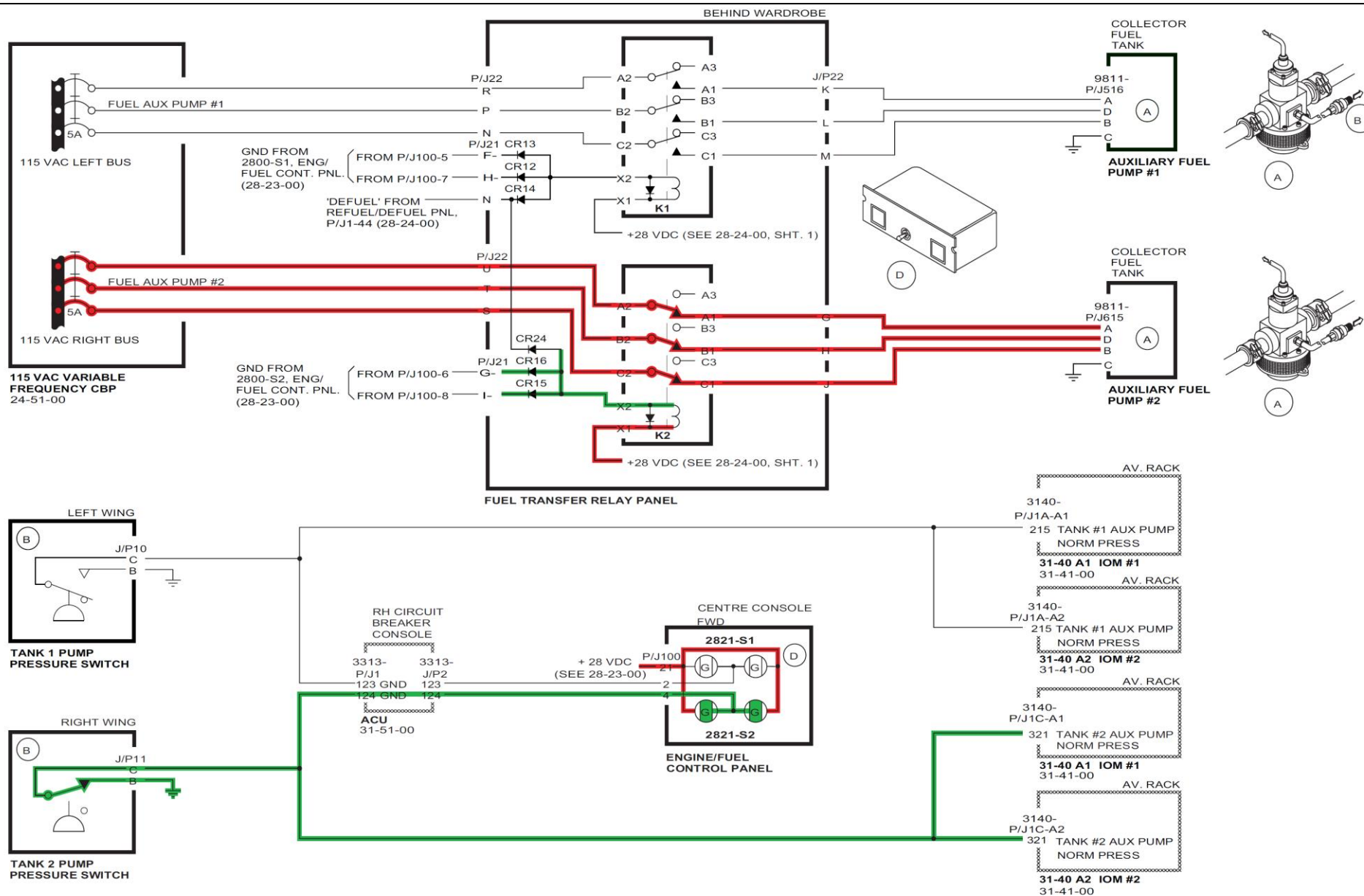


Figure 28-37 (Sheet 3 of 3) Fuel Transfer Operation – Electrical Schematic

Refuel/Defuel System

Introduction

The refuel/defuel system is used for pressure refuelling or defueling through a single point adapter.

General Description

[Refer Figure 28-38 & 28-39](#)

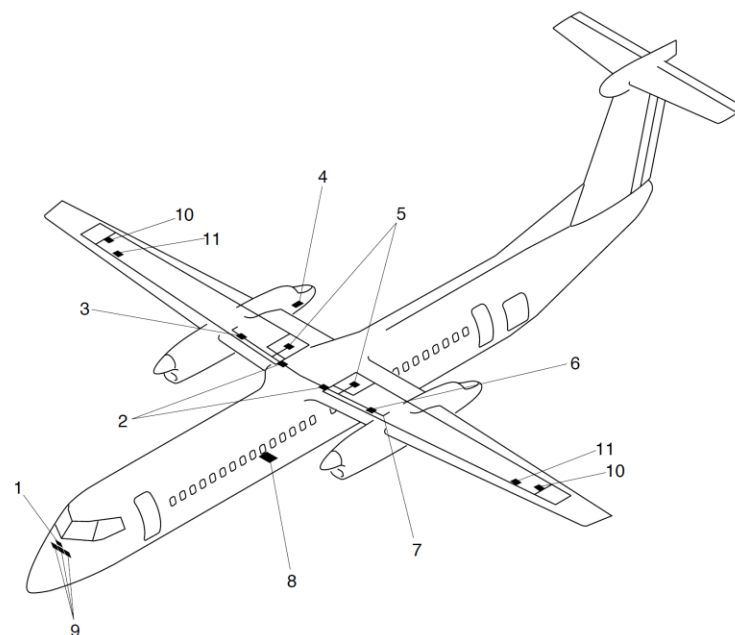
The aircraft is refuelled or defueled through the single point pressure refuelling/defueling adapter located in the right engine nacelle. Both automatic and manual refuelling modes are available. Selected and actual fuel tank quantities are displayed on the refuel/defuel control panel in the right engine nacelle. The Fuel Quantity Computer (FQC) controls the logic of the automatic refuelling and defueling process.

Gravity refuelling can be done through a gravity refuel adapter on the

top surface of the wing. Defueling can be done by external suction pressure, or by use of the auxiliary pumps in each collector bay.

The refuel/defuel system has the components that follow:

- Refuel/Defuel Control Panel
- Refuel/Defuel Indicator
- Refuel/Defuel Shut-Off Valve
- Refuel/Defuel Shut-Off Valve Actuator
- Refuel/Defuel Adapter
- No-Flow Pressure Switch
- Refuel Vent Valve
- High Level Sensor



LEGEND

1. Fuel Control (Transfer) Panel.
2. Transfer Shut-Off Valve.
3. Level Control Shut-Off Valve.
4. Refuel/Defuel Panel.
5. Auxiliary Pump.
6. Level Control Shut-Off Valve.
7. Refuel/Defuel/Transfer Manifold.
8. Fuel Quantity Computer.
9. EIS Displays.
10. Refuel Vent Valve and High Level Sensor.
11. Gravity Refill Cap.

Figure 28-38 Refuel/Defuel System

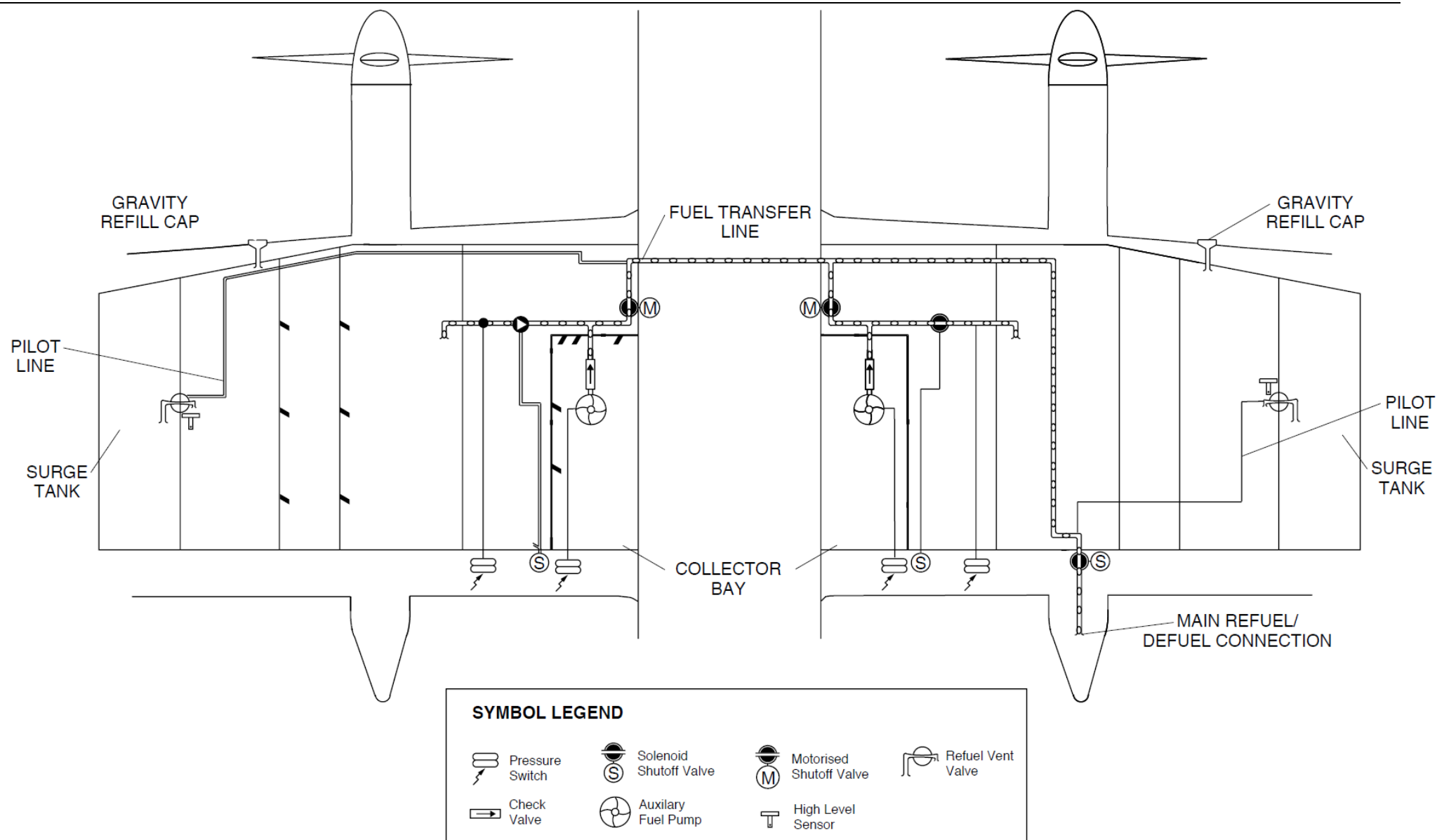


Figure 28-39 Refuel/Defuel System Block Diagram

Detailed Description

[Refer Figure 28-39](#)

The refuelling/defueling fuel flow into or from each tank is through the refuelling/defueling adapter and manifold, controlled by the refuel/defuel shut-off valve. Pressure refuelling uses the same level control shut-off valves, solenoid valves, and refuelling/defueling/transfer shut-off valves that are used for fuel transfer. The maximum refuelling pressure is 50 psi (344.7 kPa), which gives a refuelling flow rate of approximately 125 US gal/min (437 L/min). A minimum of 20 psi (138 kPa) refuelling pressure is necessary for the system to operate properly. Defueling can be done by external suction pressure or by the use of the auxiliary pumps in each collector bay.

[Refer Figure 28-40 & 28-41](#)

In the pre-select mode, the refuelling/defueling operation stops when the selected tank contents reach a pre-selected level. In the manual mode, refuelling of the aircraft continues until stopped by the operator, or until the high level sensor senses a full tank and stops refuelling. A refuel vent valve in each tank is kept open with fuel pressure during pressure refuelling. The DUMP VALVE OPEN TANK 1 or TANK 2 indicator light on the refuel/defuel panel comes on when the related refuel vent valve is open. The refuel vent valve prevents structural damage, that could occur if fuel continues to flow into full tanks.

A solenoid valve controls the operation of the level control shut-off valve in each tank. In the pre-select mode, the Fuel Quantity Computer (FQC) signals the solenoid valve to close the level control shut-off valve. The refuelling flow into each tank stops automatically at the pre-selected fuel level on the Refuel/Defuel Indicator (RDI). In the manual mode, the operator stops the refuelling process by selecting the appropriate switch on the refuel/defuel control panel.

The solenoid valve causes the level control valve to close. If the solenoid valve does not close automatically, the related level control valve stays open, and the refuelling process continues. When the tank is full, the high level sensor located at the high point of the tank, sends a signal to the High Level Control

Unit (HLCU). The HLCU de-energizes the solenoid valve and causes the level control valve to close.

Operation of the HLCU shut-off feature can be checked using the PRECHECK, OPEN, CLOSE switch, on the refuel/defuel control panel. With the switch in the PRECHECK position for the related tank, the HLCU shuts off refuelling to that tank. A no-flow pressure switch in the delivery line, located downstream of each level control shut-off valve, senses the pressure drop when the valve closes. At approximately 2 psi (13.8 kPa), the switch closes and turns on the REFUEL SHUTOFF light for the related tank on the refuel/defuel panel.

When the REFUEL/DEFUEL door is opened and electrical power is supplied to the refuel/defuel control panel, the following lights come on:

- FUELING ON caution light on the Caution and Warning Panel
- MASTER VALVE CLOSED light on refuel/defuel panel
- White floodlight for the refuel/defuel panel.

Gravity refuelling can be done through the gravity refuel adapter located on the top surface of the wing. The lightning proof cap is flush mounted.

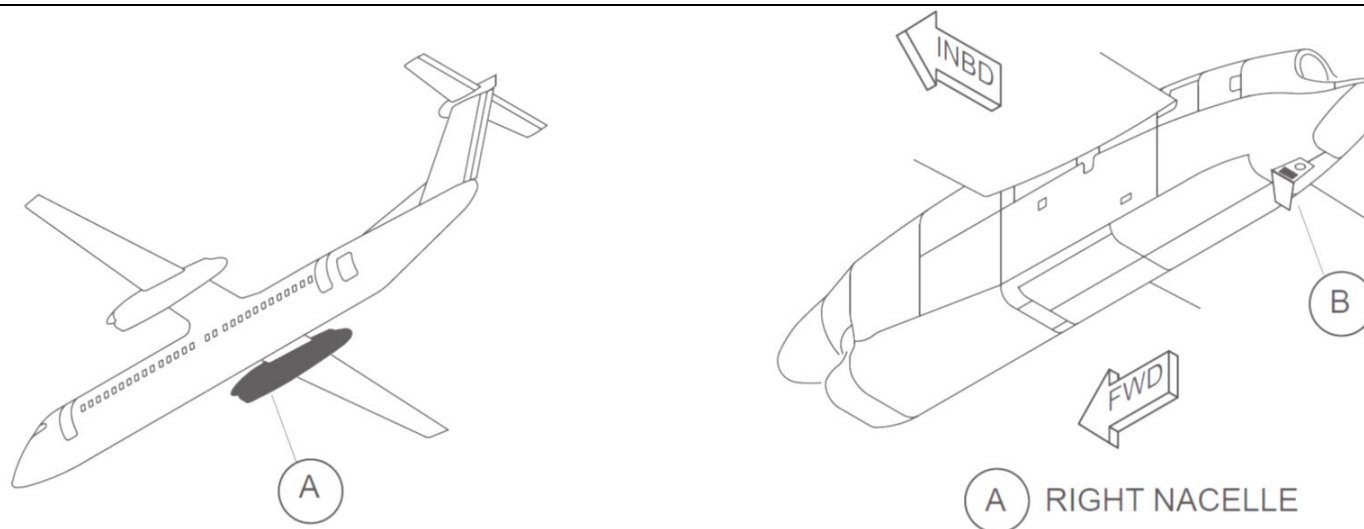


Figure 28-40 Refuel/Defuel Control Panel

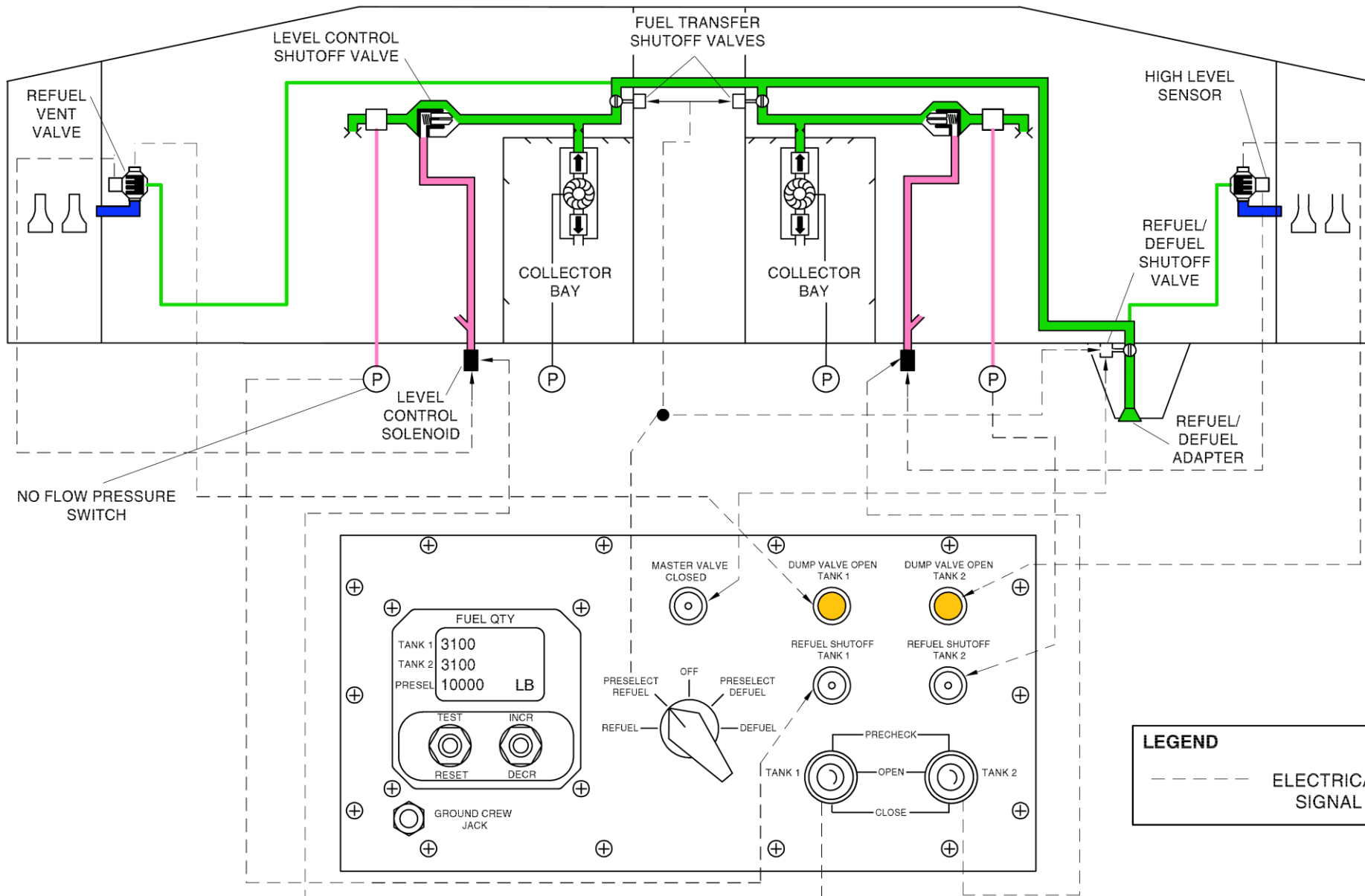


Figure 28-41 Refuelling Schematic

Refuel/ Defuel System – Component Description

Refuel/Defuel Control Panel

[Refer Figure 28-40](#)

The refuel/defuel control panel is located in the right engine nacelle, and controls pressure refuelling and defueling.

The refuel/defuel control panel has one selector switch, four toggle switches, five indicator lights and a Refuel/Defuel Indicator (RDI). The rotary selector switch is used to set either the pre-select or manual mode for refuelling or defueling. With the rotary mode switch in the PRESELECT REFUEL or PRESELECT DEFUEL position, the refuel/defuel operation stops automatically, when the pre-selected levels are reached. With the rotary mode select switch in the REFUEL or DEFUEL position, refuelling or defueling continues until the select switch is set to the OFF position. The high level sensors will always stop refuelling to the related tank, when a full tank is sensed.

The two PRECHECK/OPEN/CLOSE switches test the automatic shut-off operation of the related HLCU, when in the PRECHECK position. When in the switch is in the CLOSE position, no fuel can enter the related tank. When the switch is in the OPEN position, refuel and defuel operations can be done.

Refuel/Defuel Indicator

[Refer Figure 28-40](#)

The Refuel/Defuel Indicator (RDI) is located on the refuel/defuel control panel. The indicator shows the quantity of fuel:

- in each wing
- selected in the automatic mode for refuelling
- selected for each individual wing in the manual mode for refuelling

All quantities are shown in pounds or kilograms. The preselect values, which are shown on the RDI, are set using the INCR/DECR switch on the refuel/defuel control panel. The TEST RESET switch on the indicator is used to show and clear fault codes. Dashes shown on the RDI show a fault has been

found with the related tank. A blank shown on the display indicates a Refuel/Defuel Indicator, Fuel Quantity Computer, or refuel/defuel control panel failure.

Refuel/Defuel Shut-off Valve

[Refer Figure 28-41](#)

The refuel/defuel shut-off valve is installed in the refuel/defuel/transfer manifold, and is opened when refuelling or defueling. The valve is located on the right side aft wing spar. The valve is closed in flight to isolate the refuel adapter from the rest of the fuel system. The refuel/defuel shut-off valve is operated by an electrically driven actuator. A thermal relief valve is installed in the valve to prevent excessive pressure build-up in the fuel system lines.

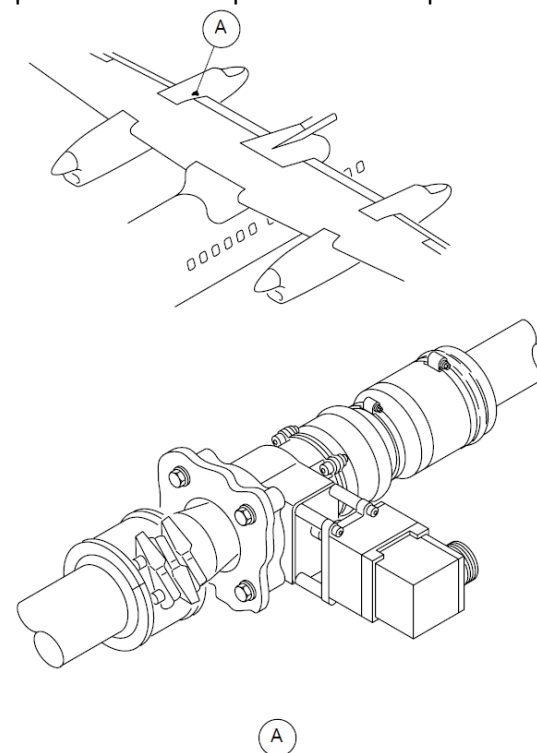


Figure 28-41 Refuel/Defuel Shut-Off Valve & Actuator

Refuel/Defuel Shut-off Valve Actuator

[Refer Figure 28-41](#)

The refuel/defuel shut-off valve actuator is attached to the refuel/defuel shut-off valve by a spline shaft. The actuator is installed on the dry side of the rear wing spar. The actuator has a physical and electrical position indication and may be replaced without draining fuel from the tank. The actuator has a lever that operates the valve and gives a position indication. The actuator is controlled by the mode selection switch on the refuel/defuel control panel. The switch opens the valve and turns off the MASTER VALVE CLOSED indicator light when any refuel/defuel mode is set.

Refuel/Defuel Adapter

[Refer Figure 28-42](#)

The refuel/defuel adapter is installed in the right engine nacelle. The adapter gives single-point pressure refuel/defuel access for the aircraft. The adapter has a metal body that has a spring-loaded closed poppet valve. A circular cap protects the poppet valve from damage and contamination.

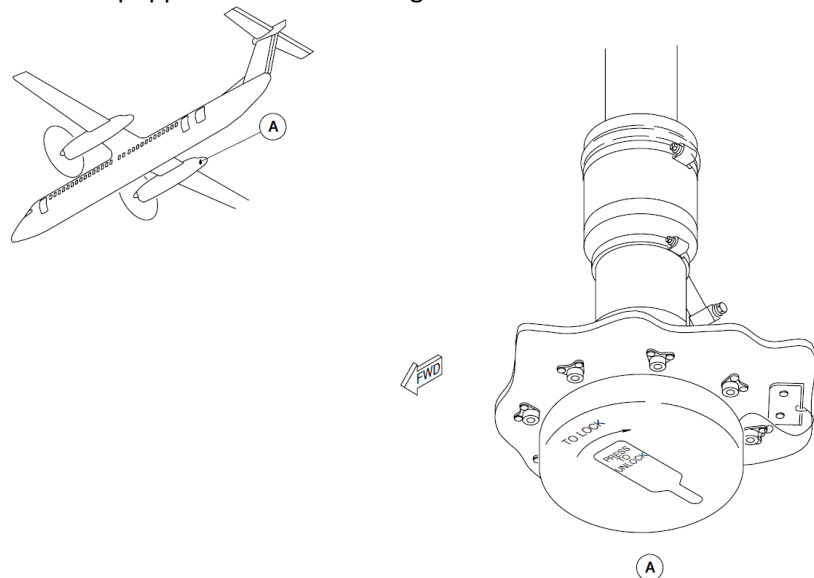


Figure 28-42 Refuel/Defuel Adapter

No-Flow Pressure Switch

[Refer Figure 28-43](#)

There are two no-flow pressure switches in the refuel/defuel system. A no-flow pressure switch is installed downstream of each level control shut-off valve. The no-flow switch senses the pressure in the fuel lines. When the level control shut-off valve closes, the pressure in the line decreases and the switch closes. This completes an electrical circuit and turns on the related white REFUEL SHUTOFF (TANK1 or TANK2) indicator on the refuel/defuel control panel. This white light indicates that related level control shut-off valve is closed and refuelling of that fuel tank is no longer possible.

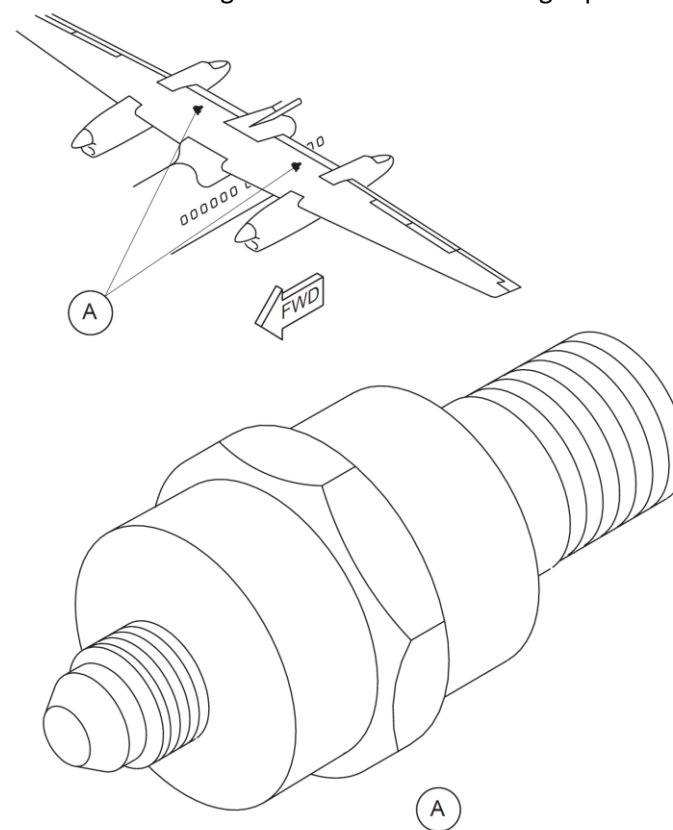


Figure 28-43 No-Flow Pressure Switch

Refuel Vent Valve

Refer Figure 28-44

A refuel vent valve is installed in each outboard bay tank wall. The valves are kept open with fuel pressure during refuelling. This protects the structure if an automatic or backup shut-off does not occur. If the tank overfills due to a malfunction of the pressure refuelling shut-off system, the open valve dumps fuel into the surge bay. If the surge bay fills up, fuel will spill overboard through the NACA vents. The refuel vent valve also releases pressure from the tank if it is more than 3.0 ± 0.25 psi (20.7 ± 1.72 kPa) above atmospheric pressure. An electrical switch on the valve senses the valve position and sends the information to the refuel/defuel control panel. The DUMP VALVE OPEN (TANK1 or TANK2) indicator lights on the refuel/defuel control panel come on when the related refuel vent valve is open. If the valve does not open during refuelling, the refuel/defuel control panel will stop the pressure refuelling.

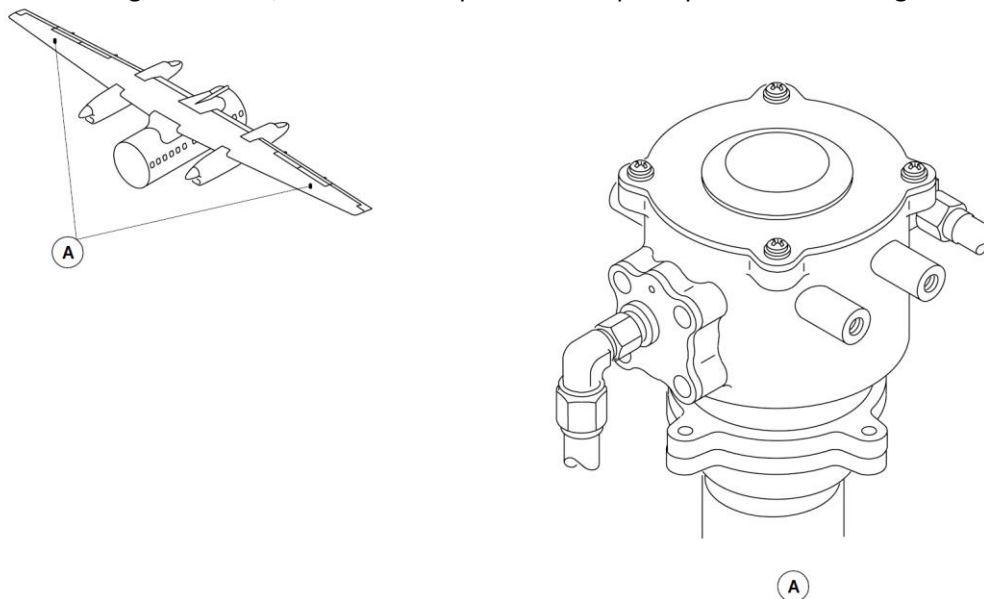


Figure 28-44 Refuel Vent (Dump) Valve

High Level Sensor

Refer Figure 28-45

A high level sensor is attached to each refuel vent valve. The high level sensor is a dual thermistor bead type. One bead gives a reference temperature while the other senses the fuel level. The sensors are monitored by the high level control unit.

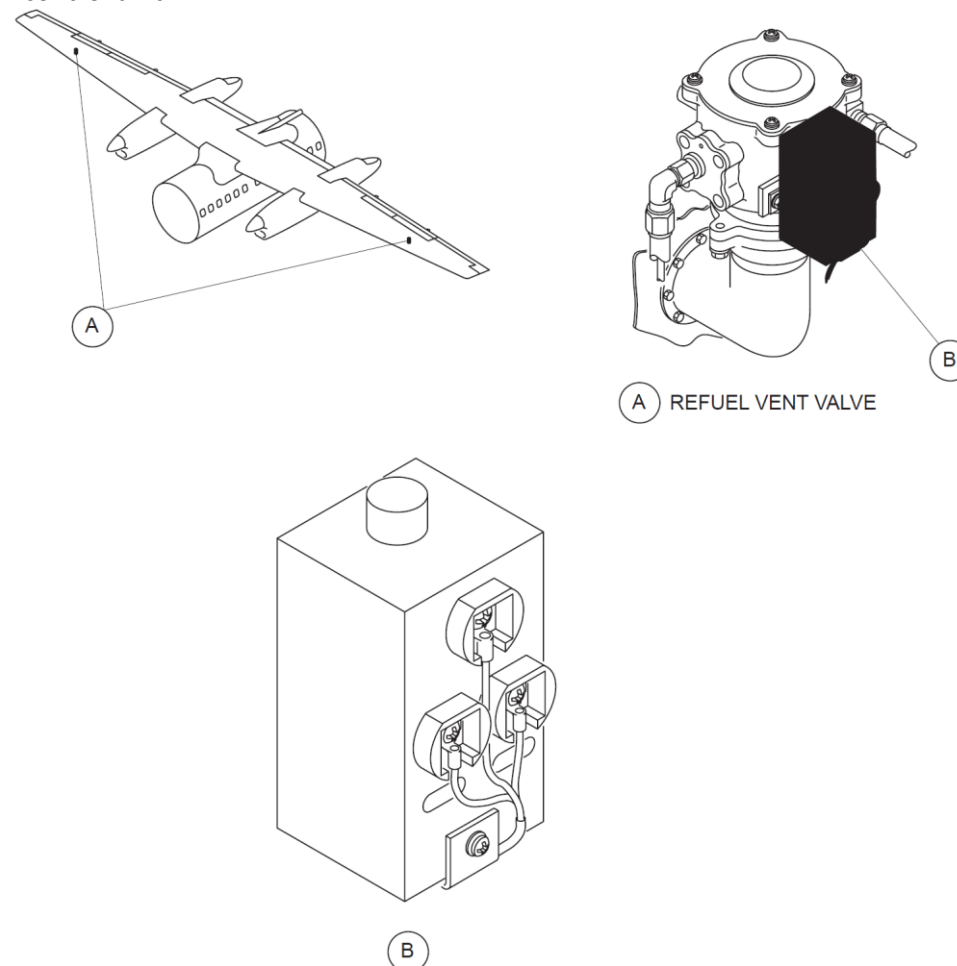


Figure 28-45 High Level Sensor

Refuel/Defuel System – Controls & Indications

[Refer Figure 28-46](#)

The RDI on the refuel/defuel panel records fault information from the FQC.

The TEST /RESET toggle switch on the RDI shows and clears fault codes.

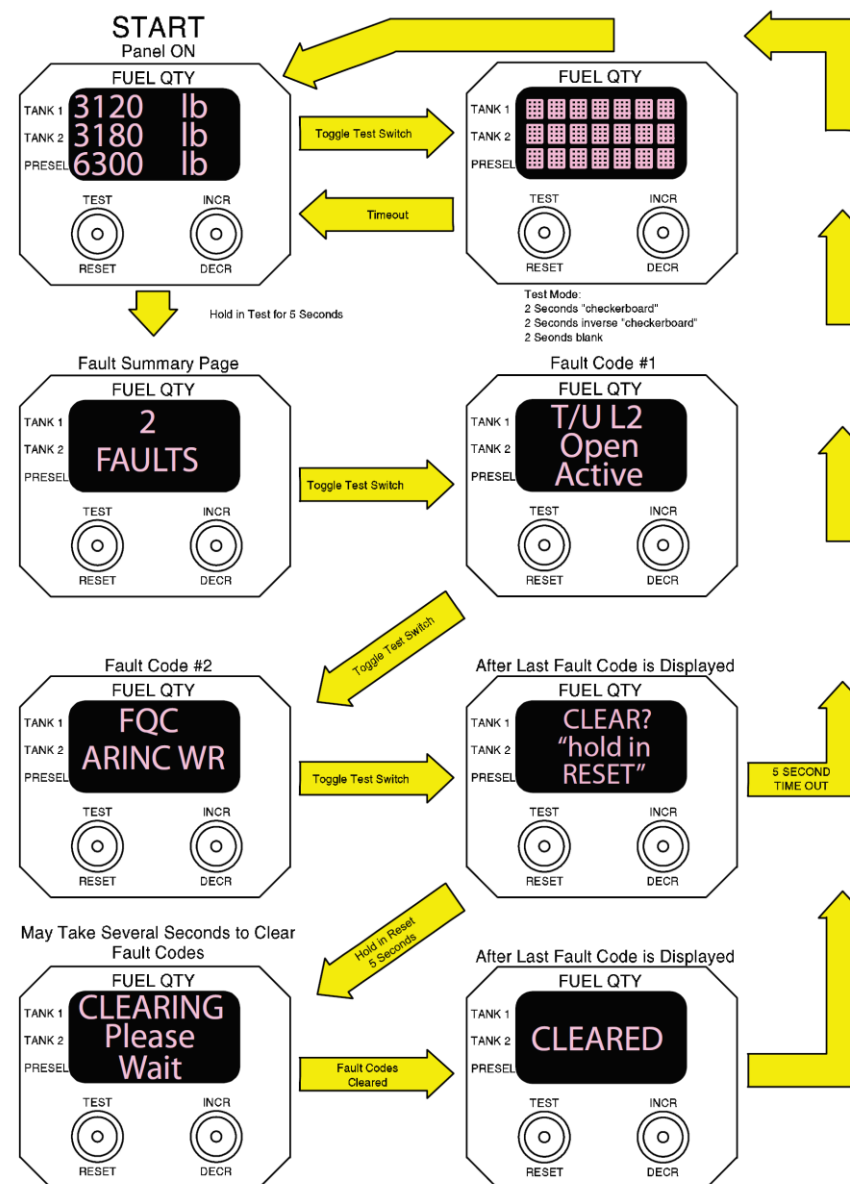


Figure 28-46 Fuel System Fault Codes

Preselect Refueling Operation

[Refer Figure 28-47 \(Sheet 1 of 2\)](#)

The INCR/DECR toggle switch on the RDI sets the desired amount of fuel for an automatic refuelling or defueling operation. When the INCR/DECR switch is toggled to the INCR or DECR position, the current PRESEL display value on the RDI will increase or decrease by 10 lbs (or 10 kg).

After this has been repeated 10 times, the PRESEL display value will increase or decrease by 100 lbs (or 100 kg). The PRESEL display value will never fall below zero or exceed 5800 kg (13780 lbs).

Four seconds after the INCR/DECR switch is returned to the centre position, the refuelling or defueling process will start. If the INCR/DECR switch is moved out of the centre position, the refuelling or defueling process will stop and will be restored after four seconds.

[Refer Figure 28-47 \(Sheet 2 of 2\)](#)

In the automatic PRESELECT refuelling mode, the desired fuel quantity is selected using the INCR/DECR switch on the RDI.

The PRESEL segment of the RDI displays the current total quantity of both tanks before the desired quantity is selected. When the desired quantity has been set, the refuelling process starts four seconds after the INCR/DECR switch is returned to the centre position.

The preselected total fuel quantity is distributed equally between the two tanks. If the tank quantities reach an imbalance of 600 lbs (273 kg), the PRESEL segment will flash the word BALANCE.

Refuelling to the heavy tank is stopped until the imbalance is less than 350 lbs (227 kg).

During pressure refuelling (automatic or manual), the REFUEL SHUTOFF (TANK 1 and TANK 2) advisory lights will extinguish to confirm that the tanks

are being refuelled.

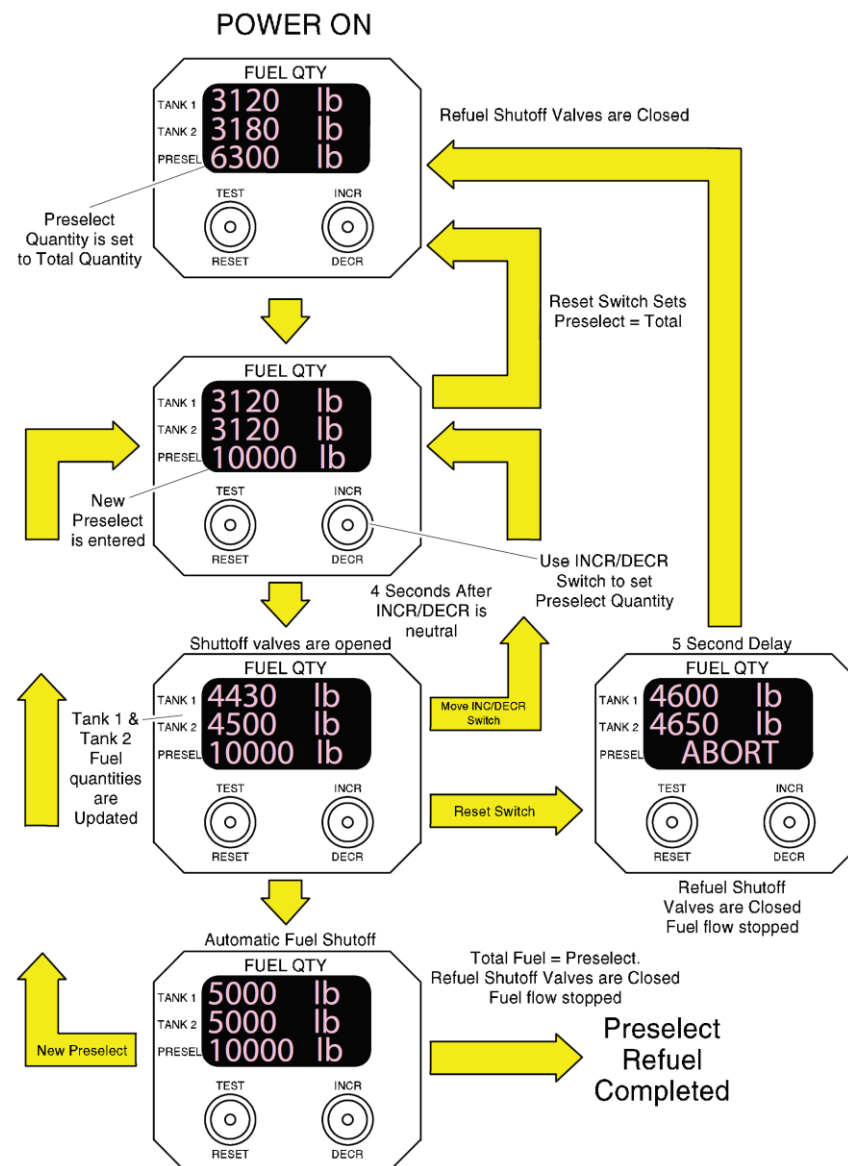
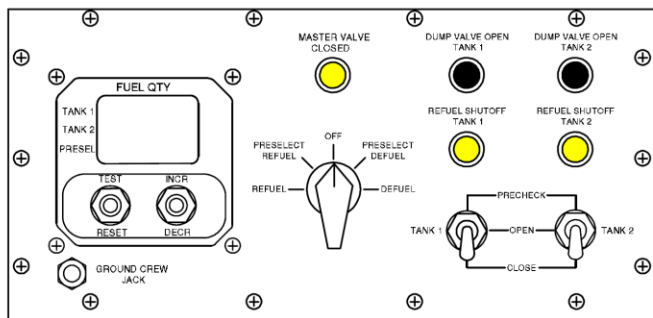
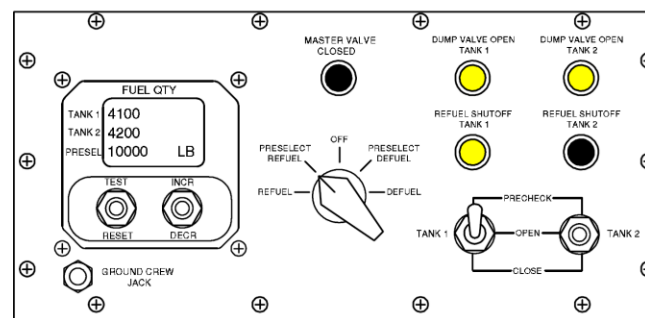


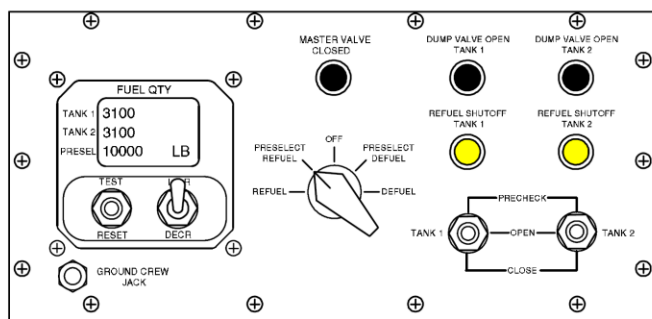
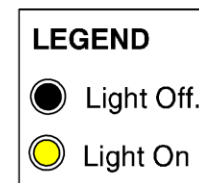
Figure 28-47 (Sheet 1 of 2) Preselect Refueling



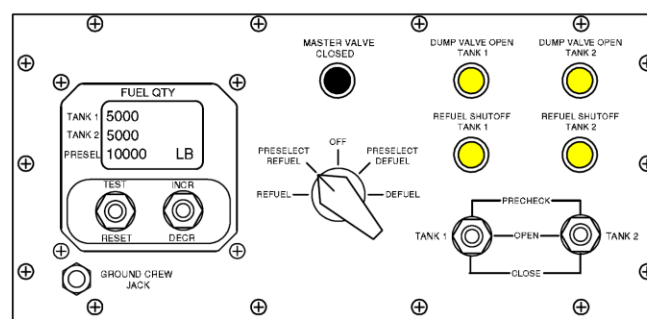
INITIAL CONFIGURATION



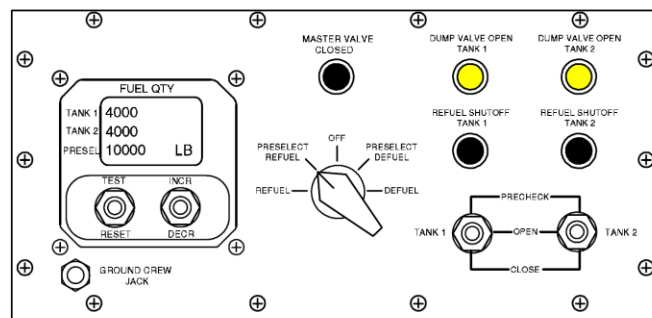
PRE-CHECK TEST (No. 1 TANK) DURING REFUELING



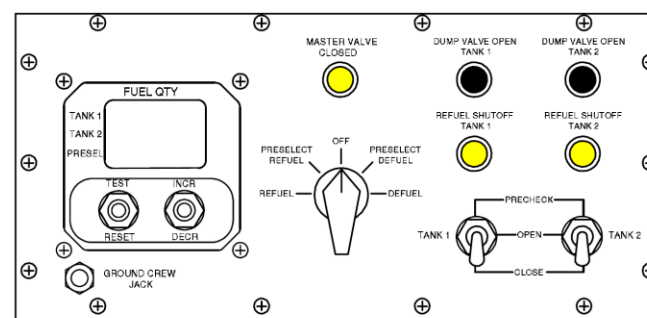
PANEL SET TO INITIATE REFUELING



TANKS FULL



REFUELING IN PROGRESS



REFUELING COMPLETE

Figure 28-47 (Sheet 2 of 2) Preselect Refueling

Preset Refueling – Electrical Schematic

[Refer Figure 28-48 \(Sheet 1 of 2\)](#)

The RDP pin 27 is powered at all times from the three REFUEL circuit breakers.

When the RDP door is opened, the RDP door switch supplies a ground signal to:

- The RDP flood light from the RDP pin 61 which causes the light to illuminate (see Sheet 2).
- The CAWP connector 3312 P5 pin 18 from the RDP pin 14 which causes the FUELING ON light to illuminate (see Sheet 2).
- To the RDP pin 7 to the fuel transfer relay panel pin “K” energizes the relay K9 closed (see Sheet 1).

The closed relay K9 supplies 28 VDC to:

- Through pin “R” to the FQC pin 49 (see Sheet 2).
- Through pin “P” to the FQC pin 55 (see Sheet 2).
- Relay K5-A2 which supplies power to energize open the Tank 1 R/D/T SOV.
- Relay K6-A2 which supplies power to energize open the Tank 2 R/D/T SOV.

Contact C2 supplies 28 VDC power:

- Through pin “H” to the RDP pin 50 to supply power to the five indicator lights (see Sheet 1).
- The RDP on pin 27 and the (open door) ground on pin 62 to energize the Refuel/ Defuel SOV closed. The closed Refuel/ Defuel SOV sends a (valve closed) ground signal to the RDP pin 41 which causes the MASTER VALVE CLOSED light to come on.
- Through pin “S” to the FQC pins 41, 42, and 52 (see Sheet 2).
- To relay K10-B3 this supplies power to the HLCU (see Sheet 1). The fuel nozzle is connected to the RDP adapter and pressurized fuel (20 psi min. to 50 psi max.) is supplied to the RDP Refuel/Defuel SOV.

LEGEND

POWER

GROUND

POWER AFTER PRE-SELECT

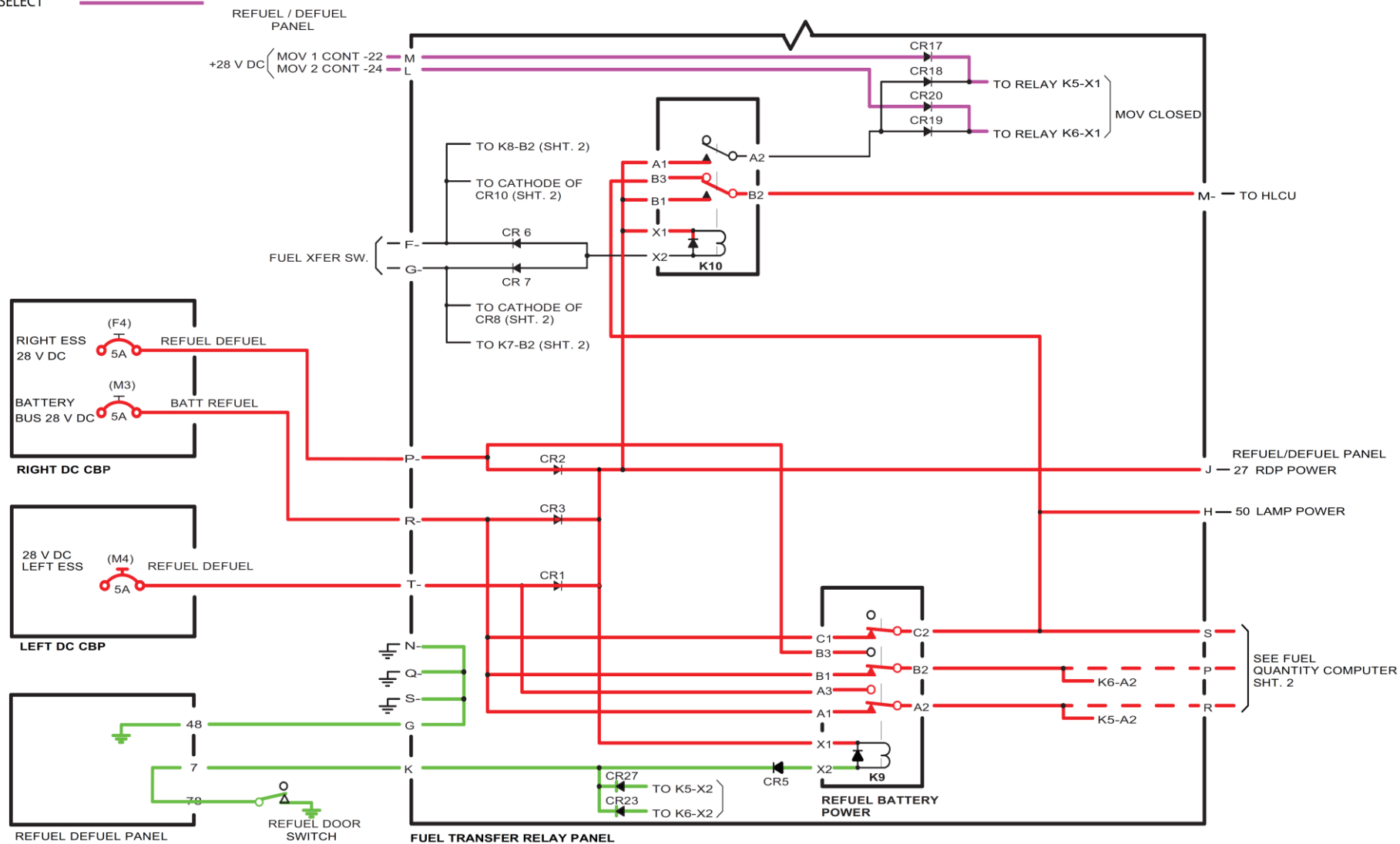


Figure 28-48 (Sheet 1 of 2) Preselect Refueling – Electrical Schematic

[Refer Figure 28-48 \(Sheet 2 of 2\)](#)

The fuel selector is set to the PRESELECT REFUEL position. This action causes:

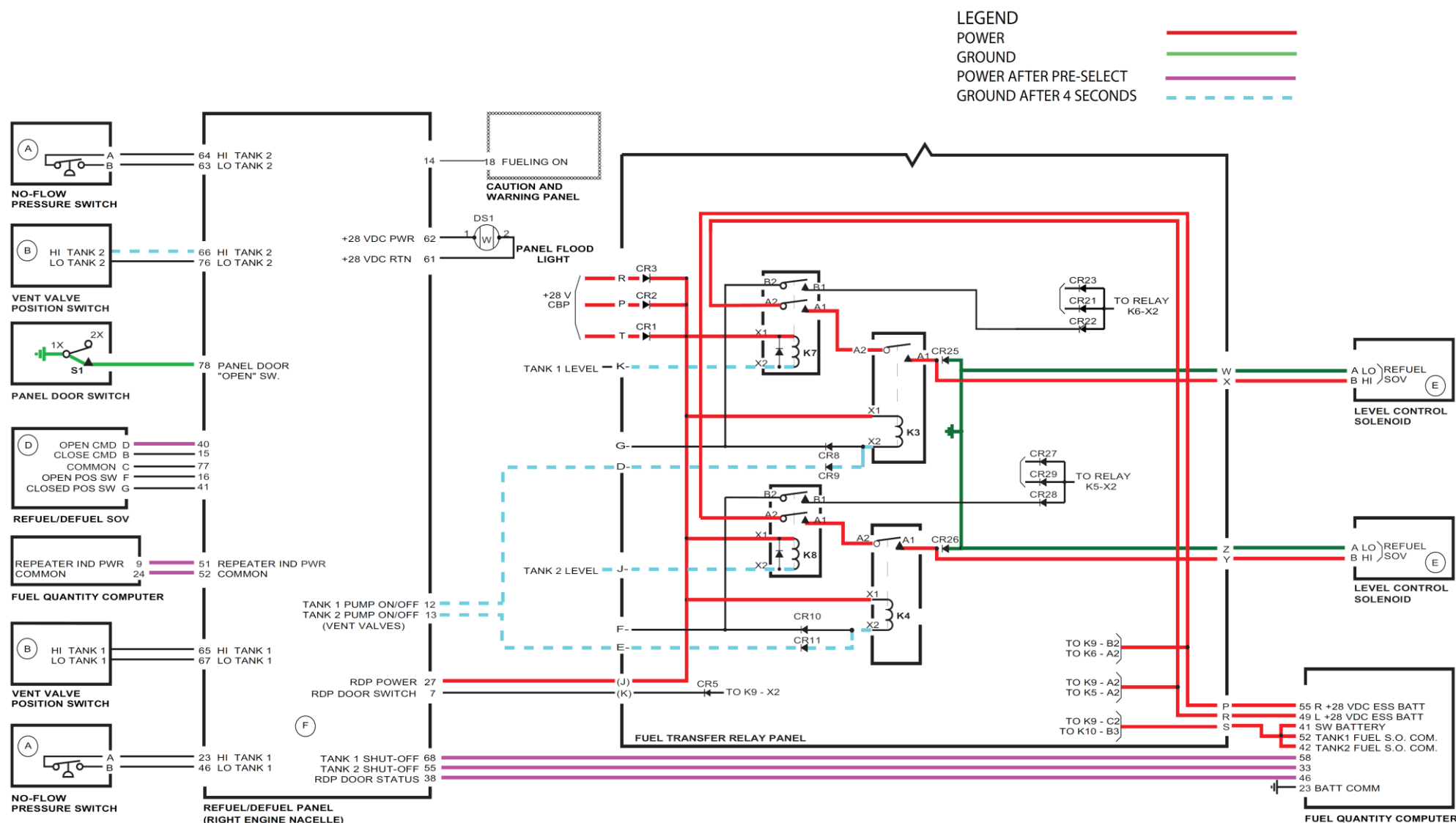
- The FQC pin 46 to receive an RDI enable signal (RDP Door Status) from the RDP pin 38.
- The FQC sends an indication power to the FQC pin 9 to the RDP pin 51 and an indication common to the FQC pin 24 to the RDP pin 52. These signals cause the Refuel/Defuel Indicator (RDI) to come on.

The INCR/DECR switch is held to the INCR position to select the fuel load.

- This sends a load request signal to the FQC which calculates the fuel load to the two fuel tanks
- Four seconds after the INCR/DECR switch is released, the FQC sends a 28 VDC (valve open) signals to the RDP pins 55 and 68.
- The RDP sends the 28 VDC (valve open) signals from pins 22 and 24 to the fuel transfer relay panel pins “L” and “M” which causes the relays K5 and K6 to energize closed to energize the two R/D/T SOV open.
- The RDP also sends a 28 VDC (valve open) signal from pin 40 directly to the Refuel/Defuel SOV which causes the valve to open and allows pressurized fuel into the fuel manifold. The open Refuel/Defuel SOV removes the ground signal from the RDP pin 41 which causes the MASTER VALVE CLOSED light to extinguish.
- The pressurized fuel opens the two vent valves (one in each tank) which send two ground signals to the RDP pins 65 and 66 and causes the DUMP VALVE OPEN 1 and 2 lights to illuminate.
- The two (vent valve open) ground signals from the RDP pins 12 and 13 to the fuel transfer relay panel pins “D-” and “E-” energize the relays K3 and K4 closed.
- The tank 1 and 2 High Level Sensor (not full) ground signals keep the relays K7 and K8 energize closed.
- The closed relays K7 and K3 energize the tank 1 Refuel Solenoid Valve open.
- The closed relays K8 and K4 energize the tank 2 Refuel Solenoid Valve open.
- The two open refuel solenoid valves allow the back-pressure fuel to be

bled off from the two level control SOV which open the two level control SOV and allow the fuel to enter the fuel tanks.

- The no-flow pressure switches sense 2 psi and open, causing the REFUEL SHUT-OFF lights to extinguish.
 - When the FQC senses the pre-selected level, the FQC stops the refuel operation to each tank. The FQC removes the 28 VDC (valve open) signals to the RDP pins 55 and 68.
 - The RDP removes the ground signals from pins 12 and 13 to the fuel transfer relay panel pins “D-” and “E-” which causes the relays K3 and K4 to de-energize open.
 - The open relays K3 and K4 remove 28 VDC from pin “X” and “Y” which causes the Tank 1 and 2 refuel solenoid valves to de-energize closed which causes the level control valves to close and stop the fuel flow.
 - The RDP also removes the ground (valve close) signals from pins 22 and 24 to the fuel transfer relay panel pins “L” and “M” which causes the relays K5 and K6 to de-energize open to energize the two R/D/T SOV closed and stop the fuel flow.
 - The decrease in fuel pressure causes the no-flow pressure switches to close which supplies ground signals to the RDP pins 23 and 64 and cause the REFUEL SHUT-OFF TANK 1 and 2 indicator lights to illuminate.
- If the FQC does not automatically stop the refueling and the tanks become full, the high level sensors remove the ground signals from the relays K7 and K8. This de-energizes the Fuel Solenoid Valves which causes the two Level Control Valves to close and stop the fuel flow.



Fuel Indicating System

Introduction

The fuel indicating system supplies information on fuel quantity, low level detection and fuel temperature for display in the flight compartment. Fuel quantity is also shown on the refuel/defuel control panel.

General Description

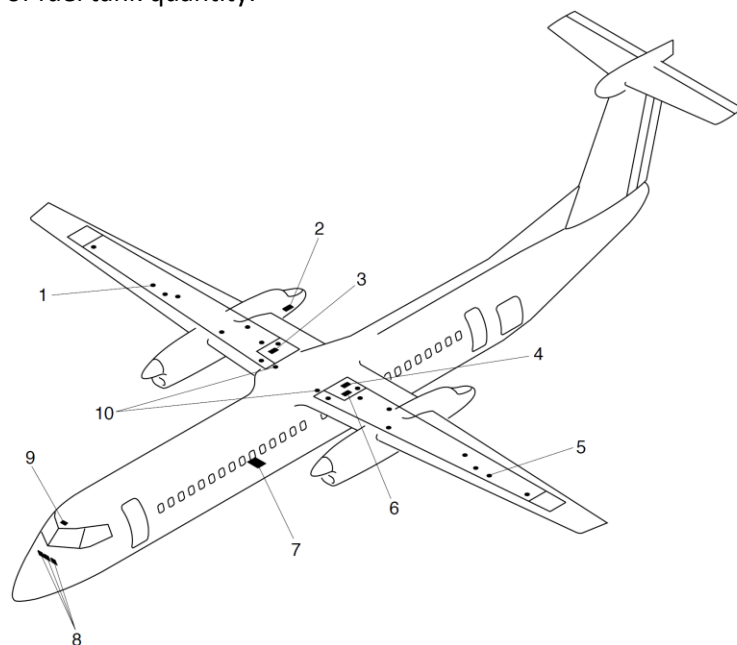
[Refer Figure 28-49](#)

The fuel indicating system provides an accurate measure of the fuel quantity in the wing tanks. Nine DC capacitance type fuel probes located in each wing send inputs to the Fuel Quantity Computer (FQC) to calculate the aircraft fuel quantity. The Engine Indicating System (EIS) in the flight compartment displays the fuel quantity information to the flight crew. The fuel quantity information is also displayed on the Refuel/Defuel Indicator (RDI) located in the refuel/defuel panel. A magnastick located in each wing provides an independent mechanical indication of fuel tank quantity.

Low fuel level is sensed by a float switch in each collector bay and is indicated by a caution light on the Caution and Warning Panel. Fuel tank temperature is measured by a Resistance Temperature Device (RTD) sensor in the left collector bay and is displayed in the flight compartment.

The fuel indicating system has these components:

- Fuel Quantity Computer
- High Level Control Unit
- High Level Switch
- Float Switch
- Temperature Sensor
- Magnastick
- Fuel Quantity Probe
- EMI Filter Adapter



LEGEND

1. Right wing fuel probe (9).
2. Refuel/defuel panel.
3. Low level float.
4. Temperature sensor.
5. Left wing fuel probe (9).
6. Low level float.
7. Fuel quantity computer.
8. EIS display.
9. Caution and warning panel.
10. EMI filter adapter.

Figure 28-49 Fuel Indicating System

Detailed Description

[Refer Figures 28-49, 28-50 & 28-51](#)

The fuel indicating system shows the aircraft fuel quantity in the flight compartment and on the refuel/defuel control panel. The fuel quantity in each tank is shown digitally on the bottom center of the Engine Display (ED). The Fuel Page of the Multi-Function Display (MFD) shows the fuel quantity in each tank on two simulated analog dials and the total fuel quantity in digital form. The fuel quantity can be shown in kilograms (kg) or pounds (lbs).

Fuel quantity and temperature data are shown in white on the Engine Display (ED) and on the Fuel Page of the Multi-Function Display (MFD). White dashes replace the display digits if the data is not valid. The display changes color to yellow if in the caution range or red if in the warning range.

If a fuel imbalance condition is detected, a yellow [BALANCE] advisory message is shown just above the FUEL legend of the ED. The message flashes continuously until a pilot action rectifies the imbalance. An imbalance condition is also shown on the MFD Fuel page by the analog pointers changing to yellow and both ED digital fuel quantities changing to yellow.

The fuel inlet temperatures to the left and right engines are shown digitally on the bottom of the ED, just below the related tank fuel quantity. The digits are shown in white with a + or – sign. The digits change to yellow if the fuel temperature is in the caution range and to red if the temperature exceeds the allowable limits. A °C unit is shown between the two inlet temperatures. The fuel temperature in the left collector tank is shown digitally on the Fuel Page, just below the left fuel quantity dial.

The Fuel Quantity Computer (FQC) supplies the fuel probes with a 1kHz electrical excitation signal that causes an average dc output proportional to the effective capacitance of each probe. The nine probes in each tank are connected to seven excitation signals. The output of each probe is wired in parallel to one of two processors in the FQC. This arrangement allows for fault isolation to a single probe and is recorded in the FQC NVM.

The FQC converts the total of all valid probe signals to fuel weight for each tank. The FQC then transmits the data for left, right, and total fuel quantity, to the Input Output Processors (IOPs) in the Integrated Flight Cabinets (IFCs). The IOPs in the IFCs transmit the fuel quantity signal to the Electronic Instrument System (EIS) through the dual ARINC 429 data busses. The information is also shown on the Refuel/Defuel Indicator (RDI) during refueling. The RDI shows the quantity of fuel in each wing and the total quantity of fuel when selected in the automatic mode for refueling. When selected in the manual mode, the RDI shows only the quantity of fuel selected for each individual wing. All quantities can be shown in pounds or kilograms.

The pre-select values for refuelling are selected using the INCR/DECR switch on the refuel/defuel control panel and are shown on the RDI. The FQC automatically starts the refuel or defuel operation four seconds after the last switch operation. The RDI also displays fault information from the FQC. The TEST/RESET switch located on the refuel/defuel control panel shows and clears fault codes.

The FQC does not process faulted probe signals because their total output data is not valid. The indicated display for data not valid is as follows:

- The ED digital quantity is replaced with white dashes.
- The MFD FUEL page QTY needle pointer, scale marks and scale numbers go out of view.
- The MFD FUEL page TOTAL FUEL digital quantity is replaced with white dashes.
- The RDI digital quantity is replaced with yellow dashes.

A blank display indicates an FQC or RDI failure.

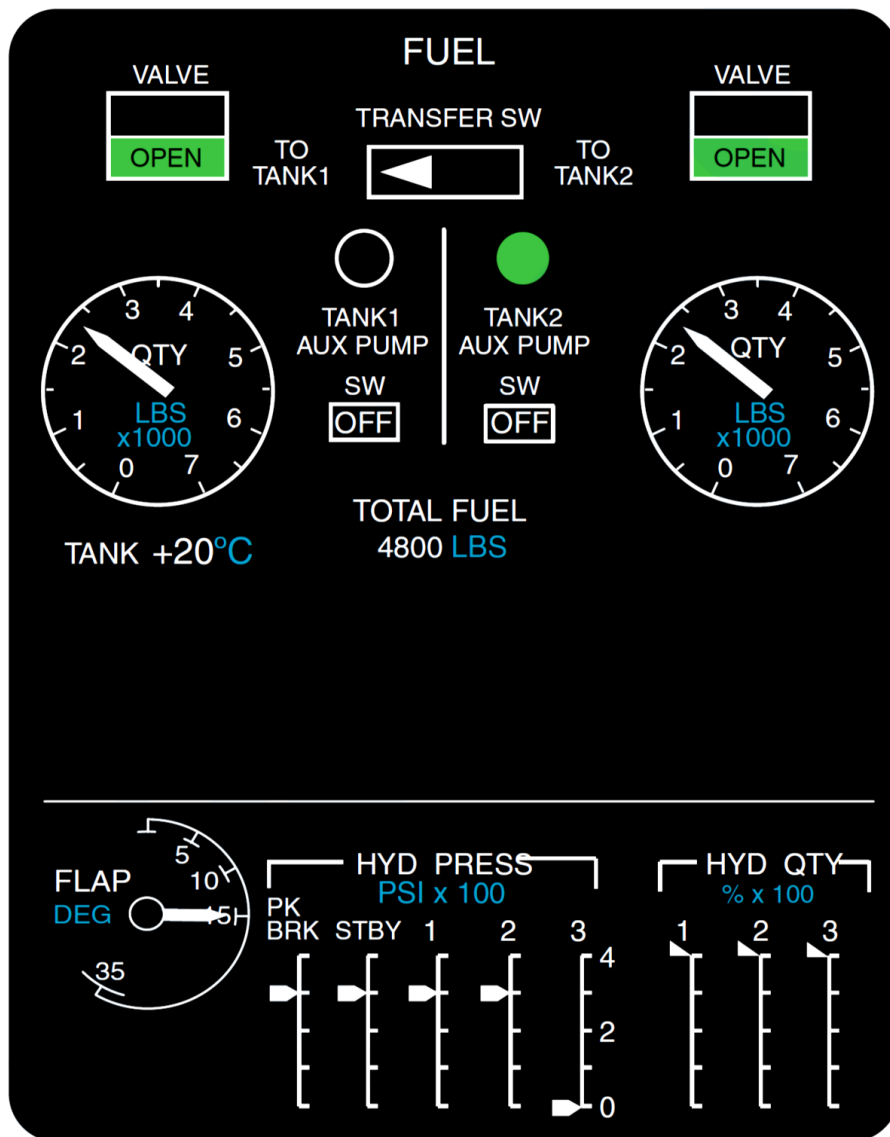
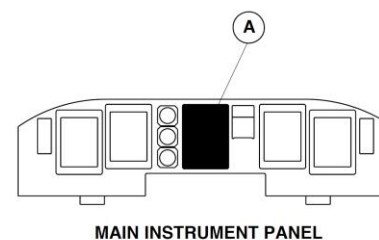


Figure 28-50 Multi-Function Display Fuel Page



LEGEND

1. Left/Right Fuel Flow Digital Value.
2. BALANCE Command Message (Fuel Limitation).
3. Left/Right Fuel Tank Quantity Digital Readout.
4. Left/Right Fuel Inlet Temperature Digital Value.

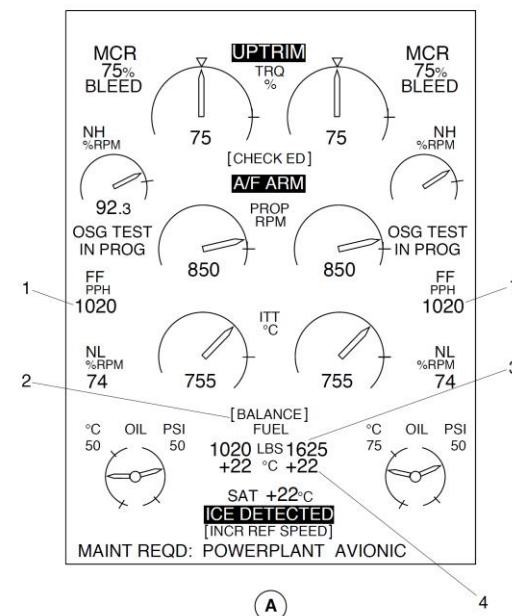


Figure 28-51 Fuel Quantity & Temperature Indications in Engine Display

Fuel Indicating System – Component Description

Fuel Quantity Computer

[Refer Figure 28-52](#)

The Fuel Quantity Computer (FQC) is located in the avionics bay below the passenger compartment floor. The FQC has two independent processors.

Each of these processors:

- calculates the fuel quantities in each tank independently
- controls the level control solenoid for pre-select refuelling
- performs a Built-In Test (BIT) of the fuel gauging components
- sends quantity, temperature compensation, and BIT data to the EIS displays.

The right processor also calculates fuel temperature measurements and transmits the information to the Refuel/Defuel Indicator (RDI).

The FQC performs an initial BIT when it starts up and a continuous BIT when it is in operation. The FQC is self-calibrating and needs no adjustments.

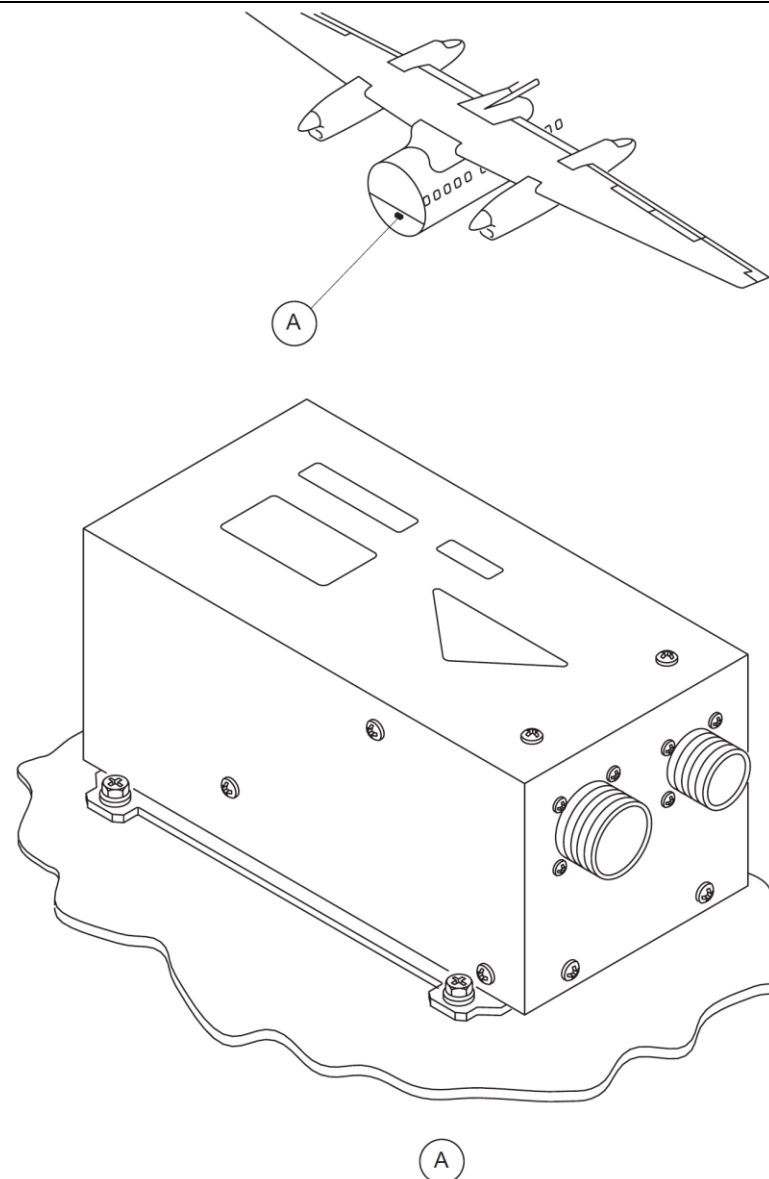


Figure 28-52 Fuel Quantity Computer

High Level Control Unit

[Refer Figure 28-53](#)

The High Level Control Unit (HLCU) monitors the high level sensors and transmits a signal to prevent overfilling the tanks during refuel or fuel transfer operations. This unit also serves as a signal conditioner for the temperature sensor. The unit is located below the cabin compartment floor.

The pre-check test switch on the refuel/defuel control panel tests the automatic shutoff operation of the high level control unit. The HLCU does a continuous BIT on itself and the high level sensor. Any fault sensed will stop the refuelling procedure.

The HLCU conditions the fuel tank temperature signal and transmits the signal to the IFC .

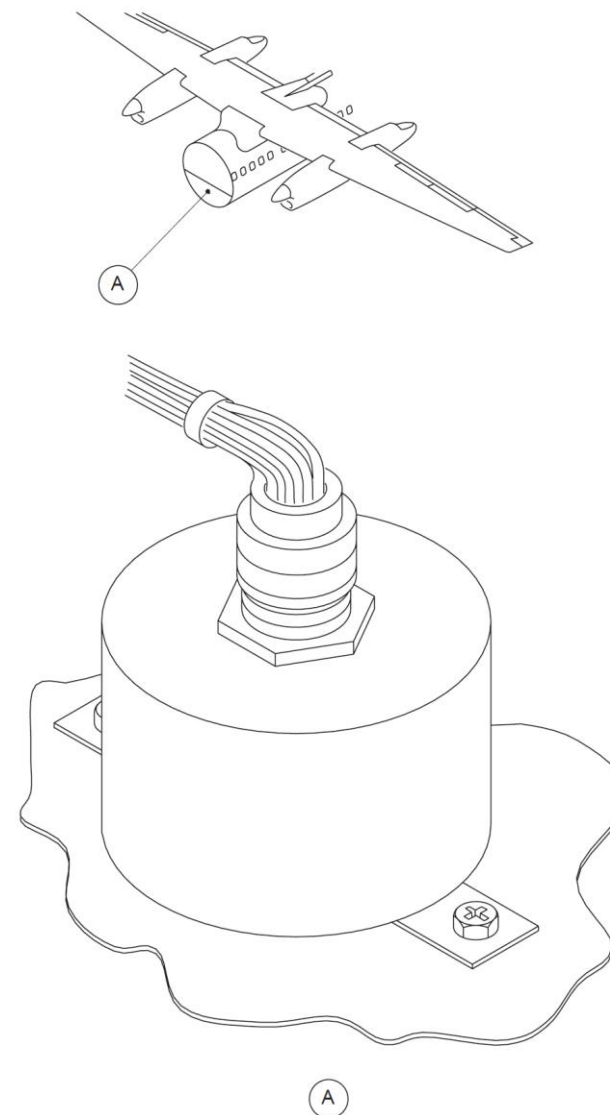


Figure 28-53 High Level Control Unit

Float Switch

[Refer Figure 28-54](#)

The low level float switch sends a signal to the caution and warning panel to indicate a low fuel level in the collector bay.

There are two low fuel level float switches, one installed in each collector bay. Each float switch is installed near the top of the collector bay on WS 60.9.

The control logic of this switch is independent of the Fuel Quantity Gauging System (FQGS). The switch will be activated when the fuel collector tank quantity drops to approximately 305 lbs and engine running with park brake OFF.

At this point the appropriate FUEL TANK LOW caution light on the caution and warning panel in the flight compartment will illuminate.

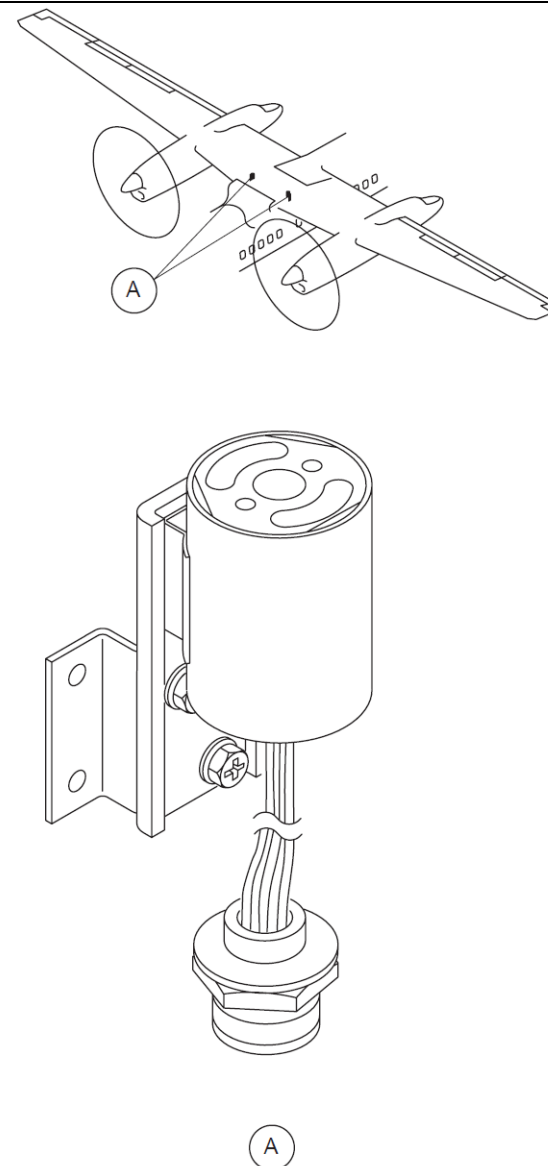


Figure 28-54 Float Switch

Temperature Sensor

[Refer Figure 28-55](#)

The temperature sensor senses the fuel temperature in the left tank and transmits the information for display in the flight compartment.

The tank temperature sensor is located on the No.1 tank collector bay wall.

The sensor uses electrical resistance to monitor the fuel temperature. The analog output is sent to the HLCU for signal conditioning, and then to the EIS for display. The system is operational when electrical power is supplied to the aircraft.

The temperature is shown on the MFD Fuel Page and ranges from -70 to +75°C (-94 to 167°F).

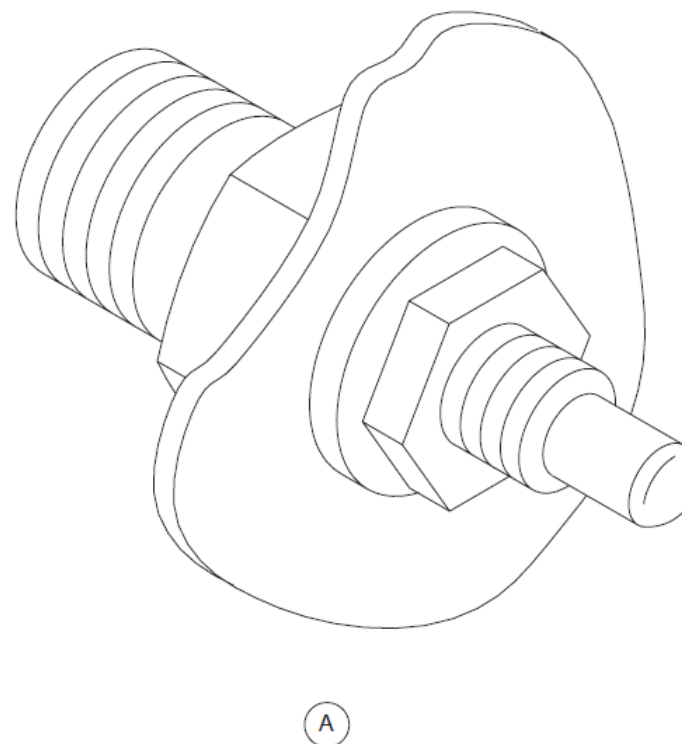
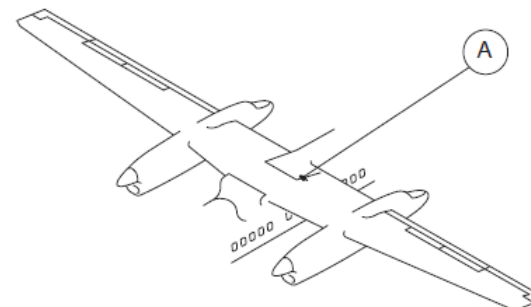


Figure 28-55 Temperature Sensor

Magnastick

[Refer Figure 28-56 & 28-57](#)

There are two magnasticks. One magnastick is installed in each main tank, outboard of the engines.

The magnasticks give an alternate means to measure the fuel quantity when the aircraft is on the ground. A magnastick is a calibrated rod with a magnet attached. It moves within a tube that extends vertically from the bottom of the fuel tank. A float moves up and down on the outside surface of this tube to match the fuel level in the tank. The float contains a magnet that attracts the magnet on the magnastick.

The magnastick is accessed at the bottom of the wing. When the rod is released, it moves slowly down until the rod magnet is attracted by the float magnet.

The fuel quantity is shown in US gallons or in litres.

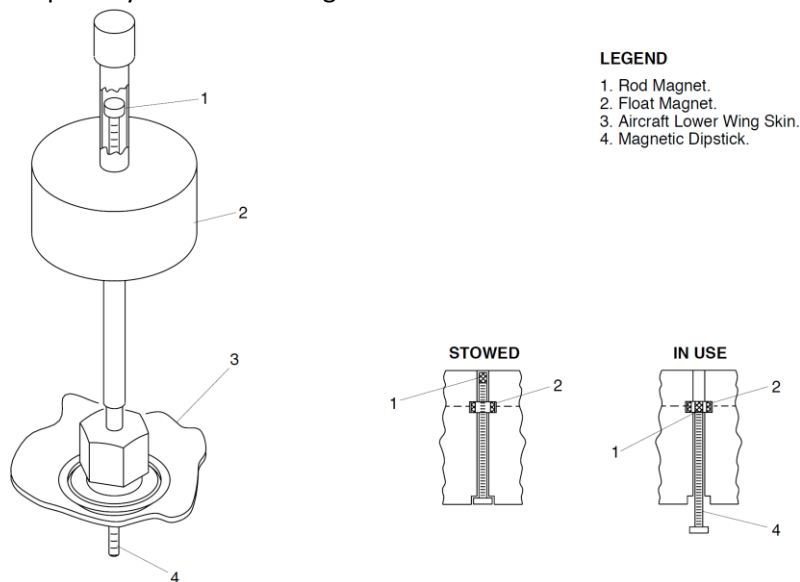


Figure 28-56 Magnastick Detail

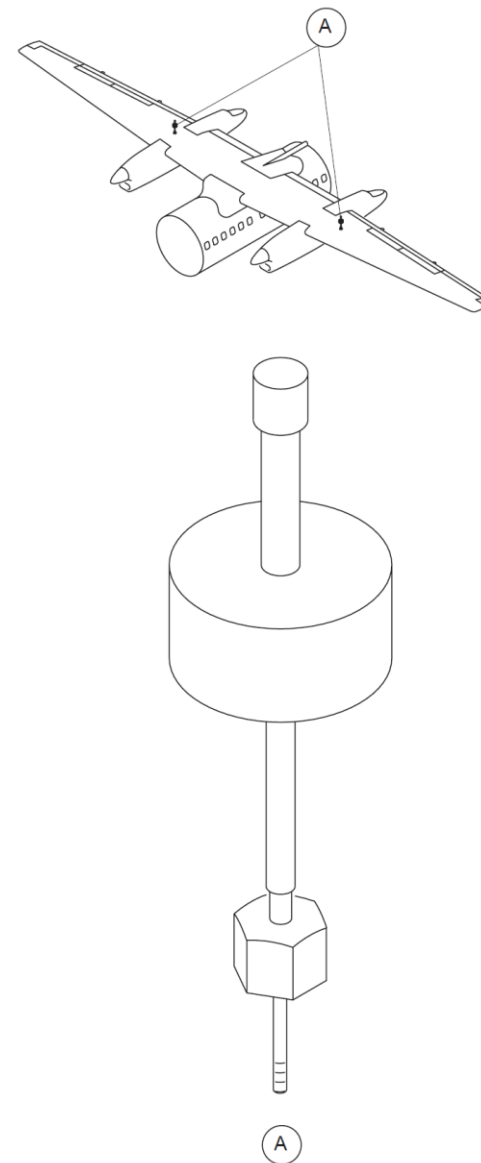


Figure 28-57 Magnastick

Fuel Quantity Probe

[Refer Figure 28-58](#)

There are 18 fuel quantity probes, nine in each wing. The probes contain two concentric metal cylinders, a terminal block, and two mounting brackets. The cylinders form the capacitor elements (plates); the inner plate is the high-impedance element and the outer plate is the low-impedance element. Changes in fuel levels around each probe cause corresponding changes in the effective capacitance of each probe. The probes are monitored by the fuel quantity computer.

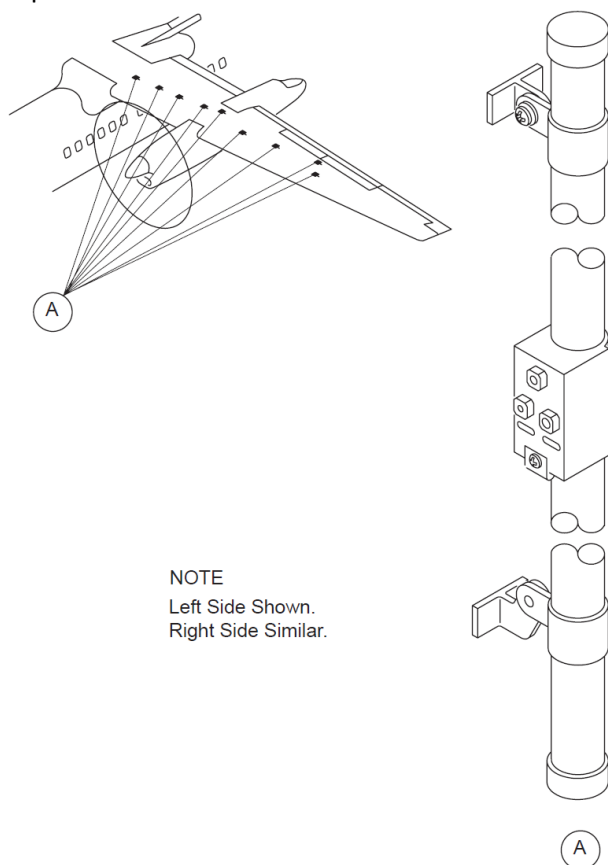


Figure 28-58 Fuel Quantity Probe

EMI Filter Adapter

There are two EMI filter adapters, one in each wing. The EMI filter adapters are installed on the dry bay connectors 2800-P4 and 2800-P5 inside the wing. The adapters contain filter capacitors to protect the fuel system from electromagnetic interference.

Fuel Quantity Indication – Electrical Schematic

[Refer Figure 28-59](#)

- The FQC supplies a 1 kHz electrical excitation to the nine probes.
- The nine fuel quantity probes in each wing sense the dielectric (capacitance) value changes caused by the changing fuel levels. As the fluid level rises the overall capacitance rises proportionately.
- The FQC measures this capacitance and converts the probe signals into a fuel weight (lbs or kg) ARINC signal.
- The FQC sends the converted quantity data to the IFC Input/output modules which converts the data for the EFIS.
- The EFIS converts the data into displays of fuel quantity the ED and MFD units.
- The FQC also sends the converted quantity data directly to the refuel/defuel panel for quantity indication.
- The fuel quantity can be shown in kilograms (kg) or pounds (lbs). This is accomplished by changing the pin programming on the FQC connector 2800-P2 pins 4, 10, 11 and 17.

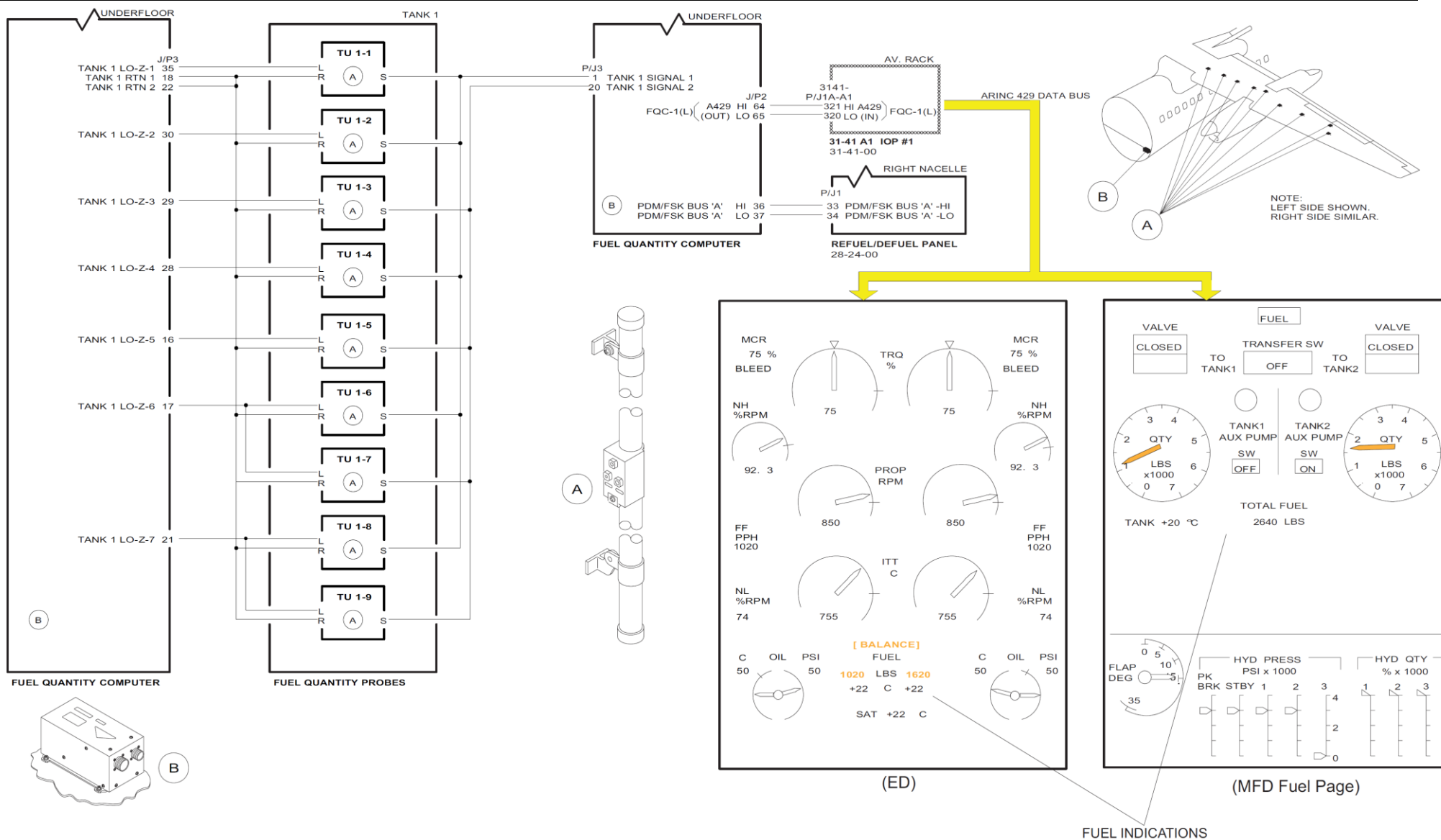


Figure 28-59 Fuel Quantity Indication – Electrical Schematic